# Discrete Mathematics

Sets, Relations, Functions, Sequences, Number Theory, and Induction

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# 1. Set Theory

Representations, Subsets, Set Operations, Inclusion–Exclusion, and Set Algebra

## 1.1 Foundations of Set Theory

Set theory is the basic language of discrete mathematics and computer science. Whether we are talking about a database table, the vertices of a network graph, a formal language, or a sample space in probability, we are working with sets.

**DEFINITION** **Set**

A **set** is an unordered collection of distinct objects, called the **elements** or **members** of the set. A set is said to **contain** its elements:

- $a \in A$ denotes that $a$ is an element of set $A$.
- $a \notin A$ denotes that $a$ is **not** an element of set $A$.

By convention, uppercase letters ($A, B, C, \ldots$) denote sets and lowercase letters ($a, b, c, \ldots$) denote individual elements:

- $A = \{2, 4, 6, 8\}$ is a finite set of four even integers ($2 \in A$, whereas $3 \notin A$).
- $B = \{a, b, c, \ldots, x, y, z\}$ denotes the lowercase letters of the English alphabet, where the ellipsis ($\ldots$) indicates a clear pattern.
- $\boldsymbol{P}$ denotes the infinite set of all positive integers.

**INTUITION & DEEP DIVE** **Why 'Unordered' and 'Distinct' Matter**

In programming, a mathematical set corresponds to a `Set` data structure (like `set` in Python or `std::unordered_set` in C++), rather than a `List` or `Array`:

1. **Unordered:** Membership ($a \in A$) does not depend on position; $\{2, 4, 6, 8\}$ and $\{8, 2, 6, 4\}$ are the exact same set.
2. **No Duplicates:** Listing an element more than once does not change the set ($\{1, 2, 2, 3\} = \{1, 2, 3\}$). When multiplicity matters, we use **multisets** (bags); when order matters, we use **tuples** or **sequences**.

## 1.2 Representations of Sets

We can describe a set in four main ways depending on whether it is finite, defined by a rule, or one of the standard number systems.

### 1.2.1 Roster (Enumeration) Method

When a set is finite (or has an obvious pattern), we can list its members inside curly braces `{ }`:

- $A = \{1, 2, 3, 4\}$ — a set of four integers.
- $B = \{a, b, \{a, b, c\}, \{a, c\}\}$ — a set with four elements, two of which are themselves sets ($a \in B$ and $\{a, c\} \in B$, though $c \notin B$).
- $H = \{\text{dog}, \text{cat}, \text{crocodile}\}$ — a set of three animals.

### 1.2.2 Standard Number Systems

Several number sets come up so often that they have dedicated boldface symbols:

| Symbol | Name | Definition / Elements |
|---|---|---|
| $\boldsymbol{P}$ | Positive integers | $\{1, 2, 3, \ldots\}$ |
| $\boldsymbol{N}$ | Natural numbers | $\{1, 2, \ldots\}$ (or $\{0, 1, 2, \ldots\}$ depending on convention) |
| $\boldsymbol{Z}$ | Integers | $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$ |
| $\boldsymbol{Q}$ | Rational numbers | $\left\{ \frac{p}{q} \mid p, q \in \boldsymbol{Z}, q \neq 0 \right\}$ |

| | | |
|---|---|---|
| $\boldsymbol{R}$ | Real numbers | All rational and irrational numbers on the number line |
| $\boldsymbol{C}$ | Complex numbers | $\{a+bi \mid a,b \in \boldsymbol{R}, i^2=-1\}$ |

**PITFALL / WARNING** **Does N Start at 0 or 1?**

Different authors disagree on whether 0 is a natural number. Here we follow the convention $\boldsymbol{N} = \{1,2,3,\ldots\}$ (matching $\boldsymbol{P}$), though many textbooks write $\boldsymbol{N} = \{0,1,2,\ldots\}$ and $\boldsymbol{Z}^+$ for $\{1,2,3,\ldots\}$. It is always good practice to check which convention a problem uses when 0 is a boundary case.

### 1.2.3 Set-Builder Notation

When listing every element is impractical, we describe a set by stating the condition an object $x$ must satisfy to belong to it:

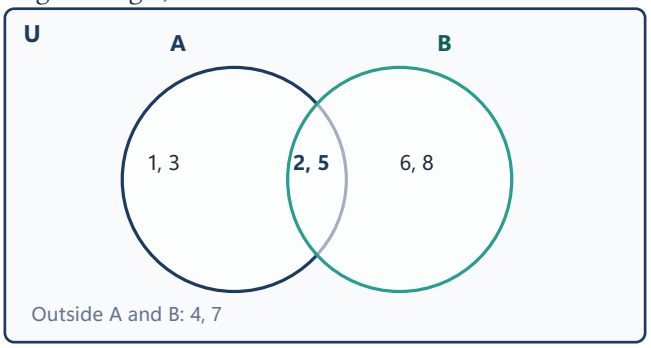
$$S = \{x \mid \text{condition on } \ x\}$$

The vertical bar $\mid$ is read as **"such that"**.

- **Example 1:** $A = \{x \mid x \in \boldsymbol{P}, x < 5\}$ gives $A = \{1,2,3,4\}$.
- **Example 2:** $B = \{x \mid \frac{x}{2} \in \boldsymbol{P}, 2 \le x \le 8\}$ requires $x$ to be a positive even integer between 2 and 8, so $B = \{2,4,6,8\}$.
- **Example 3:** $D = \{x \mid x \in \boldsymbol{N}, x < 5 \vee x > 10\}$ gives the infinite set $D = \{1,2,3,4,11,12,13,\ldots\}$.

### 1.2.4 Venn Diagrams

A **Venn diagram** represents sets visually. The **universal set** $U$—which contains all objects under consideration—is drawn as a bounding rectangle, and subsets of $U$ are drawn as circles inside it.




*Venn diagram for U = {1, 2, ..., 8}, A = {1, 2, 3, 5}, and B = {2, 5, 6, 8}.*

**EXAMPLE & WALKTHROUGH** **Roster and Set-Builder Practice**

**1. List the members of each set:**

- **(a)** $\{x \mid x \text{ is a real number such that } x^2 = 1\}$: Solving $x^2 = 1$ over $\boldsymbol{R}$ gives $\boldsymbol{\{-1, 1\}}$.
- **(b)** $\{x \mid x \text{ is a positive integer less than } 12\}$: $\boldsymbol{\{1,2,3,4,5,6,7,8,9,10,11\}}$.
- **(c)** $\{x \mid x \text{ is the square of an integer and } x < 100\}$: The integers whose squares are under 100 are $-9, \ldots, 9$, producing the squares $\boldsymbol{\{0,1,4,9,16,25,36,49,64,81\}}$.
- **(d)** $\{x \mid x \text{ is an integer such that } x^2 = 2\}$: Since $\pm\sqrt{2}$ are not integers, no integer satisfies this condition: $\varnothing$ (or $\boldsymbol{\{\}}$).

**2. Describe each set using set-builder notation:**

- **(a)** $\{0,3,6,9,12\}$: Multiples of 3 from 0 to 12: $\boldsymbol{\{3k \mid k \in Z, 0 \le k \le 4\}}$.
- **(b)** $\{-3,-2,-1,0,1,2,3\}$: Integers with absolute value at most 3: $\boldsymbol{\{x \in Z \mid -3 \le x \le 3\}}$.
- **(c)** $\{m,n,o,p\}$: $\boldsymbol{\{x \mid x \text{ is a lowercase English letter from } m \text{ to } p \text{ inclusive}\}}$.

## 1.3 Cardinality & Classification of Sets

### 1.3.1 Cardinality and the Empty Set

**DEFINITION** **Cardinality**

The **cardinality** of a finite set $A$, written $|A|$ or $n(A)$, is the number of distinct elements in $A$.

- For $A = \{1, 2, 3, ..., 10\}$, we have $|A| = 10$.
- A set with no elements has cardinality 0 and is called the **empty set** (or **null set**), written $\varnothing$ or $\{\}$:
  - $E = \{x \in \boldsymbol{N} \mid x < 1\} = \varnothing$, so $|E| = 0$.
  - $\{x \mid x < x\} = \varnothing$, since no number is strictly less than itself.
  - $\{\text{humans who have walked on Mars}\} = \varnothing$.

**PITFALL / WARNING** **The Empty Set $\varnothing$ vs. The Singleton Set $\{\varnothing\}$**

It is easy to mix up $\varnothing$ and $\{\varnothing\}$:

- $\varnothing = \{\}$ is an empty box with **zero** elements ($|\varnothing| = 0$).
- $\{\varnothing\}$ is a box containing **one** element—namely, an empty box ($|\{\varnothing\}| = 1$, since $\varnothing \in \{\varnothing\}$).

### 1.3.2 Subsets and Proper Subsets

**DEFINITION** **Subset & Proper Subset**

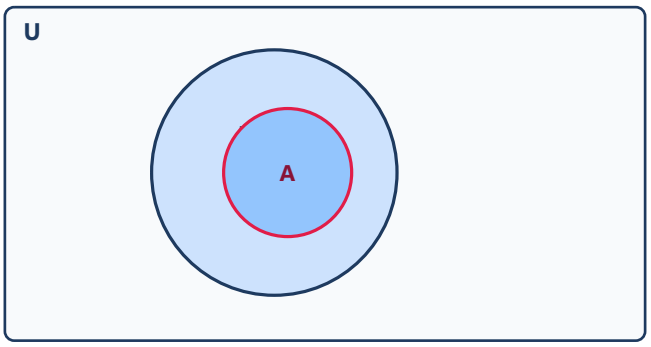
Let $A$ and $B$ be sets:

1. **Subset ($\boldsymbol{A \subseteq B}$):** $A$ is a **subset** of $B$ if every element of $A$ is also an element of $B$:

$$A \subseteq B \iff \forall x(x \in A \to x \in B)$$

   When $A \subseteq B$, we can also say $B$ is a **superset** of $A$ ($B \supseteq A$).
2. **Proper Subset ($\boldsymbol{A \subset B}$):** $A$ is a **proper subset** of $B$ if $A \subseteq B$ and $A \neq B$ (meaning $B$ has at least one element not in $A$).

For instance, $\{1, 2, 3\} \subset \{1, 2, 3, 4, 5\}$, and the standard number systems form a chain of subsets: $\boldsymbol{N} \subseteq \boldsymbol{Z} \subseteq \boldsymbol{Q} \subseteq \boldsymbol{R} \subseteq \boldsymbol{C}$.




*Venn diagram showing A as a subset of B (A ⊆ B) inside universe U.*

**CORE CONCEPT** **Basic Subset Properties**

For any sets $A$, $B$, and $C$:

1. **Reflexivity:** $A \subseteq A$ (every set is a subset of itself).
2. **Empty Set:** $\varnothing \subseteq A$ (the empty set is a subset of every set).
3. **Transitivity:** If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

**INTUITION & DEEP DIVE** **Why Is ∅ a Subset of Every Set?**

Saying $\varnothing \subseteq A$ means $\forall x(x \in \varnothing \to x \in A)$. Because $\varnothing$ has no elements, the condition $x \in \varnothing$ is always false, making the implication vacuously true. Put another way: for $\varnothing \subseteq A$ to fail, we would need to find an element inside $\varnothing$ that is missing from $A$—which is impossible.

**EXAMPLE & WALKTHROUGH** **Finding Intermediate Proper Subsets**

**Problem:** Let $X = \{1, 2, 5\}$ and $Z = \{1, 2, 5, 7, 8\}$. Find all sets $Y$ such that $X \subset Y$ and $Y \subset Z$.

**Solution:**

- Because $X \subset Y$ is a **proper** subset, $Y$ must contain $1, 2, 5$ plus at least one more element.
- Because $Y \subset Z$ is also a **proper** subset, every element of $Y$ must come from $Z$, and $Y$ cannot equal all of $Z$.
- The remaining elements in $Z$ are 7 and 8. Including just one of them gives the two valid possibilities:

$$Y_1 = \{1, 2, 5, 7\} \quad \text{and} \quad Y_2 = \{1, 2, 5, 8\}$$

### 1.3.3 Equal, Equivalent, and Disjoint Sets

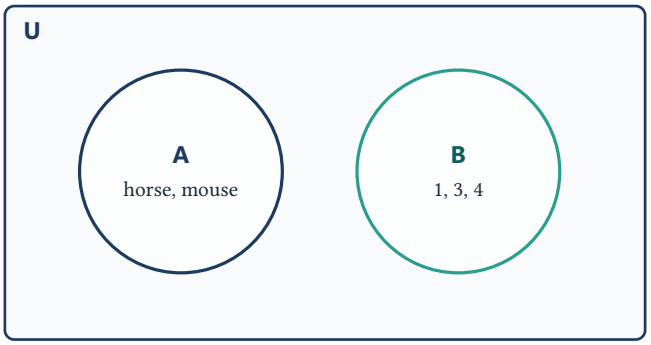
**DEFINITION** **Equal, Equivalent, and Disjoint Sets**

Let $A$ and $B$ be sets:

1. **Equal Sets ($A = B$):** $A$ and $B$ contain the exact same elements:

$$A = B \iff (A \subseteq B) \land (B \subseteq A)$$

   Set equality is reflexive ($A = A$), symmetric ($A = B \implies B = A$), and transitive ($A = B$ and $B = C \implies A = C$). Order and repetition do not affect equality: $\{3, 2, 2, 3\} = \{2, 3\}$.
2. **Equivalent Sets ($A \sim B$):** $A$ and $B$ have the same cardinality ($|A| = |B|$). For example, $\{1, 2, 3\} \sim \{a, b, c\}$ because both have 3 elements, even though the sets are not equal.
3. **Disjoint Sets ($A \cap B = \varnothing$, or `A // B`):** $A$ and $B$ share no elements in common.




*Disjoint sets A = {horse, mouse} and B = {1, 3, 4} with no elements in common.*

### 1.3.4 Power Sets

**DEFINITION** **Power Set**

The **power set** of $A$, written $P(A)$ or $2^A$, is the set of all subsets of $A$:

$$P(A) = \{S \mid S \subseteq A\}$$

If $|A| = n$, then $|P(A)| = 2^n$ (since each of the $n$ elements can either be included or excluded when forming a subset).

- If $A = \{1, 2\}$, then $|P(A)| = 2^2 = 4$ and $P(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$.
- For $\{\varnothing\}$, we have $|\{\varnothing\}| = 1$, so $P(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$ has $2^1 = 2$ elements.
- If $A = \{1, 2, 3\}$, then $P(A)$ has $2^3 = 8$ subsets, which we can list by size:

$$P(A) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

**EXAMPLE & WALKTHROUGH** **Working with Subsets and Venn Diagrams**

**1. Determine whether the first set is a subset of the second, the second is a subset of the first, or neither:**

- **(a)** People who speak English vs. people who speak English with an Australian accent: The second set is a proper subset of the first ($B \subset A$).
- **(b)** Fruits vs. citrus fruits: Every citrus fruit is a fruit, so the second set is a proper subset of the first ($B \subset A$).
- **(c)** Students studying discrete mathematics vs. students studying data structures: Neither is a subset of the other, since a student might take one course without taking the other.

**2. Months without the letter R:** Inside the universal set $U$ of all 12 months, the subset $M$ of months whose English names do not contain the letter `R` is:

$$M = \{\text{May}, \text{June}, \text{July}, \text{August}\}$$

The other 8 months ( `January` through `April` , and `September` through `December` ) lie outside $M$ in $U - M$.

**3. Visualizing subset relationships:**

- If $A \subseteq B$ and $B \subseteq C$, we draw concentric circles $A$ inside $B$ inside $C$.
- If $A \subset B$ and $A \subset C$, circle $A$ sits entirely inside the overlap region $B \cap C$ of circles $B$ and $C$.
- **Proof that $A \subseteq B$ and $B \subseteq C \implies A \subseteq C$:** Take any $x \in A$. Since $A \subseteq B$, we have $x \in B$. Then, since $B \subseteq C$, we have $x \in C$. Thus every element of $A$ is in $C$, so $A \subseteq C$. $\square$

## 1.4 Set Operations

Given sets $A$ and $B$ inside a universal set $U$, we can build new sets using several standard operations.

### 1.4.1 Intersection, Union, and Complement

**DEFINITION** **Intersection, Union, and Complement**

1. **Intersection ($A \cap B$):** Elements belonging to **both** $A$ and $B$:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

   For $A = \{1, 5, 7, 9\}$ and $B = \{1, 2, 3\}$, $A \cap B = \{1\}$.
2. **Union ($A \cup B$):** Elements belonging to $A$ **or** $B$ (or both):

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

   For $A = \{1, 2\}$ and $B = \{2, 5, 6\}$, $A \cup B = \{1, 2, 5, 6\}$.
3. **Complement ($\overline{A}$ or $A^c$):** Elements in the universe $U$ that are **not** in $A$:

$$\overline{A} = \{x \mid x \in U \wedge x \notin A\} = U - A$$

   If $U = \{1, 2, \ldots, 10\}$ and $A = \{1, 4, 5\}$, then $\overline{A} = \{2, 3, 6, 7, 8, 9, 10\}$.

**EXAMPLE & WALKTHROUGH** **Translating Everyday Statements into Set Notation**

Let $A$ be the set of red fruits, $B$ be the set of imported fruits, and $E$ be the set of fruits sold at Store X. How do we write **"the set of fruits sold at Store X that are red or imported"**?

- Being "red or imported" means belonging to $A \cup B$.
- Also being sold at Store X means belonging to $E$ as well.
- Combining the two gives:

$$E \cap (A \cup B) \quad \text{(or equivalently, } (E \cap A) \cup (E \cap B))$$

### 1.4.2 Set Difference and Symmetric Difference

**DEFINITION** **Difference & Symmetric Difference**

1. **Difference ($A - B$ or $A \setminus B$):** Elements in $A$ that are **not** in $B$:

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$$

   If $A = \{1, 2, ..., 10\}$ and $B = \{1, 2, 3, 11\}$, then $A - B = \{4, 5, ..., 10\}$ (while $B - A = \{11\}$).
2. **Symmetric Difference ($A \oplus B$):** Elements in $A$ or $B$, but **not both** (the set version of logical `XOR`):

$$A \oplus B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$$

   If $A = \{1, 2, 3\}$ and $B = \{3, 5, 6\}$, then $A \oplus B = \{1, 2, 5, 6\}$.

Symmetric difference is both **commutative** ($A \oplus B = B \oplus A$) and **associative** ($(A \oplus B) \oplus C = A \oplus (B \oplus C)$).

### 1.4.3 Cartesian Product

**DEFINITION** **Cartesian Product**

The **Cartesian product** of $A$ and $B$, written $A \times B$, is the set of all **ordered pairs** $(a, b)$ with $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

For example, if $A = \{1, 2, 3\}$ and $B = \{a, b\}$, then:

$$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$$

Key properties to keep in mind:

- **Cardinality:** For finite sets, $|A \times B| = |A| \cdot |B|$.
- **Order matters:** $(a, b) \neq (b, a)$ unless $a = b$. Consequently, $A \times B \neq B \times A$ in general (unless $A = B$ or one of the sets is empty).
- **Empty set:** If $A = \varnothing$ or $B = \varnothing$, then $A \times B = \varnothing$.

**EXAMPLE & WALKTHROUGH** **Three-Way Cartesian Product**

Let $A = \{1, 2\}$, $B = \{x, y, z\}$, and $C = \{3, 4\}$. Then $|A \times B \times C| = 2 \times 3 \times 2 = 12$, and the ordered triples in $A \times B \times C$ are:

$$A \times B \times C = \{(1, x, 3), (1, x, 4), (1, y, 3), (1, y, 4), (1, z, 3), (1, z, 4),$$
$$(2, x, 3), (2, x, 4), (2, y, 3), (2, y, 4), (2, z, 3), (2, z, 4)\}$$

**EXAMPLE & WALKTHROUGH** **Set Operations Practice**

**1. Interpreting set expressions:** Let $A$ be students who live within one mile of school, and $B$ be students who walk to classes.

- $A \cap B$: Students who live within one mile of school **and** walk to classes.

- $A \cup B$: Students who live within one mile of school **or** walk to classes.
- $A - B$: Students who live within one mile of school but do **not** walk to classes.
- $B - A$: Students who walk to classes but live more than one mile from school.

**2. Writing set expressions:** Let $A$ be sophomores and $B$ be students taking discrete mathematics.

- Sophomores taking discrete math: $\boldsymbol{A \cap B}$.
- Sophomores not taking discrete math: $\boldsymbol{A - B}$ (or $A \cap \overline{B}$).
- Students who are sophomores or taking discrete math: $\boldsymbol{A \cup B}$.
- Students who are either not sophomores or not taking discrete math: $\boldsymbol{\overline{A} \cup \overline{B}}$ (which equals $\overline{A \cap B}$).

**3. Computing set operations:**

- For $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$: $A \cup B = \{0, 1, 2, 3, 4, 5, 6\}$, $A \cap B = \{3\}$, $A - B = \{1, 2, 4, 5\}$, $B - A = \{0, 6\}$.
- For $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$ (note $A \subset B$): $A \cup B = B$, $A \cap B = A$, $A - B = \varnothing$, $B - A = \{f, g, h\}$.

## 1.5 The Principle of Inclusion–Exclusion

When two finite sets overlap, adding $|A| + |B|$ counts every element in $A \cap B$ twice. Subtracting that overlap once gives the Principle of Inclusion–Exclusion.

**CORE CONCEPT** **Principle of Inclusion–Exclusion**

For finite sets $A, B, C$:

1. **Two Disjoint Sets ($\boldsymbol{A \cap B = \varnothing}$):** $|A \cup B| = |A| + |B|$
2. **Two General Sets:** $|A \cup B| = |A| + |B| - |A \cap B|$
3. **Three General Sets:**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

**INTUITION & DEEP DIVE** **Why Do We Add Back |A ∩ B ∩ C|?**

Think about an element $x$ that belongs to all three sets $A$, $B$, and $C$:

- It is counted 3 times in $|A| + |B| + |C|$.
- It is subtracted 3 times in $-|A \cap B| - |A \cap C| - |B \cap C|$, bringing its net count down to 0.
- Adding $+|A \cap B \cap C|$ adds it back once so that $x$ is counted **exactly once** in the union.

**EXAMPLE & WALKTHROUGH** **Three-Set Survey Problem**

**Problem:** In a survey of 60 people ($|U| = 60$) regarding three magazines ($E$, $G$, and $F$):

- $|E| = 25$, $|G| = 26$, and $|F| = 26$.
- $|E \cap F| = 9$, $|E \cap G| = 11$, and $|G \cap F| = 8$.
- 8 people read none of the three magazines ($|\overline{E \cup G \cup F}| = 8$).

How many people read all three magazines?

**Solution:**

1. Since 8 of the 60 people read none of the magazines, the number who read at least one is:

$$|E \cup G \cup F| = 60 - 8 = 52$$

2. Plugging the known counts into the 3-set Inclusion–Exclusion formula:

$$52 = (25 + 26 + 26) - (11 + 9 + 8) + |E \cap G \cap F|$$

$$52 = 77 - 28 + |E \cap G \cap F| = 49 + |E \cap G \cap F|$$

$$|E \cap G \cap F| = 52 - 49 = 3$$

So **3 people** read all three magazines.

## 1.6 Laws of Set Algebra and Duality

Set operations follow algebraic rules that mirror propositional logic ($\cup$ acts like $\vee$, $\cap$ acts like $\wedge$, complement acts like $\neg$, $\varnothing$ acts like False, and $U$ acts like True). Because of this symmetry, every law comes with a **dual**: swapping $\cup \leftrightarrow \cap$ and $\varnothing \leftrightarrow U$ turns any valid set identity into another valid identity.

| Law Name | Primary Identity | Dual Identity |
|---|---|---|
| 1. Identity Laws | $A \cup \varnothing \equiv A$ | $A \cap U \equiv A$ |
| 2. Null / Domination Laws | $A \cap \varnothing \equiv \varnothing$ | $A \cup U \equiv U$ |
| 3. Complement Laws | $A \cup \overline{A} \equiv U$ | $A \cap \overline{A} \equiv \varnothing$ |
| 4. Idempotent Laws | $A \cup A \equiv A$ | $A \cap A \equiv A$ |
| 5. Involution Law | $\overline{\left(\overline{A}\right)} \equiv A$ | (Self-dual) |
| 6. Absorption Laws | $A \cup (A \cap B) \equiv A$ | $A \cap (A \cup B) \equiv A$ |
| 7. Commutative Laws | $A \cup B \equiv B \cup A$ | $A \cap B \equiv B \cap A$ |
| 8. Associative Laws | $A \cup (B \cup C) \equiv (A \cup B) \cup C$ | $A \cap (B \cap C) \equiv (A \cap B) \cap C$ |
| 9. Distributive Laws | $A \cup (B \cap C) \equiv (A \cup B) \cap (A \cup C)$ | $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$ |
| 10. De Morgan's Laws | $\overline{A \cap B} \equiv \overline{A} \cup \overline{B}$ | $\overline{A \cup B} \equiv \overline{A} \cap \overline{B}$ |
| 11. 0/1 Laws | $\overline{\varnothing} \equiv U$ | $\overline{U} \equiv \varnothing$ |

## 1.7 Proving Set Identities

There are three common ways to check or prove that two set expressions always produce the same set.

### 1.7.1 Venn Diagrams

To see why $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$, we can draw two 3-circle Venn diagrams:

- On the left, shade $B \cup C$ and keep only the part inside $A$.
- On the right, shade $A \cap B$ and $A \cap C$ and combine them.

Both give the exact same shaded region. Venn diagrams are great for building intuition, while membership tables and algebraic proofs provide formal verification.

### 1.7.2 Membership Tables

Just like truth tables in logic, a **membership table** checks every possible combination of whether an element $x$ is inside (`1`) or outside (`0`) each base set. Two expressions are equal if their columns match in every row:

| $A$ | $B$ | $C$ | $B \cup C$ | $\boldsymbol{A \cap (B \cup C)}$ | $A \cap B$ | $A \cap C$ | $\boldsymbol{(A \cap B) \cup (A \cap C)}$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | **0** | 0 | 0 | **0** |
| 0 | 0 | 1 | 1 | **0** | 0 | 0 | **0** |
| 0 | 1 | 0 | 1 | **0** | 0 | 0 | **0** |
| 0 | 1 | 1 | 1 | **0** | 0 | 0 | **0** |
| 1 | 0 | 0 | 0 | **0** | 0 | 0 | **0** |
| 1 | 0 | 1 | 1 | **1** | 0 | 1 | **1** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | **1** | 1 | 0 | **1** |
| 1 | 1 | 1 | 1 | **1** | 1 | 1 | **1** |

Since the two bold columns match across all 8 rows, $A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$.

### 1.7.3 Algebraic Proofs

We can also rewrite one side step-by-step using the laws of set algebra (plus the difference rule $X - Y \equiv X \cap \overline{Y}$).

For example, to prove $(A \cap B) \cup \left(A \cap \overline{B}\right) \equiv A$:

$$\begin{aligned}
(A \cap B) \cup \left(A \cap \overline{B}\right) &\equiv A \cap \left(B \cup \overline{B}\right) && \text{(Distributive Law)} \\
&\equiv A \cap U && \text{(Complement Law)} \\
&\equiv A && \text{(Identity Law)} \quad \square
\end{aligned}$$

**EXAMPLE & WALKTHROUGH** **Algebraic Proofs with Set Difference**

**1. Prove that $A \cap (B - C) \equiv (A \cap B) - (A \cap C)$:**

Starting from the right-hand side:

$$\begin{aligned}
(A \cap B) - (A \cap C) &\equiv (A \cap B) \cap \overline{(A \cap C)} && \text{(Definition of Difference)} \\
&\equiv (A \cap B) \cap \left(\overline{A} \cup \overline{C}\right) && \text{(De Morgan's Law)} \\
&\equiv \left((A \cap B) \cap \overline{A}\right) \cup \left((A \cap B) \cap \overline{C}\right) && \text{(Distributive Law)} \\
&\equiv \left(B \cap \left(A \cap \overline{A}\right)\right) \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Commutative \& Associative Laws)} \\
&\equiv (B \cap \varnothing) \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Complement Law)} \\
&\equiv \varnothing \cup \left(A \cap \left(B \cap \overline{C}\right)\right) && \text{(Domination Law)} \\
&\equiv A \cap \left(B \cap \overline{C}\right) && \text{(Identity Law)} \\
&\equiv A \cap (B - C) && \text{(Definition of Difference)} \quad \square
\end{aligned}$$

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**2. Prove that $(A \cup B) - C \equiv (A - C) \cup (B - C)$:**

Starting from the left-hand side:

$$\begin{aligned}
(A \cup B) - C &\equiv (A \cup B) \cap \overline{C} && \text{(Definition of Difference)} \\
&\equiv \overline{C} \cap (A \cup B) && \text{(Commutative Law)} \\
&\equiv \left(\overline{C} \cap A\right) \cup \left(\overline{C} \cap B\right) && \text{(Distributive Law)} \\
&\equiv \left(A \cap \overline{C}\right) \cup \left(B \cap \overline{C}\right) && \text{(Commutative Law)} \\
&\equiv (A - C) \cup (B - C) && \text{(Definition of Difference)} \quad \square
\end{aligned}$$

# 2. Functions and Relations

Binary Relations, Matrix Representations, Equivalence Relations, Functions, Bijections, and Composition

## 2.1 Binary Relations

In discrete mathematics and computer science, relationships between elements of sets—such as keys mapped to database records, vertices connected by edges in a network, or tasks constrained by scheduling dependencies—are formalized through the concept of a *relation*. While a Cartesian product $A \times B$ pairs every element of $A$ with every element of $B$, a relation selects the specific subset of ordered pairs that satisfy a meaningful condition.

**DEFINITION** **Binary Relation**

Let $A$ and $B$ be sets. A **binary relation** $R$ from set $A$ to set $B$ is a subset of the Cartesian product $A \times B$:

$$R \subseteq A \times B$$

- If an ordered pair $(a, b)$ belongs to $R$, we write $(a, b) \in R$ or use the infix notation $aRb$ (read as "$a$ is related to $b$ by $R$").
- If $(a, b)$ does not belong to $R$, we write $(a, b) \notin R$ or $a \not R b$.
- Set $A$ is called the **domain** of $R$, and set $B$ is called the **codomain** of $R$.

**INTUITION & DEEP DIVE** **Why Relations Are Defined as Subsets of Cartesian Products**

Defining a relation $R$ simply as a set of ordered pairs $R \subseteq A \times B$ separates *which* pairs are related from *how* the rule is phrased (such as "is a divisor of" or "is enrolled in"). Because $A \times B$ has $|A| \cdot |B|$ possible ordered pairs, there are $2^{|A| \cdot |B|}$ distinct binary relations from a finite set $A$ to a finite set $B$. In relational databases (SQL), a two-column table over domains $A$ and $B$ is a finite binary relation $R \subseteq A \times B$.

**EXAMPLE & WALKTHROUGH** **Divisibility Relation Between Two Sets**

Let $P = \{2, 3, 4\}$ and $Q = \{2, 4, 8, 9, 15\}$. Suppose we define a binary relation $R$ from $P$ to $Q$ by the rule:

$$(p, q) \in R \quad \text{if and only if} \quad p \text{ divides } q \quad (\text{written } p \mid q)$$

**Step-by-step evaluation across $P \times Q$:**

- For $p = 2$: 2 divides $2, 4, 8$ in $Q$, yielding pairs $(2, 2), (2, 4), (2, 8)$.
- For $p = 3$: 3 divides $9, 15$ in $Q$, yielding pairs $(3, 9), (3, 15)$.
- For $p = 4$: 4 divides $4, 8$ in $Q$, yielding pairs $(4, 4), (4, 8)$.

Collecting all valid ordered pairs gives:

$$R = \{(2, 2), (2, 4), (4, 4), (2, 8), (4, 8), (3, 9), (3, 15)\}$$

We can visualize this bipartite mapping between $P$ and $Q$ structurally:

| **Domain $P = \{2, 3, 4\}$** | **Outgoing Relation Arrows to Codomain $Q = \{2, 4, 8, 9, 15\}$** |
|---|---|
| • $p = 2$ (even divisor) | $2 \to \{2, 4, 8\}$ |
| • $p = 3$ (odd prime divisor) | $3 \to \{9, 15\}$ |
| • $p = 4$ (multiple of 4) | $4 \to \{4, 8\}$ |

### 2.1.1 Relations on a Single Set

In many applications—such as ordering numbers, comparing strings, or modeling transitions between states of a finite automaton—the domain and codomain are the same set.

**DEFINITION** **Relation on a Set**

A **relation on a set $A$** is a binary relation from $A$ to $A$, that is, a subset of $A \times A$:

$$R \subseteq A \times A$$

**EXAMPLE & WALKTHROUGH** **Prime Factor Relation on a Set**

Let $A = \{2, 3, 4, 8, 9\}$, and let $R$ be the relation on $A$ defined by:

$$(x, y) \in R \quad \text{if and only if} \quad x \text{ is a prime factor of } y$$

**Analysis:** For $(x, y)$ to belong to $R$, $x$ itself must be a prime number and must divide $y$. Among the elements of $A = \{2, 3, 4, 8, 9\}$, only 2 and 3 are prime ($4 = 2^2$, $8 = 2^3$, and $9 = 3^2$ are composite):

- For $x = 2$: 2 is a prime factor of 2, 4, and 8.
- For $x = 3$: 3 is a prime factor of 3 and 9.
- For $x \in \{4, 8, 9\}$: None of these are prime, so they contribute no pairs as first coordinates.

Therefore:

$$R = \{(2, 2), (2, 4), (2, 8), (3, 3), (3, 9)\}$$

## 2.2 Representation and Inverse of Relations

### 2.2.1 Zero-One Matrix Representation

When sets $A$ and $B$ are finite, a binary relation $R \subseteq A \times B$ can be represented compactly in memory as a **zero-one matrix** (boolean adjacency matrix).

**CORE CONCEPT** **Matrix Representation of a Binary Relation**

Suppose $R$ is a relation from $A = \{a_1, a_2, ..., a_m\}$ to $B = \{b_1, b_2, ..., b_n\}$, where the elements of each set are listed in a fixed order. The relation $R$ is represented by the $m \times n$ matrix $M_R = [m_{ij}]$, where:

$$M_R = \begin{pmatrix} m_{11} & m_{12} & \dots & m_{1n} \\ m_{21} & m_{22} & \dots & m_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{m1} & m_{m2} & \dots & m_{mn} \end{pmatrix} \quad \text{with} \quad m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

Each row $i$ corresponds to element $a_i \in A$, and each column $j$ corresponds to element $b_j \in B$.

**EXAMPLE & WALKTHROUGH** **Matrix of the Divisibility Relation**

Returning to $P = \{2, 3, 4\}$ and $Q = \{2, 4, 8, 9, 15\}$ with $(p, q) \in R \iff p \mid q$, we index the elements in increasing order:

$$p_1 = 2, \quad p_2 = 3, \quad p_3 = 4$$

$$q_1 = 2, \quad q_2 = 4, \quad q_3 = 8, \quad q_4 = 9, \quad q_5 = 15$$

The relation can be written in indexed form as:

$$\begin{aligned} R &= \{(2, 2), (2, 4), (4, 4), (2, 8), (4, 8), (3, 9), (3, 15)\} \\ &= \{(p_1, q_1), (p_1, q_2), (p_3, q_2), (p_1, q_3), (p_3, q_3), (p_2, q_4), (p_2, q_5)\} \end{aligned}$$

Setting entry $(i, j)$ to 1 whenever $(p_i, q_j) \in R$ and 0 otherwise yields the $3 \times 5$ zero-one matrix:

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

**EXAMPLE & WALKTHROUGH** **Constructing a Relation Matrix**

**Problem:** Given $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, with the relation:

$$R = \{(1, a), (1, b), (2, b), (3, a), (3, b), (3, c)\}$$

Represent this relation in matrix form.

**Solution:** Order the domain elements as $a_1 = 1, a_2 = 2, a_3 = 3$ (rows $1, 2, 3$) and the codomain elements as $b_1 = a, b_2 = b, b_3 = c$ (columns $1, 2, 3$):

- **Row 1 (1):** $(1, a) \in R$, $(1, b) \in R$, $(1, c) \notin R \Longrightarrow [1, 1, 0]$.
- **Row 2 (2):** $(2, a) \notin R$, $(2, b) \in R$, $(2, c) \notin R \Longrightarrow [0, 1, 0]$.
- **Row 3 (3):** $(3, a) \in R$, $(3, b) \in R$, $(3, c) \in R \Longrightarrow [1, 1, 1]$.

Combining the three rows gives the $3 \times 3$ matrix representation:

$$M_R = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

### 2.2.2 Inverse Relations

Reversing the direction of every ordered pair in a relation $R$ from $A$ to $B$ produces a relation from $B$ to $A$.

**DEFINITION** **Inverse Relation**

Let $R$ be a binary relation from set $A$ to set $B$. The **inverse relation** of $R$, denoted by $R^{-1}$, is the relation from $B$ to $A$ defined by reversing the order of every pair in $R$:

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

In matrix representation, the matrix of $R^{-1}$ is the **transpose** of the matrix of $R$:

$$M_{R^{-1}} = (M_R)^T$$

**EXAMPLE & WALKTHROUGH** **Inverse of the Divisibility Relation**

Let $P = \{2, 3, 4\}$ and $Q = \{2, 4, 8, 9, 15\}$, where $(p, q) \in R$ if $p$ divides $q$:

$$R = \{(2, 2), (2, 4), (4, 4), (2, 8), (4, 8), (3, 9), (3, 15)\}$$

The inverse relation $R^{-1} \subseteq Q \times P$ is characterized verbally by:

$$(q, p) \in R^{-1} \quad \text{if and only if} \quad q \text{ is a multiple of } p$$

Reversing each ordered pair in $R$ yields:

$$R^{-1} = \{(2, 2), (4, 2), (4, 4), (8, 2), (8, 4), (9, 3), (15, 3)\}$$

Its matrix representation is the $5 \times 3$ transpose $M^T$:

$$M_{R^{-1}} = M^T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

## 2.3 Properties of Relations on a Set

When a relation $R$ is defined on a single set $A$ ($R \subseteq A \times A$), we classify its behavior using four fundamental properties: *reflexivity*, *symmetry*, *antisymmetry*, and *transitivity*.

### 2.3.1 Reflexivity

**DEFINITION** **Reflexive Relation**

A relation $R$ on a set $A$ is called **reflexive** if every element of $A$ is related to itself:

$$\forall a \in A, \quad (a, a) \in R$$

**Matrix signature:** Every entry on the main diagonal of $M_R$ is 1 (i.e., $m_{ii} = 1$ for all $1 \leq i \leq |A|$).

**EXAMPLE & WALKTHROUGH** **Testing Reflexivity on a Finite Set**

Let $A = \{1, 2, 3\}$ and consider the relation:

$$R = \{(1,1), (1,3), (2,2), (2,1), (2,3), (3,3)\}$$

Because $A$ contains the elements $1, 2, 3$ and $R$ contains all three self-loops $(1,1) \in R$, $(2,2) \in R$, and $(3,3) \in R$, the relation $R$ is **reflexive**.

**EXAMPLE & WALKTHROUGH** **Testing Whether y = x² Is Reflexive on Positive Integers**

**Problem:** A relation $R$ is defined on the set of positive integers $\mathbb{Z}^+$ by:

$$R : y = x^2, \quad x, y \in \mathbb{Z}^+$$

Is this relation reflexive?

**Solution: No, $R$ is not reflexive.** By definition, for $R$ to be reflexive on $\mathbb{Z}^+$, the pair $(x, x)$ must belong to $R$ for *every* positive integer $x \in \mathbb{Z}^+$. That requires $x = x^2$ for all $x \in \mathbb{Z}^+$.

- For $x = 1$: $1 = 1^2 = 1$, so $(1, 1) \in R$.
- For $x = 2$: $2^2 = 4 \neq 2$, so $(2, 2) \notin R$ (instead $(2, 4) \in R$).

Since the universal requirement $\forall x \in \mathbb{Z}^+, (x, x) \in R$ fails for $x = 2$, $R$ is *not reflexive*.

### 2.3.2 Symmetry and Antisymmetry

**DEFINITION** **Symmetric Relation**

A relation $R$ on a set $A$ is called **symmetric** if whenever $a$ is related to $b$, $b$ is also related to $a$:

$$\forall a, b \in A, \quad (a, b) \in R \implies (b, a) \in R$$

**Matrix signature:** $M_R$ is a symmetric matrix ($M_R = (M_R)^T$, meaning $m_{ij} = m_{ji}$ for all $i, j$).

**EXAMPLE & WALKTHROUGH** **Symmetric Relation**

Let $A = \{1, 2, 3, 4\}$ and let $R$ be defined as:

$$R = \{(1,1), (1,2), (2,1), (2,3), (3,2), (4,4)\}$$

Every off-diagonal pair has its reverse present in $R$: $(1,2)$ is paired with $(2,1)$, and $(2,3)$ is paired with $(3,2)$. Therefore, $R$ is **symmetric**. Note that $R$ is *not* reflexive because $(2,2) \notin R$ and $(3,3) \notin R$.

**DEFINITION** **Antisymmetric Relation**

A relation $R$ on a set $A$ is called **antisymmetric** if two distinct elements are never mutually related—that is, if both $(a,b) \in R$ and $(b,a) \in R$ hold, then $a$ and $b$ must be the same element:

$$\forall a, b \in A, \quad ((a,b) \in R \land (b,a) \in R) \implies a = b$$

Equivalently, by contraposition: for all $a \neq b$, if $(a,b) \in R$, then $(b,a) \notin R$. **Matrix signature:** For all $i \neq j$, at most one of $m_{ij}$ and $m_{ji}$ can be 1 (i.e., $m_{ij} \cdot m_{ji} = 0$ for $i \neq j$). Diagonal entries $m_{ii}$ may be 0 or 1 freely.

**EXAMPLE & WALKTHROUGH** **Antisymmetric Relation**

Let $A = \{1, 2, 3\}$ and let $R = \{(1,1), (2,2), (1,2)\}$.

- For the self-loops $(1,1)$ and $(2,2)$, $a = b$ holds automatically.
- For the distinct pair $(1,2) \in R$, the reverse pair $(2,1)$ is *not* in $R$.

Hence there is no pair of distinct elements $a \neq b$ with both $(a,b) \in R$ and $(b,a) \in R$, which means $R$ is **antisymmetric**.

**PITFALL / WARNING** **Symmetric and Antisymmetric Are Not Opposites**

It is easy to assume that "not symmetric" means "antisymmetric," or that a relation cannot be both. In fact:

1. **A relation can be both symmetric and antisymmetric:** Any diagonal relation (such as the equality relation $R = \{(1,1), (2,2), (3,3)\}$ or the empty relation $R = \varnothing$) is both symmetric and antisymmetric.
2. **A relation can be neither symmetric nor antisymmetric:** For example, $R = \{(1,2), (2,1), (2,3)\}$ is not symmetric (missing $(3,2)$) and not antisymmetric (contains both $(1,2)$ and $(2,1)$).

### 2.3.3 Transitivity

**DEFINITION** **Transitive Relation**

A relation $R$ on a set $A$ is called **transitive** if whenever $a$ is related to $b$ and $b$ is related to $c$, then $a$ is also directly related to $c$:

$$\forall a, b, c \in A, \quad ((a,b) \in R \land (b,c) \in R) \implies (a,c) \in R$$

**EXAMPLE & WALKTHROUGH** **Verifying Transitivity via Two-Step Chains**

Let $A = \{1, 2, 3, 4\}$ and define $R = \{(1,2), (1,3), (2,3), (3,4), (1,4), (2,4)\}$. To verify that $R$ is transitive, we inspect every composable pair $(a,b) \in R$ and $(b,c) \in R$ and confirm that $(a,c)$ belongs to $R$:

| First Pair $(a, b)$ | Second Pair $(b, c)$ | Required $(a, c)$ | Status in $R$ |
|---|---|---|---|
| $(1, 2)$ | $(2, 3)$ | $(1, 3)$ | Present in $R$ (✓) |
| $(1, 2)$ | $(2, 4)$ | $(1, 4)$ | Present in $R$ (✓) |
| $(1, 3)$ | $(3, 4)$ | $(1, 4)$ | Present in $R$ (✓) |
| $(2, 3)$ | $(3, 4)$ | $(2, 4)$ | Present in $R$ (✓) |

Because every two-step path $a \to b \to c$ has its direct edge $a \to c$ in $R$ (in fact, $R$ is the strict ordering $<$ on $\{1, 2, 3, 4\}$), $R$ is **transitive**.

**EXAMPLE & WALKTHROUGH** **Analyzing the Relation x + y = 5 on {1, 2, 3, 4}**

**Problem:** Let $A = \{1, 2, 3, 4\}$ and let relation $R$ on $A$ be defined by:

$$R : x + y = 5 \quad \text{for } x, y \in A$$

Determine all properties of relation $R$.

**Solution:** First, list all ordered pairs $(x, y) \in A \times A$ satisfying $x + y = 5$:

$$R = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$

Now test each of the four properties:

1. *Reflexive?* **No.** For $x = 1$, $1 + 1 = 2 \neq 5$, so $(1, 1) \notin R$ (no $(a, a)$ is in $R$ because $2a = 5$ has no integer solution).
2. *Symmetric?* **Yes.** Addition is commutative ($x + y = 5 \implies y + x = 5$). Explicitly, $(1, 4), (4, 1) \in R$ and $(2, 3), (3, 2) \in R$.
3. *Antisymmetric?* **No.** Both $(1, 4) \in R$ and $(4, 1) \in R$ hold with $1 \neq 4$.
4. *Transitive?* **No.** We have $(1, 4) \in R$ and $(4, 1) \in R$, which by transitivity would require $(1, 1) \in R$, but $(1, 1) \notin R$.

**Conclusion:** The relation $R$ is **symmetric only** (it is neither reflexive, nor antisymmetric, nor transitive).

### 2.3.4 Equivalence Relations

**DEFINITION** **Equivalence Relation**

A relation $R$ on a set $A$ is called an **equivalence relation** if it simultaneously satisfies three properties:

1. **Reflexive**: $(a, a) \in R$ for all $a \in A$.
2. **Symmetric**: $(a, b) \in R \implies (b, a) \in R$ for all $a, b \in A$.
3. **Transitive**: $(a, b) \in R \land (b, c) \in R \implies (a, c) \in R$ for all $a, b, c \in A$.

**INTUITION & DEEP DIVE** **Equivalence Relations Partition a Set into Disjoint Classes**

An equivalence relation generalizes "equality" by grouping elements that share a chosen attribute (such as congruence modulo $n$, having the same string length, or residing in the same connected component of an undirected graph). Every equivalence relation $R$ on $A$ partitions $A$ into mutually disjoint subsets called *equivalence classes* $[a]_R = \{x \in A \mid (a, x) \in R\}$. Within each class, every element is related to every other element.

**EXAMPLE & WALKTHROUGH** **Equivalence Relation on A = {1, 2, 3, 4}**

Let $A = \{1, 2, 3, 4\}$ and consider the relation:

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (4,4)\}$$

Let us check why $R$ is an equivalence relation:

- **Reflexive:** $(1,1), (2,2), (3,3), (4,4) \in R$.
- **Symmetric:** Every pair among $\{1,2,3\}$ has its reverse in $R$, and $(4,4)$ is its own reverse.
- **Transitive:** Notice that $R = (\{1,2,3\} \times \{1,2,3\}) \cup \{(4,4)\}$. Any composable pairs $(a,b)$ and $(b,c)$ either both lie inside $\{1,2,3\} \times \{1,2,3\}$ (which contains all 9 pairs on $\{1,2,3\}$, so $(a,c) \in R$ automatically) or both equal $(4,4)$.

Thus $R$ is an **equivalence relation** that partitions $A$ into two equivalence classes: $[1] = [2] = [3] = \{1,2,3\}$ and $[4] = \{4\}$.

**Summary of the Four Core Relation Properties on a Set A**

**Reflexive**

**Condition:** $\forall a \in A, (a,a) \in R$.
**Graph:** Every vertex has a self-loop.
**Matrix:** All diagonal entries $m_{ii} = 1$.

**Symmetric**

**Condition:** $(a,b) \in R \implies (b,a) \in R$.
**Graph:** Every directed edge between distinct vertices is bidirectional.
**Matrix:** $M_R = (M_R)^T$.

**Antisymmetric**

**Condition:** $(a,b) \in R \wedge (b,a) \in R \implies a = b$.
**Graph:** No two *distinct* vertices have two-way arrows between them.
**Matrix:** If $i \neq j$, then $m_{ij} = 0$ or $m_{ji} = 0$.

**Transitive**

**Condition:** $(a,b) \in R \wedge (b,c) \in R \implies (a,c) \in R$.
**Graph:** Every 2-step path $a \to b \to c$ has a direct shortcut edge $a \to c$.
**Combine:** Reflexive + Symmetric + Transitive = **Equivalence Relation**.

## 2.4 Functions: Fundamentals and Specifications

A *function* is a binary relation in which every input from the domain is assigned to a single, unambiguous output in the codomain.

**DEFINITION** **Function / Mapping / Transformation**

Let $A$ and $B$ be non-empty sets. A binary relation $f$ from $A$ to $B$ is a **function** (also called a **mapping** or **transformation**) if every element in $A$ is associated with **exactly one** element in $B$. We write:

$$f : A \to B$$

which reads "$f$ maps $A$ to $B$." Formally, $f \subseteq A \times B$ is a function if and only if:

1. **Totality (Every input is mapped):** $\forall a \in A, \exists b \in B$ such that $(a,b) \in f$.
2. **Uniqueness (At most one output per input):** $\forall a \in A, \forall b_1, b_2 \in B$, if $(a,b_1) \in f$ and $(a,b_2) \in f$, then $b_1 = b_2$.

**CORE CONCEPT** **Domain, Codomain, Image, Pre-image, and Range**

When $f : A \to B$ and $(a,b) \in f$, we write $f(a) = b$:

- **Domain:** The set $A$ of all valid inputs.
- **Codomain:** The target set $B$ of potential outputs.
- **Image:** If $f(a) = b$, then $b$ is the *image* of $a$ under $f$.
- **Pre-image:** If $f(a) = b$, then $a$ is a *pre-image* of $b$ under $f$.
- **Range:** The set of all actual images attained by $f$, denoted $f(A)$ or $\text{Range}(f)$:

$$\text{Range}(f) = f(A) = \{f(a) \mid a \in A\} = \{b \in B \mid \exists a \in A, f(a) = b\} \subseteq B$$

The range of $f$ is always a subset (possibly a *proper subset*) of the codomain $B$.

### 2.4.1 Specifying a Function

A function can be specified explicitly in two main ways:

1. **Set of Ordered Pairs:** For finite sets $A = \{1, 2, 3\}$ and $B = \{u, v, w\}$, we can list the pairs directly:

$$f = \{(1, u), (2, v), (3, w)\} \iff f(1) = u, \quad f(2) = v, \quad f(3) = w$$

2. **Assignment Formula:** We can provide an algebraic rule or expression for computing $f(x)$ from $x$, such as:

$$f(x) = x^2, \quad \text{or equivalently in set-builder form:} \quad f = \{(x, x^2) \mid x \in \mathbb{R}\}$$

**EXAMPLE & WALKTHROUGH** **Determining Whether a Relation Is a Function**

**Problem:** Given $A = \{1, 2, 3\}$, $B = \{u, v, w\}$, and $f = \{(1, u), (2, u), (2, v), (3, w)\}$. Is $f$ a function?

**Solution: No, $f$ is not a function.** Let us check each element of the domain $A = \{1, 2, 3\}$:

- Element $1 \in A$ is mapped only to $u$.
- Element $2 \in A$ appears in *two distinct ordered pairs*: $(2, u) \in f$ and $(2, v) \in f$, where $u \neq v$.
- Element $3 \in A$ is mapped only to $w$.

Because the input $2 \in A$ is associated with two different outputs in $B$ ($u$ and $v$), $f$ violates the uniqueness condition. Note that having $(1, u)$ and $(2, u)$ share the output $u$ is completely fine for a function, but mapping a single input 2 to two outputs $u$ and $v$ is not allowed.

## 2.5 Classification of Functions: Injection, Surjection, Bijection

Functions $f : A \to B$ are classified by whether distinct inputs stay distinct in the output and whether the range covers the entire codomain $B$.

### 2.5.1 Injection (One-to-One Functions)

**DEFINITION** **Injection / One-to-One Function**

A function $f : A \to B$ is called **one-to-one** or **injective** (an **injection**) if distinct elements of $A$ always map to distinct elements of $B$:

$$\forall a, b \in A, \quad a \neq b \implies f(a) \neq f(b)$$

By contraposition, this is equivalent to:

$$\forall a, b \in A, \quad f(a) = f(b) \implies a = b$$

In an arrow diagram, every element of $B$ receives *at most one* incoming arrow from $A$. For finite sets, an injection $f : A \to B$ can exist only if $|A| \leq |B|$.

**EXAMPLE & WALKTHROUGH** **Testing Injectivity of f(x) = x²**

**Problem:**

1. Is the function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x^2$ injective?

2. What if we define $f = \{(x, x^2) \mid x \geq 0\}$ (restricting the domain to non-negative integers)?

**Solution:**

1. *For $f : \mathbb{Z} \to \mathbb{Z}$ with $f(x) = x^2$:* **Not injective.** Take $a = 1 \in \mathbb{Z}$ and $b = -1 \in \mathbb{Z}$. Then $a \neq b$, yet:

$$f(1) = 1^2 = 1 \quad \text{and} \quad f(-1) = (-1)^2 = 1 \implies f(1) = f(-1)$$

   Because two distinct integers share the same image, $f$ is not one-to-one on $\mathbb{Z}$.

2. *When restricted to $x \geq 0$ (i.e., $f : \mathbb{N}_0 \to \mathbb{Z}$ where $f = \{(x, x^2) \mid x \geq 0\}$):* **Yes, $f$ is injective.** Suppose $a, b \geq 0$ and $f(a) = f(b)$. Then:

$$a^2 = b^2 \implies a^2 - b^2 = 0 \implies (a - b)(a + b) = 0$$

   Since $a \geq 0$ and $b \geq 0$, $a + b = 0$ can only happen when $a = b = 0$; in all other cases $a + b > 0$, forcing $a - b = 0$, so $a = b$. Restricting the domain removes the negative duplicates and makes $f$ one-to-one.

### 2.5.2 Surjection (Onto Functions)

**DEFINITION** **Surjection / Onto Function**

A function $f : A \to B$ is called **onto** or **surjective** (a **surjection**) if every element of the codomain $B$ is the image of *at least one* element of the domain $A$:

$$\forall b \in B, \quad \exists a \in A \quad \text{such that} \quad f(a) = b$$

Equivalently, the range of $f$ equals the entire codomain: $\text{Range}(f) = f(A) = B$. In an arrow diagram, every element of $B$ receives *one or more* incoming arrows from $A$. For finite sets, a surjection $f : A \to B$ can exist only if $|A| \geq |B|$.

**EXAMPLE & WALKTHROUGH** **Testing Surjectivity of f(x) = x⁴ + 3**

**Problem:** Is the function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x^4 + 3$ onto?

**Solution: No, $f$ is not onto.** For $f : \mathbb{Z} \to \mathbb{Z}$ to be onto, for every integer $y \in \mathbb{Z}$ there would need to exist an integer $x \in \mathbb{Z}$ such that $x^4 + 3 = y$, or $x^4 = y - 3$.

- *Lower bound on the range:* Since $x^4 \geq 0$ for all $x \in \mathbb{Z}$, we have $f(x) = x^4 + 3 \geq 3$ for all $x \in \mathbb{Z}$. Thus, no integer $y < 3$ (for example, $y = 0$, which would require $x^4 = -3$) is in the range of $f$.
- *Gaps above 3:* Even for $y \geq 3$, take $y = 5 \in \mathbb{Z}$. Then $x^4 + 3 = 5 \implies x^4 = 2$, which has no integer solution $x \in \mathbb{Z}$.

Because $\text{Range}(f) = \{3, 4, 19, 84, \ldots\} \subsetneq \mathbb{Z}$, $f$ is not onto.

### 2.5.3 Bijection and Inverse Functions

**DEFINITION** **Bijection / One-to-One Correspondence**

A function $f : A \to B$ is a **bijection** (or **one-to-one correspondence**) if it is simultaneously:

1. **Injective (one-to-one)**: $f(a) = f(b) \implies a = b$, and
2. **Surjective (onto)**: $\forall b \in B, \exists a \in A$ with $f(a) = b$.

Every element in $B$ is the image of a *unique* element in $A$. For finite sets, a bijection between $A$ and $B$ exists if and only if $|A| = |B|$.

**DEFINITION** **Inverse Function**

Let $f : A \to B$ be a **bijection**. The **inverse function** of $f$, denoted by $f^{-1} : B \to A$, is the function that assigns to each element $b \in B$ the unique element $a \in A$ such that $f(a) = b$:

$$f^{-1}(b) = a \iff f(a) = b$$

A function $f : A \to B$ is *invertible* (has an inverse function $f^{-1} : B \to A$) if and only if $f$ is a bijection.

**PITFALL / WARNING** **Inverse Relation $R^{-1}$ vs. Inverse Function $f^{-1}$**

Any binary relation $R \subseteq A \times B$ always has an *inverse relation* $R^{-1} \subseteq B \times A$. However, if $f : A \to B$ is a function, its inverse relation $f^{-1} = \{(b, a) \mid (a, b) \in f\}$ is a valid *function* from $B$ to $A$ **if and only if $f$ is a bijection**:

- If $f$ is *not injective*, then some $b \in B$ has two pre-images $a_1 \neq a_2$, so $f^{-1}(b)$ would have two outputs (violating uniqueness).
- If $f$ is *not surjective*, then some $b \in B$ has *no* pre-image in $A$, so $f^{-1}(b)$ would be undefined on $b$ (violating totality).

**EXAMPLE & WALKTHROUGH** **Computing Inverse Functions**

1. **Finite Set of Ordered Pairs:** Let $A = \{1, 2, 3\}$, $B = \{u, v, w\}$, and $f = \{(1, u), (2, v), (3, w)\}$. Since $f$ is both one-to-one and onto, its inverse function is obtained by swapping each pair:
$$f^{-1} = \{(u, 1), (v, 2), (w, 3)\}$$
2. **Algebraic Formula:** Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x + 1$. Since $f$ is a bijection, we find $f^{-1}$ by solving $y = f(x)$ for $x$:
$$y = x + 1 \implies x = y - 1 \implies f^{-1}(y) = y - 1 \quad (\text{or } f^{-1}(x) = x - 1)$$

## 2.6 Function Composition

When the codomain of one function matches (or is contained within) the domain of a second function, we can chain the two mappings together into a pipeline.

**DEFINITION** **Composition of Functions**

Let $g : A \to B$ be a function from set $A$ to set $B$, and let $f : B \to C$ be a function from set $B$ to set $C$. The **composition** of $f$ and $g$, denoted by $f \circ g$, is the function from $A$ to $C$ defined by:

$$(f \circ g)(a) = f(g(a)) \quad \text{for all } a \in A$$




*Function Composition Pipeline: g is applied first (A → B), followed by f (B → C)*

**EXAMPLE & WALKTHROUGH** **Composing Two and Three Functions**

**Problem:** Given $f(x) = x + 1$ and $g(x) = x^2$ on $\mathbb{R}$:

1. Compute $(f \circ g)(x)$ and compare it with $(g \circ f)(x)$.
2. Given $h(x) = x + 1$, determine the triple composition $(f \circ g \circ h)(x)$.

**Solution:**

1. **Two-function compositions:** Applying $g$ first and then $f$:

$$(f \circ g)(x) = f(g(x)) = f(x^2) = x^2 + 1$$

By contrast, applying $f$ first and then $g$:

$$(g \circ f)(x) = g(f(x)) = g(x+1) = (x+1)^2 = x^2 + 2x + 1$$

Notice that $(f \circ g)(x) \neq (g \circ f)(x)$—function composition is **not commutative** in general.

2. **Triple composition $(f \circ g \circ h)(x)$ with $h(x) = x + 1$:** Function composition is *associative*, and we evaluate from right to left (innermost function first):

$$\begin{aligned}(f \circ g \circ h)(x) &= f(g(h(x))) \\ &= f(g(x+1)) \quad (\text{since } h(x) = x+1) \\ &= f((x+1)^2) \quad (\text{since } g(u) = u^2) \\ &= (x+1)^2 + 1 \quad (\text{since } f(v) = v+1) \\ &= x^2 + 2x + 2\end{aligned}$$

## 2.7 Worked Practice Problems

Below are complete step-by-step solutions to practice problems on injectivity, surjectivity, and bijections.

**EXAMPLE & WALKTHROUGH** **Injectivity and Surjectivity on $\mathbb{Z} \to \mathbb{Z}$**

**Problem:**

- **Part 1:** Determine whether each of these functions from $\mathbb{Z}$ to $\mathbb{Z}$ is one-to-one (injective): **(a)** $f(n) = n - 1$ **(b)** $f(n) = n^2 + 1$ **(c)** $f(n) = n^3$ **(d)** $f(n) = \lceil \frac{n}{2} \rceil$
- **Part 2:** Which of the functions above are onto (surjective)?

**Solutions:**

- **(a) $f(n) = n - 1$:**
  - *One-to-one?* **Yes.** Suppose $f(a) = f(b)$ for $a, b \in \mathbb{Z}$. Then $a - 1 = b - 1 \implies a = b$.
  - *Onto?* **Yes.** Let $y \in \mathbb{Z}$ be any target integer. Choose $n = y + 1 \in \mathbb{Z}$. Then $f(y+1) = (y+1) - 1 = y$. Since $f$ is both one-to-one and onto, $f(n) = n - 1$ is a *bijection* on $\mathbb{Z}$.
- **(b) $f(n) = n^2 + 1$:**
  - *One-to-one?* **No.** Counterexample: $f(1) = 1^2 + 1 = 2$ and $f(-1) = (-1)^2 + 1 = 2$, so $f(1) = f(-1)$ with $1 \neq -1$.
  - *Onto?* **No.** Since $n^2 \geq 0$ for all $n \in \mathbb{Z}$, $f(n) = n^2 + 1 \geq 1$. Thus no integer $y \leq 0$ (e.g., $y = 0$) is in the range of $f$ (nor is $y = 3$, since $n^2 = 2$ has no integer solution).
- **(c) $f(n) = n^3$:**
  - *One-to-one?* **Yes.** The cube function is strictly increasing on $\mathbb{R}$ (and hence on $\mathbb{Z}$): if $a^3 = b^3$, taking real cube roots yields $a = b$.
  - *Onto?* **No.** Only perfect cubes belong to the range. For example, $y = 2 \in \mathbb{Z}$ has no integer pre-image because $n^3 = 2$ has no solution in $\mathbb{Z}$ ($1^3 = 1 < 2 < 8 = 2^3$).
- **(d) $f(n) = \lceil \frac{n}{2} \rceil$:**
  - *One-to-one?* **No.** Every consecutive pair of odd and even integers maps to the same ceiling value: for example, $f(1) = \lceil \frac{1}{2} \rceil = 1$ and $f(2) = \lceil \frac{2}{2} \rceil = 1$, yet $1 \neq 2$.
  - *Onto?* **Yes.** Given any target integer $y \in \mathbb{Z}$, choose the even integer $n = 2y \in \mathbb{Z}$ (or $n = 2y - 1$). Then $f(2y) = \lceil \frac{2y}{2} \rceil = \lceil y \rceil = y$.

| Function $f : \mathbb{Z} \to \mathbb{Z}$ | One-to-One | Onto | Bijection? |
|---|---|---|---|
| **(a)** $f(n) = n - 1$ | Yes (✓) | Yes (✓) | Yes |
| **(b)** $f(n) = n^2 + 1$ | No (⊗, $f(1) = f(-1)$) | No (⊗, $f(n) \geq 1$) | No |
| **(c)** $f(n) = n^3$ | Yes (✓) | No (⊗, $2 \notin$ Range) | No |
| **(d)** $f(n) = \lceil \frac{n}{2} \rceil$ | No (⊗, $f(1) = f(2)$) | Yes (✓, $f(2y) = y$) | No |

**EXAMPLE & WALKTHROUGH** **Surjectivity of Two-Variable Integer Functions f : $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ (Set I)**

**Problem:** Determine whether $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto if: **(a)** $f(m, n) = 2m - n$ **(b)** $f(m, n) = m^2 - n^2$ **(c)** $f(m, n) = m + n + 1$ **(d)** $f(m, n) = |m| - |n|$ **(e)** $f(m, n) = m^2 - 4$

**Solutions:** To show $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is *onto*, we check whether every integer $k \in \mathbb{Z}$ can be written as $f(m, n) = k$ for some $(m, n) \in \mathbb{Z} \times \mathbb{Z}$.

- **(a) $f(m, n) = 2m - n$: Onto.** For any $k \in \mathbb{Z}$, set $m = 0$ and $n = -k$ (both in $\mathbb{Z}$). Then $f(0, -k) = 2(0) - (-k) = k$.
- **(b) $f(m, n) = m^2 - n^2$: Not onto.** Factoring gives $m^2 - n^2 = (m - n)(m + n)$. Notice that $(m - n)$ and $(m + n)$ always have the *same parity* (their difference is $2n$, which is even).
  - If both factors are even, their product is a multiple of 4.
  - If both factors are odd, their product is odd.

  Therefore, $m^2 - n^2$ can never equal an integer congruent to 2 modulo 4. Specifically, $k = 2 \in \mathbb{Z}$ has no integer solution because the only factorizations of 2 are $1 \times 2$ and $(-1) \times (-2)$, where the factors have opposite parity. Thus $2 \notin \text{Range}(f)$.
- **(c) $f(m, n) = m + n + 1$: Onto.** For any $k \in \mathbb{Z}$, choose $m = k - 1 \in \mathbb{Z}$ and $n = 0 \in \mathbb{Z}$. Then $f(k - 1, 0) = (k - 1) + 0 + 1 = k$.
- **(d) $f(m, n) = |m| - |n|$: Onto.** Let $k \in \mathbb{Z}$ be any integer:
  - If $k \geq 0$, choose $(m, n) = (k, 0)$, giving $f(k, 0) = |k| - |0| = k$.
  - If $k < 0$, choose $(m, n) = (0, k)$, giving $f(0, k) = |0| - |k| = -(-k) = k$.

  Every integer $k \in \mathbb{Z}$ is achieved.
- **(e) $f(m, n) = m^2 - 4$: Not onto.** Since $m^2 \geq 0$ for all $m \in \mathbb{Z}$, $f(m, n) = m^2 - 4 \geq -4$. No integer $k < -4$ (such as $k = -5$) is in the range (nor is $k = -2$, since $m^2 = 2$ has no integer solution).

**EXAMPLE & WALKTHROUGH** **Surjectivity of Two-Variable Integer Functions f : $\mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ (Set II)**

**Problem:** Determine whether the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ is onto if: **(a)** $f(m, n) = m + n$ **(b)** $f(m, n) = m^2 + n^2$ **(c)** $f(m, n) = m$ **(d)** $f(m, n) = |n|$ **(e)** $f(m, n) = m - n$

**Solutions:**

- **(a) $f(m, n) = m + n$: Onto.** For any $k \in \mathbb{Z}$, choose $(m, n) = (k, 0) \in \mathbb{Z} \times \mathbb{Z}$; then $f(k, 0) = k + 0 = k$.

- **(b) $f(m,n) = m^2 + n^2$: Not onto.** Since $m^2 + n^2 \geq 0$, negative integers (e.g., $k = -1$) are never attained, and positive integers like $k = 3$ cannot be written as a sum of two integer squares.
- **(c) $f(m,n) = m$: Onto.** This is the projection onto the first coordinate. For any $k \in \mathbb{Z}$, $f(k,0) = k$.
- **(d) $f(m,n) = |n|$: Not onto.** Since $|n| \geq 0$ for all $n \in \mathbb{Z}$, no negative integer (such as $k = -1$) is in the range of $f$.
- **(e) $f(m,n) = m - n$: Onto.** For any $k \in \mathbb{Z}$, choose $(m,n) = (k,0) \in \mathbb{Z} \times \mathbb{Z}$; then $f(k,0) = k - 0 = k$.

**EXAMPLE & WALKTHROUGH** **Testing Bijections from $\mathbb{R}$ to $\mathbb{R}$**

**Problem:** Determine whether each of these functions is a bijection from $\mathbb{R}$ to $\mathbb{R}$:

- **Set 1: (a)** $f(x) = -3x + 4$ **(b)** $f(x) = -3x^2 + 7$ **(c)** $f(x) = \frac{x+1}{x+2}$ **(d)** $f(x) = x^5 + 1$
- **Set 2: (a)** $f(x) = 2x + 1$ **(b)** $f(x) = x^2 + 1$ **(c)** $f(x) = x^3$ **(d)** $f(x) = \frac{x^2+1}{x^2+2}$

**Solutions for Set 1:**

- **1(a) $f(x) = -3x + 4$: Bijection.**
  - *One-to-one:* $-3a + 4 = -3b + 4 \implies -3a = -3b \implies a = b$.
  - *Onto:* For any $y \in \mathbb{R}$, solving $y = -3x + 4$ gives $x = \frac{4-y}{3} \in \mathbb{R}$, so $f\left(\frac{4-y}{3}\right) = y$.
- **1(b) $f(x) = -3x^2 + 7$: Not a bijection** (neither one-to-one nor onto).
  - *Not one-to-one:* $f(1) = 4 = f(-1)$ with $1 \neq -1$.
  - *Not onto:* Since $x^2 \geq 0$, $f(x) \leq 7$, so $y = 8 \in \mathbb{R}$ has no real pre-image.
- **1(c) $f(x) = \frac{x+1}{x+2}$: Not a bijection from $\mathbb{R}$ to $\mathbb{R}$.**
  - *Not defined on all of* $\mathbb{R}$*:* At $x = -2$, the denominator is 0, so $f(-2)$ is undefined.
  - *Not onto* $\mathbb{R}$*:* Rewriting $f(x) = \frac{(x+2)-1}{x+2} = 1 - \frac{1}{x+2}$ shows that $f(x) \neq 1$ for all $x \in \mathbb{R} \setminus \{-2\}$, so $y = 1$ has no pre-image. (Note: $f$ is a bijection from $\mathbb{R} \setminus \{-2\}$ to $\mathbb{R} \setminus \{1\}$.)
- **1(d) $f(x) = x^5 + 1$: Bijection.**
  - *One-to-one:* $a^5 + 1 = b^5 + 1 \implies a^5 = b^5 \implies a = b$ (since odd powers are strictly increasing on $\mathbb{R}$).
  - *Onto:* For any $y \in \mathbb{R}$, every real number has a unique real fifth root $x = \sqrt[5]{y-1} \in \mathbb{R}$, satisfying $f(\sqrt[5]{y-1}) = (y-1) + 1 = y$.

**Solutions for Set 2:**

- **2(a) $f(x) = 2x + 1$: Bijection.**
  - *One-to-one:* $2a + 1 = 2b + 1 \implies a = b$.
  - *Onto:* For any $y \in \mathbb{R}$, $x = \frac{y-1}{2} \in \mathbb{R}$ satisfies $f\left(\frac{y-1}{2}\right) = y$ (with inverse $f^{-1}(y) = \frac{y-1}{2}$).
- **2(b) $f(x) = x^2 + 1$: Not a bijection** (neither one-to-one nor onto).
  - *Not one-to-one:* $f(1) = 2 = f(-1)$.
  - *Not onto:* $f(x) = x^2 + 1 \geq 1$, so $y = 0 \in \mathbb{R}$ has no real pre-image.
- **2(c) $f(x) = x^3$: Bijection.**
  - *One-to-one:* $a^3 = b^3 \implies a = b$ in $\mathbb{R}$.
  - *Onto:* For any $y \in \mathbb{R}$, $x = \sqrt[3]{y} \in \mathbb{R}$ satisfies $f(\sqrt[3]{y}) = (\sqrt[3]{y})^3 = y$. (Unlike on $\mathbb{Z}$, $f(x) = x^3$ *is* onto on $\mathbb{R}$.)
- **2(d) $f(x) = \frac{x^2+1}{x^2+2}$: Not a bijection** (neither one-to-one nor onto).
  - *Not one-to-one:* $f(1) = \frac{2}{3} = f(-1)$ since $(-x)^2 = x^2$.
  - *Not onto:* Writing $f(x) = 1 - \frac{1}{x^2+2}$ and using $x^2 \geq 0 \implies x^2 + 2 \geq 2 \implies 0 < \frac{1}{x^2+2} \leq \frac{1}{2}$, we see that $\text{Range}(f) = \left[\frac{1}{2}, 1\right) \subsetneq \mathbb{R}$.

# 3. Sequences and Series

Discrete Sequences, Geometric and Arithmetic Progressions, Sigma Notation, and Closed-Form Summations

## 3.1 Sequences

In computer science and discrete mathematics, ordered lists of elements—such as array entries indexed in memory, step counts of iterative algorithms across input sizes, or states produced at discrete clock ticks—are modeled mathematically as *sequences*.

**DEFINITION** **Sequence and Term**

A **sequence** is a function from a subset of the set of integers—usually either the set of positive integers $\{1, 2, 3, \ldots\}$ or the set of non-negative integers $\{0, 1, 2, \ldots\}$—to a target set $S$:

$$a : \{1, 2, 3, \ldots\} \to S \quad \text{or} \quad a : \{0, 1, 2, \ldots\} \to S$$

- The image $a(n)$ of an integer $n$ is denoted by subscript notation $a_n$ and is called the $\boldsymbol{n}$**-th term** of the sequence.
- The entire sequence is denoted using braces as $\{a_n\}$ (or $\{a_n\}_{n=1}^{\infty}$).

**INTUITION & DEEP DIVE** **Sequences Are Functions with an Integer Domain**

Why do we define a sequence as a *function* rather than a *set*?

1. **Order matters:** In a set, $\{1, 2, 3\} = \{3, 1, 2\}$, whereas the sequence $1, 2, 3, \ldots$ is distinct from $3, 1, 2, \ldots$ because the domain index $n$ fixes each position.
2. **Repetition is allowed:** In a set, $\{1, -1, 1, -1, \ldots\} = \{1, -1\}$ has only two elements, whereas the alternating sequence $a_n = (-1)^n$ has infinitely many indexed terms $a_1, a_2, a_3, \ldots$ that take values in $\{-1, 1\}$.

**EXAMPLE & WALKTHROUGH** **The Harmonic Sequence**

Consider the sequence $\{a_n\}$ defined for $n \geq 1$ by the explicit formula:

$$a_n = \frac{1}{n}$$

Evaluating $a_n$ at $n = 1, 2, 3, 4, \ldots$ produces the sequence of terms $a_1, a_2, a_3, a_4, \ldots$, namely:

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \ldots, \quad \frac{1}{n}, \quad \ldots$$

This sequence is known in mathematics and algorithm analysis as the *harmonic sequence*.

**EXAMPLE & WALKTHROUGH** **Finding the First Five Terms of a Sequence**

**Problem:** Determine the first 5 terms of the following sequences:

1. $a_n = n + 1$
2. $a_n = n^2$

**Solution:** When the starting index is not explicitly stated, the standard convention in discrete mathematics is $n = 1, 2, 3, 4, 5$ (though 0-based indexing $n = 0, 1, 2, 3, 4$ is also common in computer science). Both conventions are shown below:

1. **For $\boldsymbol{a_n = n + 1}$:**
   - Starting at $n = 1$:

$$a_1 = 1 + 1 = 2, \quad a_2 = 2 + 1 = 3, \quad a_3 = 3 + 1 = 4, \quad a_4 = 4 + 1 = 5, \quad a_5 = 5 + 1 = 6$$

First 5 terms: $\mathbf{2, 3, 4, 5, 6}$.

- (If starting at $n = 0$: $a_0 = 1, a_1 = 2, a_2 = 3, a_3 = 4, a_4 = 5$, i.e., $\mathbf{1, 2, 3, 4, 5}$.)

2. **For $a_n = n^2$:**
   - Starting at $n = 1$:

$$a_1 = 1^2 = 1, \quad a_2 = 2^2 = 4, \quad a_3 = 3^2 = 9, \quad a_4 = 4^2 = 16, \quad a_5 = 5^2 = 25$$

   First 5 terms: $\mathbf{1, 4, 9, 16, 25}$.
   - (If starting at $n = 0$: $a_0 = 0, a_1 = 1, a_2 = 4, a_3 = 9, a_4 = 16$, i.e., $\mathbf{0, 1, 4, 9, 16}$.)

## 3.2 Geometric and Arithmetic Progressions

Two families of sequences appear throughout computer science: *geometric progressions* (which model multiplicative growth or decay, divide-and-conquer recursion trees, and binary representations) and *arithmetic progressions* (which model constant additive stepping and nested loop counters).

### 3.2.1 Geometric Progressions

**DEFINITION** **Geometric Progression**

A **geometric progression** is a sequence of the form:

$$a, \quad ar, \quad ar^2, \quad ar^3, \quad \ldots, \quad ar^n, \quad \ldots$$

where the **initial term** $a$ and the **common ratio** $r$ are real numbers ($a, r \in \mathbb{R}$).

- For 0-based indexing ($n = 0, 1, 2, \ldots$), the general term is $a_n = ar^n$.
- For 1-based indexing ($n = 1, 2, 3, \ldots$), the $n$-th term is $a_n = ar^{n-1}$.
- For every $n \geq 1$ (with $a \neq 0, r \neq 0$), the ratio of consecutive terms is constant: $\frac{a_n}{a_{n-1}} = r$.

**EXAMPLE & WALKTHROUGH** **Geometric Progression $c_n = 2 \cdot 5^n$**

Consider the sequence $\{c_n\}$ defined for $n = 0, 1, 2, 3, \ldots$ by:

$$c_n = 2 \cdot 5^n$$

This is a geometric progression with initial term $a = c_0 = 2$ and common ratio $r = 5$. Its terms are:

$$c_0 = 2 \cdot 5^0 = 2, \quad c_1 = 2 \cdot 5^1 = 10, \quad c_2 = 2 \cdot 5^2 = 50, \quad c_3 = 2 \cdot 5^3 = 250, \quad \ldots$$

Each term is obtained by multiplying the preceding term by $r = 5$.

### 3.2.2 Arithmetic Progressions

**DEFINITION** **Arithmetic Progression**

An **arithmetic progression** is a sequence of the form:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad \ldots, \quad a + nd, \quad \ldots$$

where the **initial term** $a$ and the **common difference** $d$ are real numbers ($a, d \in \mathbb{R}$).

- For 0-based indexing ($n = 0, 1, 2, \ldots$), the general term is $a_n = a + nd$.
- For 1-based indexing ($n = 1, 2, 3, \ldots$), the $n$-th term is $a_n = a + (n - 1)d$.
- For every $n \geq 1$, the difference between consecutive terms is constant: $a_n - a_{n-1} = d$.

**EXAMPLE & WALKTHROUGH** **Arithmetic Progression $b_n = 1 + 3n$**

Consider the sequence $\{b_n\}$ defined for $n = 0, 1, 2, 3, \ldots$ by:

$$b_n = 1 + n \cdot 3$$

This is an arithmetic progression with initial term $a = b_0 = 1$ and common difference $d = 3$. Its terms are:

$$b_0 = 1, \quad b_1 = 1 + 3 = 4, \quad b_2 = 1 + 6 = 7, \quad b_3 = 1 + 9 = 10, \quad \ldots$$

Each successive term increases by the constant difference $d = 3$.

**INTUITION & DEEP DIVE** **Discrete Analogues of Linear and Exponential Functions**

- An **arithmetic progression** $a_n = a + nd$ is the discrete counterpart of the continuous *linear function* $f(x) = dx + a$ (constant first difference $\Delta a_n = a_{n+1} - a_n = d$, analogous to a constant derivative $f'(x) = d$).
- A **geometric progression** $a_n = ar^n$ is the discrete counterpart of the continuous *exponential function* $f(x) = ar^x$ (constant consecutive ratio $\frac{a_{n+1}}{a_n} = r$).

**EXAMPLE & WALKTHROUGH** **Classifying Geometric and Arithmetic Sequences**

**Problem:** Identify which of the following sequences are geometric progressions and which are arithmetic progressions:

1. $2, 4, 8, 16, \ldots$
2. $1, 4, 7, 10, \ldots$
3. $1, -1, 1, -1, \ldots$
4. $5, 9, 13, 17, \ldots$

**Solution:**

1. **$2, 4, 8, 16, \ldots$ — Geometric Progression.**
   - *Consecutive ratios:* $\frac{4}{2} = 2$, $\frac{8}{4} = 2$, $\frac{16}{8} = 2$.
   - *Parameters:* Initial term $a = 2$, common ratio $r = 2$.
   - *Explicit formula:* $a_n = 2 \cdot 2^n = 2^{n+1}$ (for $n \geq 0$), or $a_n = 2^n$ (for $n \geq 1$).
2. **$1, 4, 7, 10, \ldots$ — Arithmetic Progression.**
   - *Consecutive differences:* $4 - 1 = 3$, $7 - 4 = 3$, $10 - 7 = 3$.
   - *Parameters:* Initial term $a = 1$, common difference $d = 3$.
   - *Explicit formula:* $a_n = 1 + 3n$ (for $n \geq 0$), or $a_n = 3n - 2$ (for $n \geq 1$).
3. **$1, -1, 1, -1, \ldots$ — Geometric Progression.**
   - *Consecutive ratios:* $\frac{-1}{1} = -1$, $\frac{1}{-1} = -1$, $\frac{-1}{1} = -1$. (Differences alternate between $-2$ and $+2$, so it is *not* arithmetic.)
   - *Parameters:* Initial term $a = 1$, common ratio $r = -1$.
   - *Explicit formula:* $a_n = (-1)^n$ (for $n \geq 0$), or $a_n = (-1)^{n-1}$ (for $n \geq 1$).
4. **$5, 9, 13, 17, \ldots$ — Arithmetic Progression.**
   - *Consecutive differences:* $9 - 5 = 4$, $13 - 9 = 4$, $17 - 13 = 4$.
   - *Parameters:* Initial term $a = 5$, common difference $d = 4$.
   - *Explicit formula:* $a_n = 5 + 4n$ (for $n \geq 0$), or $a_n = 4n + 1$ (for $n \geq 1$).

### 3.2.3 Reference Table of Common Sequences

Recognizing standard integer sequences by their first several terms is helpful for pattern recognition, solving recurrences, and analyzing algorithms.

| $n$-th Term ($n \geq 1$) | First 10 Terms | Sequence Name / Context |
|---|---|---|
| $n^2$ | $1, 4, 9, 16, 25, 36, 49, 64, 81, 100, \ldots$ | Perfect squares (quadratic growth) |
| $n^3$ | $1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, \ldots$ | Perfect cubes (cubic growth) |
| $n^4$ | $1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, \ldots$ | Fourth powers (quartic growth) |
| $2^n$ | $2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, \ldots$ | Powers of 2 (binary subsets) |
| $3^n$ | $3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, \ldots$ | Powers of 3 (ternary growth) |
| $n!$ | $1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, \ldots$ | Factorials (permutations of $n$ items) |
| $f_n$ | $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, (89), \ldots$ | Fibonacci ($f_1 = 1, f_2 = 1, f_n = f_{n-1} + f_{n-2}$) |

## 3.3 Series and Sigma Summation Notation

Adding together consecutive terms of a sequence $\{a_n\}$ produces a *series*, which we express concisely using capital Greek sigma ($\Sigma$) notation.

**DEFINITION** **Series and Summation Notation**

The sum of the terms $a_m, a_{m+1}, \ldots, a_k$ from a sequence $\{a_n\}$ is written in **sigma notation** as:

$$\sum_{i=m}^{k} a_i = a_m + a_{m+1} + \ldots + a_{k-1} + a_k$$

- The variable $i$ is called the **index of summation**. It is a bound (dummy) variable: replacing $i$ with $j$ or $k$ does not change the value of the sum.
- The integer $m$ is the **lower limit**, and the integer $k$ (with $k \geq m$) is the **upper limit**.
- The total number of terms in the sum is $(k - m + 1)$.

**EXAMPLE & WALKTHROUGH** **Expanding and Evaluating Summations**

1. **Harmonic Sum ($\sum_{j=1}^{n} \frac{1}{j}$):** In expanded form:

$$\sum_{j=1}^{n} \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$$

Evaluating for $n = 5$ using the least common denominator $\text{lcm}(1, 2, 3, 4, 5) = 60$:

$$\sum_{j=1}^{5} \frac{1}{j} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{60 + 30 + 20 + 15 + 12}{60} = \frac{137}{60}$$

2. **Sum of Squares ($\sum_{i=1}^{k} i^2$):** In expanded form:

$$\sum_{i=1}^{k} i^2 = 1 + 4 + 9 + \ldots + k^2$$

Evaluating for $k = 10$:

$$\sum_{i=1}^{10} i^2 = 1 + 4 + 9 + 16 + 25 + 36 + 49 + 64 + 81 + 100 = 385$$

3. **Shifted Linear Sum ($\sum_{i=2}^{n}(i+1)$):** In expanded form:

$$\sum_{i=2}^{n}(i+1) = 3 + 4 + 5 + \ldots + (n+1)$$

Evaluating for $n = 7$ (which has $7 - 2 + 1 = 6$ terms):

$$\sum_{i=2}^{7}(i+1) = 3 + 4 + 5 + 6 + 7 + 8 = 33$$

## 3.4 Geometric Series and Closed-Form Summation Formulae

In algorithm analysis, rather than adding terms one by one, we aim to convert a summation $\sum_{k=m}^{n} a_k$ into an equivalent *closed-form expression* that can be evaluated in $O(1)$ arithmetic operations.

### 3.4.1 The Geometric Series Theorem

**CORE CONCEPT** **Theorem: Sum of a Finite Geometric Series**

If $a$ and $r$ are real numbers and $r \neq 0$, then:

$$\sum_{j=0}^{n} ar^j = \begin{cases} \frac{ar^{n+1}-a}{r-1} = \frac{a(r^{n+1}-1)}{r-1} = \frac{a(1-r^{n+1})}{1-r} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1 \end{cases}$$

**INTUITION & DEEP DIVE** **Derivation of the Geometric Series Formula (Shift-and-Subtract)**

Why does $\sum_{j=0}^{n} ar^j$ simplify to $\frac{ar^{n+1}-a}{r-1}$?

- **Case $r = 1$:** Each of the $n+1$ terms equals $a \cdot 1^j = a$, so the sum is $(n+1)a$.
- **Case $r \neq 1$:** Let $S_n = \sum_{j=0}^{n} ar^j = a + ar + ar^2 + \ldots + ar^n$. Multiply both sides by $r$:

$$rS_n = ar + ar^2 + ar^3 + \ldots + ar^n + ar^{n+1}$$
$$S_n = a + ar + ar^2 + \ldots + ar^{n-1} + ar^n$$

Subtracting the second equation from the first cancels all intermediate terms (*telescoping cancellation*):

$$(r-1)S_n = ar^{n+1} - a \implies S_n = \frac{ar^{n+1} - a}{r-1}$$

**EXAMPLE & WALKTHROUGH** **Sum of the First 10 Terms of a Geometric Series**

**Problem:** Given the geometric series $2 + 6 + 18 + 54 + \ldots$, what is the sum of the first 10 terms of this series?

**Solution:**

1. Identify the parameters:
   - First term: $a = 2$.
   - Common ratio: $r = \frac{6}{2} = \frac{18}{6} = 3$.
   - Upper index limit: Because the summation formula $\sum_{j=0}^{n} ar^j$ starts at $j = 0$, the first 10 terms correspond to indices $j = 0, 1, 2, \ldots, 9$, so $\boldsymbol{n = 9}$ (yielding $n + 1 = 10$ terms).
2. Apply the Geometric Series Theorem:

$$\sum_{j=0}^{9} 2 \cdot 3^j = \frac{2 \cdot 3^{9+1} - 2}{3-1} = \frac{2 \cdot 3^{10} - 2}{2} = 3^{10} - 1 = 59049 - 1 = 59048$$

**PITFALL / WARNING** **Off-by-One Trap in Geometric Series: Counting Terms vs. Upper Limit n**

In the formula $\sum_{j=0}^{n} ar^j = \frac{ar^{n+1}-a}{r-1}$, the exponent in the numerator is $\boldsymbol{n+1}$, which equals the **total number of terms** when the sum starts at $j = 0$.

- If a problem asks for the *first* 10 *terms* starting at $j = 0$, the upper limit is $n = 9$, and the exponent is $n + 1 = 10$.
- If a summation instead starts at $j = 1$, such as $\sum_{j=1}^{n} ar^j$, the first term of the sum is $ar$ (not $a$), and there are $n$ terms, so the sum is $\frac{(ar)r^n - ar}{r-1} = \frac{ar^{n+1}-ar}{r-1}$.

### 3.4.2 Table of Useful Summation Formulae

The following closed-form identities are standard tools for simplifying finite and infinite series:

| Summation (Sum) | Closed Form | Name & Applicability |
|---|---|---|
| $\sum_{k=0}^{n} ar^k \quad (r \neq 0)$ | $\frac{ar^{n+1} - a}{r-1}, \quad r \neq 1$ | Finite Geometric Series ($n+1$ terms) |
| $\sum_{k=1}^{n} k$ | $\frac{n(n+1)}{2}$ | Sum of first $n$ positive integers (Gauss's formula) |
| $\sum_{k=1}^{n} k^2$ | $\frac{n(n+1)(2n+1)}{6}$ | Sum of first $n$ perfect squares |
| $\sum_{k=1}^{n} k^3$ | $\frac{n^2(n+1)^2}{4} = \left(\frac{n(n+1)}{2}\right)^2$ | Sum of first $n$ perfect cubes (Nicomachus's identity) |
| $\sum_{k=0}^{\infty} x^k, \quad \lvert x\rvert < 1$ | $\frac{1}{1-x}$ | Infinite Geometric Series (converges iff $\lvert x\rvert < 1$) |
| $\sum_{k=1}^{\infty} kx^{k-1}, \quad \lvert x\rvert < 1$ | $\frac{1}{(1-x)^2}$ | Differentiated Infinite Geometric Series |

### 3.4.3 Index Splitting for Non-Unit Lower Limits

When a summation starts at $k = m > 1$ instead of $k = 1$, we can apply the standard closed-form formulas from the table above by subtracting the missing prefix sum $\sum_{k=1}^{m-1} a_k$:

$$\sum_{k=m}^{n} a_k = \sum_{k=1}^{n} a_k - \sum_{k=1}^{m-1} a_k$$

**EXAMPLE & WALKTHROUGH** **Evaluating Shifted Summations**

**Problem:** Evaluate the following two summations:

$$(1) \quad \sum_{i=2}^{10} i \qquad (2) \quad \sum_{k=10}^{15} k^2$$

**Solutions:**

1. **Evaluating $\sum_{i=2}^{10} i$:** Using index splitting and Gauss's formula $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$:

$$\sum_{i=2}^{10} i = \sum_{i=1}^{10} i - \sum_{i=1}^{1} i = \frac{10 \cdot (10+1)}{2} - 1 = 55 - 1 = 54$$

(*Verification by arithmetic series:* There are $10 - 2 + 1 = 9$ terms with first term 2 and last term 10, so $S = 9 \cdot \frac{2+10}{2} = 9 \cdot 6 = 54$.)

2. **Evaluating $\sum_{k=10}^{15} k^2$:** Using index splitting with $m = 10$ (so $m - 1 = 9$) and the sum-of-squares formula $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\begin{aligned}\sum_{k=10}^{15} k^2 &= \sum_{k=1}^{15} k^2 - \sum_{k=1}^{9} k^2 \\ &= \frac{15 \cdot (15+1) \cdot (2(15)+1)}{6} - \frac{9 \cdot (9+1) \cdot (2(9)+1)}{6} \\ &= \frac{15 \cdot 16 \cdot 31}{6} - \frac{9 \cdot 10 \cdot 19}{6} \\ &= 1240 - 285 = 955\end{aligned}$$

(*Direct term-by-term verification:* $10^2 + 11^2 + 12^2 + 13^2 + 14^2 + 15^2 = 100 + 121 + 144 + 169 + 196 + 225 = 955$.)

## 3.5 Worked Practice Exercises

Below are worked solutions to additional practice problems on sequences, recurrences, pattern recognition, and summations.

**EXAMPLE & WALKTHROUGH** **Finding Sequence Terms from Explicit and Recursive Rules**

**Part 1:** Find the first four terms of each of the following sequences whose $n$-th term ($n = 1, 2, 3, 4$) is given by:

- **(a) $a_n = 2^n - n$:**

$$a_1 = 2^1 - 1 = 1, \quad a_2 = 2^2 - 2 = 2, \quad a_3 = 2^3 - 3 = 5, \quad a_4 = 2^4 - 4 = 12$$

- **(b) $a_n = 3$:** (Constant sequence)

$$a_1 = 3, \quad a_2 = 3, \quad a_3 = 3, \quad a_4 = 3$$

- **(c) $a_n = 7 + 4^n$:**

$$a_1 = 7 + 4 = 11, \quad a_2 = 7 + 16 = 23, \quad a_3 = 7 + 64 = 71, \quad a_4 = 7 + 256 = 263$$

- **(d) $a_n = 2^n + (-2)^n$:**

$$a_1 = 2 - 2 = 0, \quad a_2 = 4 + 4 = 8, \quad a_3 = 8 - 8 = 0, \quad a_4 = 16 + 16 = 32$$

**Part 2:** Find the first five terms ($n = 0, 1, 2, 3, 4$ or $n = 1, \ldots, 5$) of sequences defined by recurrence relations:

- **(a) $a_n = 6a_{n-1}, \quad a_0 = 2$:** Geometric progression with $a = 2, r = 6$:

$$a_0 = 2, \quad a_1 = 12, \quad a_2 = 72, \quad a_3 = 432, \quad a_4 = 2592$$

- **(b) $a_n = a_{n-1}^2, \quad a_1 = 2$:** Each term squares the previous term ($a_n = 2^{2^{n-1}}$):

$$a_1 = 2, \quad a_2 = 2^2 = 4, \quad a_3 = 4^2 = 16, \quad a_4 = 16^2 = 256, \quad a_5 = 256^2 = 65536$$

- **(c) $a_n = a_{n-1} + 3a_{n-2}, \quad a_0 = 1, a_1 = 2$:**

$$a_0 = 1, \quad a_1 = 2, \quad a_2 = 2 + 3(1) = 5, \quad a_3 = 5 + 3(2) = 11, \quad a_4 = 11 + 3(5) = 26$$

- **(d) $a_n = na_{n-1} + n^2 a_{n-2}, \quad a_0 = 1, a_1 = 1$:**
  - $a_0 = 1, \quad a_1 = 1$
  - $a_2 = 2a_1 + 2^2 a_0 = 2(1) + 4(1) = 6$
  - $a_3 = 3a_2 + 3^2 a_1 = 3(6) + 9(1) = 27$
  - $a_4 = 4a_3 + 4^2 a_2 = 4(27) + 16(6) = 108 + 96 = 204$

**EXAMPLE & WALKTHROUGH** **Constructing the First 10 Terms from Verbal Rules**

**Problem:** List the first 10 terms of each of these sequences:

- **(a) Begins with 2 and each successive term is 3 more than the preceding term:** Arithmetic progression ($a = 2, d = 3$): $\mathbf{2, 5, 8, 11, 14, 17, 20, 23, 26, 29}$.
- **(b) Lists each positive integer three times, in increasing order:** $\mathbf{1, 1, 1, 2, 2, 2, 3, 3, 3,} 4$.
- **(c) Lists the odd positive integers in increasing order, listing each odd integer twice:** $\mathbf{1, 1, 3, 3, 5, 5, 7, 7, 9, 9}$.
- **(d) Whose $n$-th term is $n! - 2^n$ (for $n = 1, 2, ..., 10$):**
  - $n = 1$: $1 - 2 = -1$
  - $n = 2$: $2 - 4 = -2$
  - $n = 3$: $6 - 8 = -2$
  - $n = 4$: $24 - 16 = 8$
  - $n = 5$: $120 - 32 = 88$
  - $n = 6$: $720 - 64 = 656$
  - $n = 7$: $5040 - 128 = 4912$
  - $n = 8$: $40320 - 256 = 40064$
  - $n = 9$: $362880 - 512 = 362368$
  - $n = 10$: $3628800 - 1024 = 3627776$
- **(e) Begins with 3, where each succeeding term is twice the preceding term:** Geometric progression ($a = 3, r = 2$): $\mathbf{3, 6, 12, 24, 48, 96, 192, 384, 768, 1536}$.
- **(f) First term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms:** Fibonacci-like recurrence ($a_n = a_{n-1} + a_{n-2}$): $\mathbf{2, 4, 6, 10, 16, 26, 42, 68, 110, 178}$.
- **(g) Whose $n$-th term is the number of bits in the binary expansion of positive integer $n$ ($\lfloor \log_2 n \rfloor + 1$):** For $n = 1; 2, 3; 4, 5, 6, 7; 8, 9, 10$: $\mathbf{1, 2, 2, 3, 3, 3, 3, 4, 4, 4}$.
- **(h) Where the $n$-th term is the number of letters in the English word for the index $n$:** For *one (3), two (3), three (5), four (4), five (4), six (3), seven (5), eight (5), nine (4), ten (3)*: $\mathbf{3, 3, 5, 4, 4, 3, 5, 5, 4, 3}$.

**EXAMPLE & WALKTHROUGH** **Finding Explicit Formulas for Sequence Patterns**

**Problem:** For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the given list ($n = 1, 2, 3, ...$):

- **(a) $\mathbf{3, 6, 11, 18, 27, 38, 51, 66, 83, 102, ...}$** The first differences are $3, 5, 7, 9, 11, ...$ (consecutive odd numbers), or comparing directly with $n^2 = 1, 4, 9, 16, 25, ...$, each term is 2 more than $n^2$. **Formula:** $a_n = n^2 + 2$ (for $n \geq 1$).
- **(b) $\mathbf{7, 11, 15, 19, 23, 27, 31, 35, 39, 43, ...}$** Constant difference $d = 4$ with initial term $a_1 = 7$. **Formula:** $a_n = 7 + 4(n - 1) = 4n + 3$ (for $n \geq 1$).
- **(c) $\mathbf{1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, ...}$** Each term is the *binary (base-2) representation* of the positive integer $n$ ($1_2 = 1, 2_2 = 10, 3_2 = 11, 4_2 = 100, ...$).
- **(d) $\mathbf{1, 2, 2, 2, 3, 3, 3, 3, 3, 5, 5, 5, 5, 5, 5, 5, ...}$** The integer 1 appears 1 time, 2 appears 3 times, 3 appears 5 times, 5 appears 7 times: the $k$-th Fibonacci number (starting from $f_2 = 1, f_3 = 2, f_4 = 3, f_5 = 5, ...$) is repeated $(2k - 1)$ times.

**EXAMPLE & WALKTHROUGH** **Evaluating Summations and Closed Forms**

**Part 1:** Find the value of each of these sums:

- **(a) $\sum_{k=1}^{5} (k + 1)$:**

$$\sum_{k=1}^{5}(k+1) = 2+3+4+5+6 = 20$$

- **(b) $\sum_{j=0}^{4}(-2)^j$:** Geometric series with $a=1, r=-2, n=4$:

$$\sum_{j=0}^{4}(-2)^j = \frac{(-2)^5-1}{-2-1} = \frac{-32-1}{-3} = \frac{-33}{-3} = 11$$

- **(c) $\sum_{i=1}^{10} 3$:** Sum of 10 identical copies of 3: $10 \times 3 = 30$.
- **(d) $\sum_{j=0}^{8}\left(2^{j+1}-2^j\right)$:** This is a *telescoping sum*. Since $2^{j+1}-2^j = 2^j$, or by telescoping cancellation $(2^1-2^0)+(2^2-2^1)+\ldots+(2^9-2^8)$:

$$\sum_{j=0}^{8}(2^{j+1}-2^j) = 2^9-2^0 = 512-1 = 511$$

**Part 2:** Find the value of each of these sums of terms of a geometric progression:

- **(a) $\sum_{j=0}^{8} 3\cdot 2^j$:** With $a=3, r=2, n=8$:

$$\sum_{j=0}^{8}3\cdot 2^j = \frac{3\cdot 2^9-3}{2-1} = 3(512-1) = 3\cdot 511 = 1533$$

- **(b) $\sum_{j=1}^{8} 2^j$:** Subtract the $j=0$ term ($2^0=1$) from $\sum_{j=0}^{8}2^j = 2^9-1 = 511$:

$$\sum_{j=1}^{8}2^j = 511-1 = 510$$

- **(c) $\sum_{j=2}^{8}(-3)^j$:** First term at $j=2$ is $a=(-3)^2=9$, common ratio $r=-3$, and there are $8-2+1=7$ terms:

$$\sum_{j=2}^{8}(-3)^j = \frac{9((-3)^7-1)}{-3-1} = \frac{9(-2187-1)}{-4} = \frac{9(-2188)}{-4} = 9\cdot 547 = 4923$$

- **(d) $\sum_{j=0}^{8} 2\cdot(-3)^j$:** With $a=2, r=-3, n=8$:

$$\sum_{j=0}^{8}2\cdot(-3)^j = \frac{2((-3)^9-1)}{-3-1} = \frac{2(-19683-1)}{-4} = \frac{-19684}{-2} = 9842$$

**Part 3:** Find a closed-form value for shifted power sums:

- **(a) Evaluate $\sum_{k=50}^{100}k^2$:** Using index splitting ($m=50 \Longrightarrow m-1=49$) and $\sum_{k=1}^{n}k^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\begin{aligned}\sum_{k=50}^{100}k^2 &= \sum_{k=1}^{100}k^2 - \sum_{k=1}^{49}k^2 \\ &= \frac{100\cdot 101\cdot 201}{6} - \frac{49\cdot 50\cdot 99}{6} \\ &= 338350 - 40425 = 297925\end{aligned}$$

- **(b) Evaluate $\sum_{k=99}^{200}k^3$:** Using index splitting ($m=99 \Longrightarrow m-1=98$) and $\sum_{k=1}^{n}k^3 = \frac{n^2(n+1)^2}{4}$:

$$\begin{aligned}\sum_{k=99}^{200}k^3 &= \sum_{k=1}^{200}k^3 - \sum_{k=1}^{98}k^3 \\ &= \left(\frac{200\cdot 201}{2}\right)^2 - \left(\frac{98\cdot 99}{2}\right)^2 \\ &= (20100)^2 - (4851)^2 = 404010000 - 23532201 = 380477799\end{aligned}$$

# 4. Number Theory

Divisibility, the Division Algorithm, Modular Arithmetic, Prime Numbers, and the Euclidean Algorithm

## 4.1 Divisibility

Number theory is the branch of mathematics devoted to the study of the integers $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$ and their arithmetic properties. While historically regarded as a purely theoretical discipline, number theory forms the backbone of modern computer science—powering public-key cryptography (such as RSA and Diffie-Hellman), cryptographic hash functions, pseudorandom number generation, error-correcting codes, and hash table indexing.

The most basic relationship between two integers is whether one divides the other evenly without leaving a remainder.

**DEFINITION** **Divisibility**

Let $a$ and $b$ be integers with $a \neq 0$. We say that $\boldsymbol{a}$ **divides** $\boldsymbol{b}$ if there exists an integer $c \in \mathbb{Z}$ such that:

$$b = ac$$

- When $a$ divides $b$, we call $a$ a **factor** or **divisor** of $b$, and we call $b$ a **multiple** of $a$.
- Notation:
  - $a \mid b$ denotes that $a$ divides $b$.
  - $a \nmid b$ denotes that $a$ does not divide $b$.

**INTUITION & DEEP DIVE** **Why Divisibility Uses Multiplication Instead of Fractions**

Notice that the definition $b = ac$ (for some $c \in \mathbb{Z}$) is written entirely using integer multiplication rather than division ($\frac{b}{a}$). When working with integers, we want to stay inside $\mathbb{Z}$ without introducing fractions or floating-point numbers. Asking whether $a \mid b$ is simply asking whether the equation $ax = b$ has an integer solution $x \in \mathbb{Z}$.

**EXAMPLE & WALKTHROUGH** **Basic Examples of Divisibility**

1. **Does 5 divide 10?** Yes, $5 \mid 10$ because there is an integer $c = 2 \in \mathbb{Z}$ such that $10 = 5 \cdot 2$.
2. **Does 3 divide 6?** Yes, $3 \mid 6$ because there is an integer $c = 2 \in \mathbb{Z}$ such that $6 = 3 \cdot 2$.
3. **Does 2 divide 9?** No, $2 \nmid 9$ because no integer $c \in \mathbb{Z}$ satisfies $9 = 2c$ (since $2 \cdot 4 = 8 < 9 < 10 = 2 \cdot 5$).

### 4.1.1 Algebraic Properties of Divisibility

Divisibility obeys three foundational rules that let us combine, scale, and chain divisibility relationships without computing explicit quotients.

**CORE CONCEPT** **Theorem: Fundamental Properties of Divisibility**

Let $a$, $b$, and $c$ be integers, where $a \neq 0$. Then:

1. **Additive Closure**: If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
2. **Scaling by Any Integer**: If $a \mid b$, then $a \mid bc$ for all integers $c \in \mathbb{Z}$.
3. **Transitivity**: If $a \mid b$ and $b \mid c$ (with $b \neq 0$), then $a \mid c$.

**Proof:**

- **(i)** Suppose $a \mid b$ and $a \mid c$. By definition, there exist integers $s, t \in \mathbb{Z}$ such that $b = as$ and $c = at$. Adding these two equations gives:

$$b + c = as + at = a(s + t)$$

Since $s + t \in \mathbb{Z}$, it follows that $a \mid (b + c)$.

- **(ii)** Suppose $a \mid b$. Then there exists $k \in \mathbb{Z}$ such that $b = ak$. Multiplying both sides by any integer $c \in \mathbb{Z}$ gives:

$$bc = (ak)c = a(kc)$$

Since $kc \in \mathbb{Z}$, we conclude that $a \mid bc$.

- **(iii)** Suppose $a \mid b$ and $b \mid c$. Then there exist integers $s, t \in \mathbb{Z}$ such that $b = as$ and $c = bt$. Substituting $b = as$ into the second equation gives:

$$c = (as)t = a(st)$$

Since $st \in \mathbb{Z}$, we conclude that $a \mid c$. □

Combining parts (i) and (ii) immediately gives a useful **Linear Combination Property**: if $a \mid b$ and $a \mid c$, then $a \mid (mb + nc)$ for any integers $m, n \in \mathbb{Z}$.

**EXAMPLE & WALKTHROUGH** **Applying the Divisibility Properties**

- **Property (i)**: Since $2 \mid 6$ and $2 \mid 4$, we immediately know that $2 \mid (6 + 4)$, or $2 \mid 10$.
- **Property (ii)**: Since $5 \mid 15$, multiplying 15 by $c = 4$ guarantees that $5 \mid (15 \cdot 4)$, or $5 \mid 60$.
- **Property (iii)**: Since $7 \mid 21$ and $21 \mid 42$, transitivity guarantees that $7 \mid 42$.

**EXAMPLE & WALKTHROUGH** **Practice Problems: Testing and Applying Divisibility**

**Part A — Determine whether $a \mid b$ for the following pairs:**

1. $a = 3, b = 10$: We check whether there is an integer $c \in \mathbb{Z}$ such that $10 = 3c$. Since $3 \cdot 3 = 9$ and $3 \cdot 4 = 12$, no such integer exists ($\frac{10}{3} = 3.33\ldots \notin \mathbb{Z}$). Therefore, $\mathbf{3 \nmid 10}$.
2. $a = 19, b = 95$: We check whether 95 is an integer multiple of 19. Since $95 = 19 \cdot 5$ and $5 \in \mathbb{Z}$, we conclude that $\mathbf{19 \mid 95}$.

---

**Part B — Verify the following using the divisibility properties:**

1. **If $8 \mid 32$ and $8 \mid 64$, does $8 \mid 96$? Yes.** By Property (i) (with $a = 8$, $b = 32$, and $c = 64$), since $8 \mid 32$ and $8 \mid 64$, we must have $8 \mid (32 + 64)$. Since $32 + 64 = 96$, it follows that $\mathbf{8 \mid 96}$ (indeed, $96 = 8 \cdot 12$).
2. **If $16 \mid 80$, does $16 \mid 240$? Yes.** Notice that $240 = 80 \cdot 3$. By Property (ii) (with $a = 16$, $b = 80$, and multiplier $c = 3 \in \mathbb{Z}$), since $16 \mid 80$, we have $16 \mid (80 \cdot 3)$, which means $\mathbf{16 \mid 240}$ (indeed, $240 = 16 \cdot 15$).
3. **If $8 \mid 24$ and $24 \mid 72$, does $8 \mid 72$? Yes.** By Property (iii) (transitivity, with $a = 8$, $b = 24$, and $c = 72$), since $8 \mid 24$ and $24 \mid 72$, it follows directly that $\mathbf{8 \mid 72}$ (indeed, $72 = 8 \cdot 9$).

**PITFALL / WARNING** **The Converse of Properties (i) and (ii) Does Not Hold**

Be careful not to reverse the implications in the divisibility theorem:

- **Sum trap**: $a \mid (b + c)$ does **not** imply that $a \mid b$ and $a \mid c$. For instance, $5 \mid (2 + 3)$, yet $5 \nmid 2$ and $5 \nmid 3$.
- **Product trap**: $a \mid bc$ does **not** generally imply $a \mid b$ or $a \mid c$ when $a$ is composite. For example, $6 \mid (3 \cdot 4)$ because $6 \mid 12$, yet $6 \nmid 3$ and $6 \nmid 4$. (This implication only holds when $a$ is a prime number, a result known as Euclid's Lemma.)

## 4.2 The Division Algorithm & Modulo Concept

When a positive integer $d > 0$ does not divide $a$ evenly, integer division yields both an integer quotient and a strictly smaller non-negative remainder.

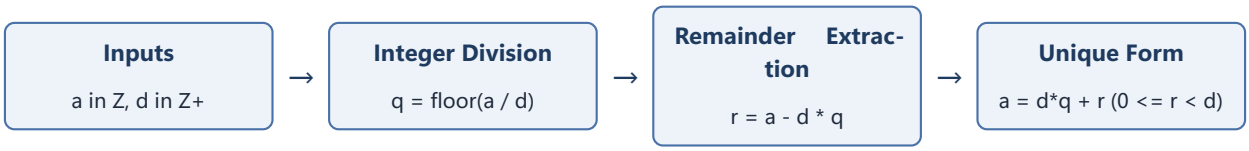
**CORE CONCEPT** **Theorem: The Division Algorithm**

Let $a$ be an integer and $d$ be a positive integer ($d \in \mathbb{Z}^+$). Then there exist **unique** integers $q$ and $r$, with $0 \leq r < d$, such that:

$$a = dq + r$$

The four parts of this equation have standard names:

- $a$: **dividend**
- $d$: **divisor** (strictly positive: $d > 0$)
- $q$: **quotient**, denoted $q = a \div d = \left\lfloor \frac{a}{d} \right\rfloor$
- $r$: **remainder**, denoted $r = a \bmod d$, satisfying $0 \leq r < d$




*Decomposition of an integer dividend a by positive divisor d via the Division Algorithm.*

**EXAMPLE & WALKTHROUGH** **Division Algorithm Decompositions**

1. $10 = 4 \cdot 2 + 2$: Dividing $a = 10$ by $d = 4$ gives quotient $q = 2$ and remainder $r = 2$ (with $0 \leq 2 < 4$).
2. $90 = 7 \cdot 12 + 6$: Dividing $a = 90$ by $d = 7$ gives quotient $q = 12$ and remainder $r = 6$ (with $0 \leq 6 < 7$).
3. $25 = 6 \cdot 4 + 1$: Dividing $a = 25$ by $d = 6$ gives quotient $q = 4$ and remainder $r = 1$ (with $0 \leq 1 < 6$).

**EXAMPLE & WALKTHROUGH** **Practice Problems: Expressing Numbers as a = d * q + r**

**Problem:** Write each pair in the form $a = d \cdot q + r$ with $0 \leq r < d$:

1. $\boldsymbol{a = 91, d = 36}$: Dividing 91 by 36, the largest multiple of 36 not exceeding 91 is $36 \cdot 2 = 72$. Subtracting gives $r = 91 - 72 = 19$ (and $0 \leq 19 < 36$). Thus:
$$91 = 36 \cdot 2 + 19 \quad (q = 2, r = 19)$$
2. $\boldsymbol{a = 65, d = 12}$: Dividing 65 by 12, the largest multiple of 12 not exceeding 65 is $12 \cdot 5 = 60$. Subtracting gives $r = 65 - 60 = 5$ (and $0 \leq 5 < 12$). Thus:
$$65 = 12 \cdot 5 + 5 \quad (q = 5, r = 5)$$
3. $\boldsymbol{a = 121, d = 10}$: Dividing 121 by 10, the largest multiple of 10 not exceeding 121 is $10 \cdot 12 = 120$. Subtracting gives $r = 121 - 120 = 1$ (and $0 \leq 1 < 10$). Thus:
$$121 = 10 \cdot 12 + 1 \quad (q = 12, r = 1)$$

**PITFALL / WARNING** **Negative Dividends in the Division Algorithm**

In the Division Algorithm, the remainder $r$ must **always** be non-negative: $0 \leq r < d$. For example, if $a = -11$ and $d = 3$, it is **incorrect** to write $-11 = 3(-3) - 2$ because $r = -2 < 0$ violates $0 \leq r < d$. Instead, we round the quotient **down** to $q = \left\lfloor -\frac{11}{3} \right\rfloor = -4$:

$$-11 = 3(-4) + 1 \implies q = -4, r = -11 \bmod 3 = 1$$

Note that in C, C++, and Java, `-11 % 3` evaluates to `-2` (truncated division toward zero), whereas Python's `-11 % 3` evaluates to `1` (true mathematical modulo). In discrete mathematics, $r$ is always in the range $0 \leq r < d$.

### 4.2.1 The Modulo Operator and Congruence Relation

From the Division Algorithm $a = dq + r$ $(0 \leq r < d)$, the remainder $r$ can be written using modulo notation:

$$r = a \bmod d \quad \text{or as a congruence} \quad r \equiv a (\bmod d)$$

**EXAMPLE & WALKTHROUGH** **Modulo Remainder Examples**

- $2 \equiv 5 \bmod 3$ because $5 = 3 \cdot 1 + 2$, so the remainder of 5 divided by 3 is $r = 2$.
- $1 \equiv 10 \bmod 3$ because $10 = 3 \cdot 3 + 1$, so the remainder of 10 divided by 3 is $r = 1$.
- $15 \equiv 35 \bmod 20$ because $35 = 20 \cdot 1 + 15$, so the remainder of 35 divided by 20 is $r = 15$.

## 4.3 Modular Arithmetic

Often in computer science and mathematics, we care only about the remainder left when an integer is divided by a fixed positive integer $m$, called the **modulus**. For example, a 12-hour or 24-hour clock wraps around modulo 12 or 24, and 32-bit unsigned integer registers wrap around modulo $2^{32}$.

**DEFINITION** **Congruence Modulo m**

Let $a$ and $b$ be integers, and let $m$ be a positive integer ($m \in \mathbb{Z}^+$). We say that $a$ **is congruent to** $b$ **modulo** $m$, written:

$$a \equiv b (\bmod m)$$

if and only if $m$ divides their difference, i.e., $m \mid (a - b)$ (or equivalently, $a = b + km$ for some $k \in \mathbb{Z}$). If $m \nmid (a - b)$, we write $a \not\equiv b (\bmod m)$.

**CORE CONCEPT** **Theorem: Congruence and Equal Remainders**

Let $a$ and $b$ be integers, and let $m$ be a positive integer. Then:

$$a \equiv b (\bmod m) \iff a \bmod m = b \bmod m$$

That is, two integers are congruent modulo $m$ if and only if they leave the **exact same remainder** when divided by $m$.

**EXAMPLE & WALKTHROUGH** **Congruence Examples**

- $5 \equiv 15 (\bmod 10)$ because $5 \bmod 10 = 5$ and $15 \bmod 10 = 5$ (and $10 \mid (5 - 15) = -10$).
- $6 \equiv 21 (\bmod 15)$ because $6 \bmod 15 = 6$ and $21 \bmod 15 = 6$ (and $15 \mid (6 - 21) = -15$).

**INTUITION & DEEP DIVE** **Operator vs. Relation: a mod m vs. a ≡ b (mod m)**

It helps to distinguish between the binary **operator** $a \bmod m$ and the equivalence **relation** $a \equiv b (\bmod m)$:

- $a \bmod m$ **is a function** from $\mathbb{Z} \times \mathbb{Z}^+$ to $\{0, 1, \ldots, m - 1\}$ that returns a **single remainder** in $[0, m - 1]$. For example, $15 \bmod 10 = 5$.

- $a \equiv b \pmod{m}$ **is a relation** on $\mathbb{Z}$ stating that $a$ and $b$ belong to the same remainder class. Thus $5 \equiv 15 \equiv 25 \equiv -5 \pmod{10}$ are all true at the same time.

### 4.3.1 Arithmetic Operations Modulo $m$

A very useful property of congruence modulo $m$ is that it is compatible with addition, subtraction, and multiplication. This allows us to reduce intermediate results modulo $m$ **before** adding or multiplying them, keeping numbers small and avoiding overflow.

**CORE CONCEPT** **Theorem & Corollary: Modular Addition and Multiplication**

**Theorem:** Let $m$ be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

$$a + c \equiv b + d \pmod{m} \quad \text{and} \quad ac \equiv bd \pmod{m}$$

**Proof:** Since $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, there exist integers $s, t \in \mathbb{Z}$ such that $a = b + sm$ and $c = d + tm$. Then:

- $a + c = (b + d) + (s + t)m \implies a + c \equiv b + d \pmod{m}$.
- $ac = (b + sm)(d + tm) = bd + m(bt + sd + stm) \implies ac \equiv bd \pmod{m}$. $\square$

---

**Corollary:** Let $m$ be a positive integer and let $a$ and $b$ be integers. Then:

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m$$

**EXAMPLE & WALKTHROUGH** **Applying Modular Arithmetic**

1. **Combining Congruences**: Given $2 \equiv 5 \pmod{3}$ and $1 \equiv 4 \pmod{3}$:
   - Adding both sides gives $2 + 1 \equiv 5 + 4 \pmod{3}$, i.e., $\mathbf{3 \equiv 9 \pmod{3}}$ (both leave remainder 0).
   - Multiplying both sides gives $2 \cdot 1 \equiv 5 \cdot 4 \pmod{3}$, i.e., $\mathbf{2 \equiv 20 \pmod{3}}$ (both leave remainder 2, since $20 = 3 \cdot 6 + 2$).
2. **Modular Sum Reduction**: Compute $123 \bmod 11$ by splitting $123 = 100 + 23$:

$$\begin{aligned} 123 \bmod 11 &= ((100 \bmod 11) + (23 \bmod 11)) \bmod 11 \\ &= (1 + 1) \bmod 11 = 2 \end{aligned}$$

   (Check: $123 = 11 \cdot 11 + 2$.)
3. **Modular Product Reduction**: Compute $450 \bmod 21$ by factoring $450 = 90 \cdot 5$:

$$\begin{aligned} 450 \bmod 21 &= ((90 \bmod 21)(5 \bmod 21)) \bmod 21 \\ &= (6 \cdot 5) \bmod 21 = 30 \bmod 21 = 9 \end{aligned}$$

   (Check: $450 = 21 \cdot 21 + 9$.)

**PITFALL / WARNING** **Modular Division Cannot Be Done Directly**

While addition, subtraction, and multiplication work cleanly modulo $m$, **dividing (canceling a common factor) does not generally work**:

$$ac \equiv bc \pmod{m} \quad \not\Rightarrow \quad a \equiv b \pmod{m}$$

For example, $2 \cdot 3 \equiv 4 \cdot 3 \pmod{6}$ (since $6 \equiv 12 \pmod{6}$), but dividing both sides by 3 gives $2 \equiv 4 \pmod{6}$, which is **false**. You can only cancel a common factor $c$ modulo $m$ when $\gcd(c, m) = 1$.

## 4.4 Prime Numbers & Primality Testing

Every integer greater than 1 is classified as either prime or composite based on its positive divisors.

**DEFINITION** **Prime and Composite Numbers**

- An integer $p > 1$ is called **prime** if and only if its only positive factors are 1 and $p$ itself:

$$p \text{ is prime} \iff (p > 1) \wedge \forall a \in \mathbb{Z}^+ (a \mid p \implies (a = 1 \vee a = p))$$

- An integer $n > 1$ that is not prime is called **composite**. Equivalently, $n > 1$ is composite if and only if there exist integers $a, b$ such that $n = ab$ with $1 < a < n$ and $1 < b < n$.
- **Note**: The integer 1 is **neither prime nor composite**. The first several primes are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, \ldots$

### 4.4.1 The Fundamental Theorem of Arithmetic

Prime numbers serve as the multiplicative building blocks for all positive integers.

**CORE CONCEPT** **Theorem: The Fundamental Theorem of Arithmetic**

Every integer greater than 1 can be written **uniquely** as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size:

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \quad \text{with } p_1 < p_2 < \ldots < p_k \text{ primes and } a_i \in \mathbb{Z}^+$$

**EXAMPLE & WALKTHROUGH** **Prime Factorizations of Integers**

Find the prime factorizations of 100, 641, 999, and 1024:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$ (a prime number)
- $999 = 27 \cdot 37 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

### 4.4.2 Primality Testing via the $\sqrt{n}$ Bound

To check whether an integer $n$ (such as 641 above) is prime, a naive approach would test all integers from 2 up to $n - 1$. However, the following theorem cuts the work from $O(n)$ down to $O(\sqrt{n})$: we only need to test **prime** divisors up to $\lfloor \sqrt{n} \rfloor$.

**CORE CONCEPT** **Theorem: Prime Divisor Bound for Composite Numbers**

**Theorem:** If $n$ is a composite integer, then $n$ has a prime divisor less than or equal to $\sqrt{n}$.

**Proof:** Since $n$ is composite, we can write $n = ab$ where $1 < a < n$ and $1 < b < n$. Without loss of generality, assume $a \leq b$. Multiplying both sides of $a \leq b$ by $a$ gives:

$$a^2 \leq ab = n \quad \implies \quad a \leq \sqrt{n}$$

Since $a > 1$, by the Fundamental Theorem of Arithmetic, $a$ is either prime itself or has a prime factor $p \leq a$. By transitivity of divisibility ($p \mid a$ and $a \mid n$), $p \mid n$ and $p \leq a \leq \sqrt{n}$. $\square$

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**Contrapositive — Primality Test:** If an integer $n > 1$ has **no prime factor** less than or equal to $\sqrt{n}$, then $n$ is **prime**.

**EXAMPLE & WALKTHROUGH** **Primality Testing Walkthrough**

1. **Testing $n = 21$**: We have $\sqrt{21} \approx 4.6$. The primes less than or equal to 4.6 are 2 and 3. Checking divisibility: $2 \nmid 21$, but $\mathbf{3 \mid 21}$ (since $21 = 3 \cdot 7$). Thus $n = 21$ is composite and has a prime factor $3 \leq \sqrt{21}$.
2. **Testing $n = 19$**: We have $\sqrt{19} \approx 4.36$. The only primes less than or equal to 4.36 are 2 and 3.
   - Does $2 \mid 19$? No ($19 = 2 \cdot 9 + 1$).
   - Does $3 \mid 19$? No ($19 = 3 \cdot 6 + 1$).

   Because 19 has no prime factor $\leq \sqrt{19}$, **19 is prime**.
3. **Testing $n = 641$**: Since $\sqrt{641} \approx 25.31$, we test all primes $p \leq 25$: $p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23\}$. None of these nine primes divide 641, confirming that **641 is prime**.

## 4.5 Greatest Common Divisor & The Euclidean Algorithm

**DEFINITION** **Greatest Common Divisor (GCD)**

Let $a$ and $b$ be integers, not both zero. The largest integer $d$ such that $d \mid a$ and $d \mid b$ is called the **greatest common divisor** of $a$ and $b$, denoted $\gcd(a, b)$.

- **Relatively Prime (Coprime) Integers**: Two integers $a$ and $b$ are called **relatively prime** if their greatest common divisor is 1, i.e., $\gcd(a, b) = 1$.

**EXAMPLE & WALKTHROUGH** **GCD and Relatively Prime Examples**

- $\mathbf{\gcd(6, 9) = 3}$: The positive divisors of 6 are $\{1, 2, 3, 6\}$ and of 9 are $\{1, 3, 9\}$. Their common divisors are $\{1, 3\}$, and the largest is 3.
- $\mathbf{\gcd(24, 36) = 12}$: The common positive divisors of 24 and 36 are $\{1, 2, 3, 4, 6, 12\}$, so the greatest is 12.
- **5 and 7 are relatively prime**: $\gcd(5, 7) = 1$.
- **12 and 25 are relatively prime**: Even though both $12 = 2^2 \cdot 3$ and $25 = 5^2$ are composite, they share no prime factors, so $\gcd(12, 25) = 1$.

**INTUITION & DEEP DIVE** **Finding GCD and LCM from Prime Factorizations**

When the prime factorizations of $a$ and $b$ are already known:

$$a = p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}, \quad b = p_1^{b_1} p_2^{b_2} \cdots p_n^{b_n} \quad (a_i, b_i \geq 0)$$

we can find both the Greatest Common Divisor (gcd) and the Least Common Multiple (lcm) by taking the minimum and maximum exponents of each prime:

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_n^{\min(a_n, b_n)}, \quad \operatorname{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_n^{\max(a_n, b_n)}$$

Since $\min(x, y) + \max(x, y) = x + y$, these satisfy the useful identity $\boldsymbol{ab = \gcd(a, b) \cdot \operatorname{lcm}(a, b)}$. However, factoring large integers is computationally expensive. Instead, the **Euclidean Algorithm** computes $\gcd(a, b)$ in $O(\log \min(a, b))$ division steps without factoring either number.

### 4.5.1 The Euclidean Algorithm

The Euclidean Algorithm is based on a simple observation: if $a = bq + r$, then any common divisor of $a$ and $b$ also divides $r = a - bq$, and vice versa. Therefore:

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

**Algorithm: The Euclidean Algorithm** `O(log b) division steps`

**Input:** Positive integers $a$ and $b$ with $a > b > 0$.
**Output:** The greatest common divisor $\gcd(a, b)$.

1. Write $a$ and $b$ in the Division Algorithm form:
$$a = b \cdot q + r \quad \text{where } 0 \leq r < b$$
2. If $r = 0$, terminate and output $\gcd = b$.
3. Otherwise, replace $a \leftarrow b$ and $b \leftarrow r$, and repeat from Step 1 until $r$ becomes 0.
4. The $\gcd(a, b)$ is the value of $b$ when $r = 0$ (equivalently, the **last nonzero remainder**).

**EXAMPLE & WALKTHROUGH** **Euclidean Algorithm Trace: gcd(414, 662)**

**Problem:** Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

**Solution:** Setting $a = 662$ and $b = 414$, we repeatedly apply the Division Algorithm:

| Step | Division Equation ($a = b \cdot q + r$) | $a$ | $b$ | Remainder $r$ & Next Step |
|---|---|---|---|---|
| 1 | $662 = 414 \cdot 1 + 248$ | 662 | 414 | $r = 248 \neq 0 \Longrightarrow (a, b) \leftarrow (414, 248)$ |
| 2 | $414 = 248 \cdot 1 + 166$ | 414 | 248 | $r = 166 \neq 0 \Longrightarrow (a, b) \leftarrow (248, 166)$ |
| 3 | $248 = 166 \cdot 1 + 82$ | 248 | 166 | $r = 82 \neq 0 \Longrightarrow (a, b) \leftarrow (166, 82)$ |
| 4 | $166 = 82 \cdot 2 + 2$ | 166 | 82 | $r = 2 \neq 0 \Longrightarrow (a, b) \leftarrow (82, 2)$ |
| 5 | $82 = 2 \cdot 41 + 0$ | 82 | 2 | $r = 0 \Longrightarrow$ Stop! Output $b = 2$ |

Hence, $\mathbf{gcd(414, 662) = 2}$, because 2 is the last nonzero remainder.

## 4.6 Worked Practice Exercises

Below are complete worked solutions to the practice problems from Kenneth H. Rosen's **Discrete Mathematics and Its Applications** (Section 4.1 on Divisibility and Modular Arithmetic, and Section 4.3 on Primes and Greatest Common Divisors).

**EXAMPLE & WALKTHROUGH** **Divisibility and Modular Arithmetic (Section 4.1, Nos. 1, 9, 10, 11, 12, 20, 30)**

**1. Does 17 divide each of these numbers?**

- **a) 68**: Yes, $17 \mid 68$ because $68 = 17 \cdot 4$.
- **b) 84**: No, $17 \nmid 84$ because $84 = 17 \cdot 4 + 16$ ($r = 16 \neq 0$).
- **c) 357**: Yes, $17 \mid 357$ because $357 = 17 \cdot 21$.
- **d) 1001**: No, $17 \nmid 1001$ because $1001 = 17 \cdot 58 + 15$ ($r = 15 \neq 0$).

**9–12. Find the quotient $q$ and remainder $r$ when:**

| No. | Problem ($a$ divided by $d$) | Division Algorithm ($a = dq + r, 0 \leq r < d$) | Quotient $q$ | Remainder $r$ |
|---|---|---|---|---|
| 9a | 19 divided by 7 | $19 = 7 \cdot 2 + 5$ | 2 | 5 |
| 9b | $-111$ divided by 11 | $-111 = 11 \cdot (-11) + 10$ | $-11$ | 10 |

| | | | | |
|---|---|---|---|---|
| 9c | 789 divided by 23 | $789 = 23 \cdot 34 + 7$ | 34 | 7 |
| 9d | 1001 divided by 13 | $1001 = 13 \cdot 77 + 0$ | 77 | 0 |
| 9e | 0 divided by 19 | $0 = 19 \cdot 0 + 0$ | 0 | 0 |
| 9f | 3 divided by 5 | $3 = 5 \cdot 0 + 3$ | 0 | 3 |
| 9g | $-1$ divided by 3 | $-1 = 3 \cdot (-1) + 2$ | $-1$ | 2 |
| 9h | 4 divided by 1 | $4 = 1 \cdot 4 + 0$ | 4 | 0 |
| 10a | 44 divided by 8 | $44 = 8 \cdot 5 + 4$ | 5 | 4 |
| 10b | 777 divided by 21 | $777 = 21 \cdot 37 + 0$ | 37 | 0 |
| 10c | $-123$ divided by 19 | $-123 = 19 \cdot (-7) + 10$ | $-7$ | 10 |
| 10d | $-1$ divided by 23 | $-1 = 23 \cdot (-1) + 22$ | $-1$ | 22 |
| 10e | $-2002$ divided by 87 | $-2002 = 87 \cdot (-24) + 86$ | $-24$ | 86 |
| 10f | 0 divided by 17 | $0 = 17 \cdot 0 + 0$ | 0 | 0 |
| 10g | $1,234,567$ divided by 1001 | $1,234,567 = 1001 \cdot 1233 + 334$ | 1233 | 334 |
| 10h | $-100$ divided by 101 | $-100 = 101 \cdot (-1) + 1$ | $-1$ | 1 |

- **11. Find $-101 \bmod 13$ and $199 \bmod 19$**: Since $-101 = 13 \cdot (-8) + 3$, we have $-101 \bmod 13 = 3$. Since $199 = 19 \cdot 10 + 9$, we have $199 \bmod 19 = 9$.
- **12. Find $-88 \bmod 13$ and $288 \bmod 17$**: Since $-88 = 13 \cdot (-7) + 3$, we have $-88 \bmod 13 = 3$. Since $288 = 17 \cdot 16 + 16$, we have $288 \bmod 17 = 16$.

---

**20. Determine whether each of these integers is congruent to 5 modulo 17:**

- **a) 80**: $80 - 5 = 75$, and $17 \nmid 75$ ($75 = 17 \cdot 4 + 7$). Thus $80 \not\equiv 5 \pmod{17}$ (in fact, $80 \bmod 17 = 12$).
- **b) 103**: $103 - 5 = 98$, and $17 \nmid 98$ ($98 = 17 \cdot 5 + 13$). Thus $103 \not\equiv 5 \pmod{17}$ (in fact, $103 \bmod 17 = 1$).
- **c) $-29$**: $-29 - 5 = -34 = 17 \cdot (-2)$, so $17 \mid (-29 - 5)$. Thus $\mathbf{-29 \equiv 5 \pmod{17}}$.
- **d) $-122$**: $-122 - 5 = -127$, and $17 \nmid -127$ ($-127 = 17(-8) + 9$). Thus $-122 \not\equiv 5 \pmod{17}$.

---

**30. Evaluate each of these quantities using modular arithmetic properties:**

- **a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$**: This simplifies to $(-133 + 261) \bmod 23 = 128 \bmod 23$. Since $128 = 23 \cdot 5 + 13$, the result is $\mathbf{13}$.
- **b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$**: First reduce each factor: $457 = 23 \cdot 19 + 20 \implies 457 \bmod 23 = 20 \equiv -3 \pmod{23}$, and $182 = 23 \cdot 7 + 21 \implies 182 \bmod 23 = 21 \equiv -2 \pmod{23}$. Multiplying gives $(20 \cdot 21) \bmod 23 = ((-3)(-2)) \bmod 23 = 6 \bmod 23 = \mathbf{6}$.
- **c) $(99^2 \bmod 32)^3 \bmod 15$**: First compute $99 \bmod 32 = 3$ (since $99 = 32 \cdot 3 + 3$). Thus $99^2 \bmod 32 = 3^2 \bmod 32 = 9$. Next, evaluate $9^3 \bmod 15$: since $9^2 = 81 = 15 \cdot 5 + 6 \implies 9^2 \bmod 15 = 6$, we have $9^3 \bmod 15 = (6 \cdot 9) \bmod 15 = 54 \bmod 15 = \mathbf{9}$.
- **d) $((34 \cdot 102) \bmod 27)^2 \bmod 13$**: Reduce modulo 27: $34 \bmod 27 = 7$ and $102 = 27 \cdot 3 + 21 \implies 102 \bmod 27 = 21 \equiv -6 \pmod{27}$. Thus $(34 \cdot 102) \bmod 27 = (7 \cdot (-6)) \bmod 27 = -42 \bmod 27 = 12$. Finally, $12^2 \bmod 13 = (-1)^2 \bmod 13 = \mathbf{1}$.

**EXAMPLE & WALKTHROUGH** **Primes and Greatest Common Divisors (Section 4.3, Nos. 1, 14, 15, 32)**

**1. Determine whether each of these integers is prime:**

- **a) 21**: Composite, since $3 \mid 21$ ($21 = 3 \cdot 7$).
- **b) 29**: We test primes $p \leq \sqrt{29} \approx 5.38$, i.e., $p \in \{2, 3, 5\}$. Since $2 \nmid 29$, $3 \nmid 29$, and $5 \nmid 29$, **29 is prime**.
- **c) 71**: We test primes $p \leq \sqrt{71} \approx 8.42$, i.e., $p \in \{2, 3, 5, 7\}$. None divide 71, so **71 is prime**.
- **d) 97**: We test primes $p \leq \sqrt{97} \approx 9.85$, i.e., $p \in \{2, 3, 5, 7\}$. None divide 97, so **97 is prime**.
- **e) 111**: Composite. The sum of its digits is $1 + 1 + 1 = 3$, so $3 \mid 111$ ($111 = 3 \cdot 37$).
- **f) 143**: We test primes $p \leq \sqrt{143} \approx 11.95$, i.e., $p \in \{2, 3, 5, 7, 11\}$. Checking 11: $143 = 11 \cdot 13$, so **143 is composite**.

---

**14. Which positive integers less than 12 are relatively prime to 12?** Since $12 = 2^2 \cdot 3$, an integer $k$ satisfies $\gcd(k, 12) = 1$ if and only if $2 \nmid k$ and $3 \nmid k$. Checking $k \in \{1, 2, \ldots, 11\}$:

- Multiples of 2: $2, 4, 6, 8, 10$ (excluded)
- Multiples of 3: $3, 6, 9$ (excluded)

The remaining integers relatively prime to 12 are $\mathbf{1, 5, 7, 11}$.

---

**15. Determine whether the following sets of integers are pairwise relatively prime:**

- **a) 21, 34, 55**: Prime factorizations are $21 = 3 \cdot 7$, $34 = 2 \cdot 17$, and $55 = 5 \cdot 11$. Since no two numbers share a prime factor, $\gcd(21, 34) = \gcd(21, 55) = \gcd(34, 55) = 1$. **Yes, pairwise relatively prime.**
- **b) 14, 17, 85**: Although $\gcd(14, 17) = 1$ and $\gcd(14, 85) = 1$, we have $\gcd(17, 85) = 17 \neq 1$ (since $85 = 5 \cdot 17$). **No, not pairwise relatively prime.**
- **c) 25, 41, 49, 64**: Prime factorizations are $25 = 5^2$, $41 = 41$, $49 = 7^2$, and $64 = 2^6$. All prime bases $(5, 41, 7, 2)$ are distinct. **Yes, pairwise relatively prime.**
- **d) 17, 18, 19, 23**: Since $17, 19, 23$ are distinct primes that do not divide $18 = 2 \cdot 3^2$, every pair has $\gcd = 1$. **Yes, pairwise relatively prime.**

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**32. Use the Euclidean algorithm to find the greatest common divisor of each pair:**

- **a) $\gcd(1, 5)$**: $5 = 1 \cdot 5 + 0 \Longrightarrow \mathbf{\gcd(1, 5) = 1}$.
- **b) $\gcd(100, 101)$**: $101 = 100 \cdot 1 + 1$, then $100 = 1 \cdot 100 + 0 \Longrightarrow \mathbf{\gcd(100, 101) = 1}$.
- **c) $\gcd(123, 277)$**:
  - $277 = 123 \cdot 2 + 31$
  - $123 = 31 \cdot 3 + 30$
  - $31 = 30 \cdot 1 + 1$
  - $30 = 1 \cdot 30 + 0$

  Hence $\mathbf{\gcd(123, 277) = 1}$.
- **d) $\gcd(1529, 14039)$**:
  - $14039 = 1529 \cdot 9 + 278$
  - $1529 = 278 \cdot 5 + 139$
  - $278 = 139 \cdot 2 + 0$

  Hence $\mathbf{\gcd(1529, 14039) = 139}$.
- **e) $\gcd(1529, 14038)$**:
  - $14038 = 1529 \cdot 9 + 277$
  - $1529 = 277 \cdot 5 + 144$

- $277 = 144 \cdot 1 + 133$
- $144 = 133 \cdot 1 + 11$
- $133 = 11 \cdot 12 + 1$
- $11 = 1 \cdot 11 + 0$

Hence $\mathbf{gcd(1529, 14038) = 1}$.

- **f)** $\mathbf{gcd(11111, 111111)}$:
  - $111111 = 11111 \cdot 10 + 1$
  - $11111 = 1 \cdot 11111 + 0$

  Hence $\mathbf{gcd(11111, 111111) = 1}$.

# 5. Important Functions in Computer Science

Ceiling and Floor Functions, Step Graphs, Recursive Functions, and Sequence Recurrences

## 5.1 Ceiling and Floor Functions

In computer science, we frequently bridge the gap between continuous real-valued quantities ($x \in \mathbb{R}$) and discrete integer resources ($n \in \mathbb{Z}$). For example, storing a file of $x$ bytes into disk blocks of 4096 bytes requires an integer number of blocks, and splitting an array of $n$ elements in binary search requires an integer midpoint index. Two foundational rounding functions handle this conversion: the **ceiling function** and the **floor function**.

**DEFINITION** **Ceiling Function**

The **ceiling function** assigns to the real number $x \in \mathbb{R}$ the **smallest integer that is greater than or equal to** $x$.

- **Notation**:

$$f(x) = \lceil x \rceil$$

- **Characterization**: For any $x \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$\lceil x \rceil = n \iff n - 1 < x \leq n$$

**DEFINITION** **Floor Function**

The **floor function** assigns to the real number $x \in \mathbb{R}$ the **largest integer that is less than or equal to** $x$. (The floor function is also known as the **greatest integer function**.)

- **Notation**:

$$f(x) = \lfloor x \rfloor$$

- **Characterization**: For any $x \in \mathbb{R}$ and $n \in \mathbb{Z}$,

$$\lfloor x \rfloor = n \iff n \leq x < n + 1$$

**EXAMPLE & WALKTHROUGH** **Ceiling and Floor Function Evaluations**

- **Ceiling Function Examples**:
  - $\lceil 6.76 \rceil = 7$ because 7 is the smallest integer $\geq 6.76$ ($6 < 6.76 \leq 7$).
  - $\lceil 5.1 \rceil = 6$ because 6 is the smallest integer $\geq 5.1$ ($5 < 5.1 \leq 6$).
- **Floor Function Examples**:
  - $\lfloor 6.76 \rfloor = 6$ because 6 is the largest integer $\leq 6.76$ ($6 \leq 6.76 < 7$).
  - $\lfloor 5.1 \rfloor = 5$ because 5 is the largest integer $\leq 5.1$ ($5 \leq 5.1 < 6$).

### 5.1.1 Step-Function Graphs and Algebraic Properties

Both $y = \lfloor x \rfloor$ and $y = \lceil x \rceil$ are **piecewise-constant step functions** mapping $\mathbb{R}$ to $\mathbb{Z}$. On each unit interval between consecutive integers $n$ and $n + 1$:

- The **floor function** $y = \lfloor x \rfloor$ stays constant at height $n$ on the **left-closed, right-open** interval $[n, n + 1)$, jumping up by 1 at the right endpoint $x = n + 1$.
- The **ceiling function** $y = \lceil x \rceil$ stays constant at height $n + 1$ on the **left-open, right-closed** interval $(n, n + 1]$, jumping up by 1 immediately to the right of $x = n$.

## Comparison: Floor Function vs. Ceiling Function

### Floor Function: y = floor(x)

- **Direction**: Rounds **down** toward $-\infty$ on the number line.
- **Constant Steps**: $\lfloor x \rfloor = n$ for $x \in [n, n+1)$ (closed point on the left at $(n, n)$, open point on the right at $(n+1, n)$).
- **Bounds**: $x - 1 < \lfloor x \rfloor \leq x$.

### Ceiling Function: y = ceil(x)

- **Direction**: Rounds **up** toward $+\infty$ on the number line.
- **Constant Steps**: $\lceil x \rceil = n$ for $x \in (n-1, n]$ (open point on the left at $(n-1, n)$, closed point on the right at $(n, n)$).
- **Bounds**: $x \leq \lceil x \rceil < x + 1$.

### Behavior at Exact Integers (x in Z)

If and only if $x \in \mathbb{Z}$, both functions equal $x$:

$$\lfloor x \rfloor = \lceil x \rceil = x \iff x \in \mathbb{Z}$$

When $x \notin \mathbb{Z}$, we always have $\lceil x \rceil - \lfloor x \rfloor = 1$.

### Useful Algebraic Identities (for x in R, k in Z)

- **Integer Shift**: $\lfloor x + k \rfloor = \lfloor x \rfloor + k$ and $\lceil x + k \rceil = \lceil x \rceil + k$.
- **Negation Duality**:

$$\lfloor -x \rfloor = -\lceil x \rceil \quad \text{and} \quad \lceil -x \rceil = -\lfloor x \rfloor$$

| Interval of $x$ | Floor $\lfloor x \rfloor$ | Ceiling $\lceil x \rceil$ | Step-Graph Endpoint Behavior |
|---|---|---|---|
| $-3 \leq x < -2$ | $-3$ | $-3$ (at $x = -3$) or $-2$ | Floor closed at $(-3, -3)$, open at $(-2, -3)$ |
| $-2 \leq x < -1$ | $-2$ | $-2$ (at $x = -2$) or $-1$ | Floor closed at $(-2, -2)$, open at $(-1, -2)$ |
| $-1 \leq x < 0$ | $-1$ | $-1$ (at $x = -1$) or $0$ | Floor closed at $(-1, -1)$, open at $(0, -1)$ |
| $0 \leq x < 1$ | $0$ | $0$ (at $x = 0$) or $1$ | Floor closed at $(0, 0)$, open at $(1, 0)$ |
| $1 \leq x < 2$ | $1$ | $1$ (at $x = 1$) or $2$ | Floor closed at $(1, 1)$, open at $(2, 1)$ |
| $2 \leq x < 3$ | $2$ | $2$ (at $x = 2$) or $3$ | Floor closed at $(2, 2)$, open at $(3, 2)$ |

**PITFALL / WARNING** **Negative Numbers in Floor and Ceiling Functions**

A common mistake is confusing `floor(x)` with simply dropping the fractional part (truncation toward zero).

- For **positive** numbers, $\lfloor 2.99 \rfloor = 2$ does match dropping .99.
- For **negative** numbers, $\lfloor -0.1 \rfloor = -1$ (not 0) because $-1 \leq -0.1 < 0$, and $-1$ lies to the **left** of $-0.1$ on the number line. Conversely, $\lceil -0.1 \rceil = 0$ (not $-1$) because 0 lies to the **right** of $-0.1$.

When in doubt, picture the number line: **floor moves left**, **ceiling moves right**.

**EXAMPLE & WALKTHROUGH** **Practice Problems: Evaluating Floor and Ceiling Expressions**

**Set 1 — Find these values:**

- **a)** $\lfloor \mathbf{1.1} \rfloor$: Since $1 \leq 1.1 < 2$, we have $\lfloor \mathbf{1.1} \rfloor = \mathbf{1}$.
- **b)** $\lceil \mathbf{1.1} \rceil$: Since $1 < 1.1 \leq 2$, we have $\lceil \mathbf{1.1} \rceil = \mathbf{2}$.
- **c)** $\lfloor \mathbf{-0.1} \rfloor$: Since $-1 \leq -0.1 < 0$, rounding down (left) gives $\lfloor \mathbf{-0.1} \rfloor = \mathbf{-1}$.
- **d)** $\lceil \mathbf{-0.1} \rceil$: Since $-1 < -0.1 \leq 0$, rounding up (right) gives $\lceil \mathbf{-0.1} \rceil = \mathbf{0}$.
- **e)** $\lceil \mathbf{2.99} \rceil$: Since $2 < 2.99 \leq 3$, we have $\lceil \mathbf{2.99} \rceil = \mathbf{3}$.
- **f)** $\lceil \mathbf{-2.99} \rceil$: Since $-3 < -2.99 \leq -2$, rounding up (right) toward zero gives $\lceil \mathbf{-2.99} \rceil = \mathbf{-2}$.

- **g)** $\left\lfloor \frac{1}{2} + \left\lceil \frac{1}{2} \right\rceil \right\rfloor$: Evaluate from the inside out: first $\left\lceil \frac{1}{2} \right\rceil = \lceil 0.5 \rceil = 1$. Substituting into the outer floor gives:

$$\left\lfloor \frac{1}{2} + \left\lceil \frac{1}{2} \right\rceil \right\rfloor = \left\lfloor \frac{1}{2} + 1 \right\rfloor = \lfloor 1.5 \rfloor = \mathbf{1}$$

- **h)** $\left\lceil \left\lfloor \frac{1}{2} \right\rfloor + \left\lceil \frac{1}{2} \right\rceil + \frac{1}{2} \right\rceil$: First evaluate the inner terms: $\left\lfloor \frac{1}{2} \right\rfloor = 0$ and $\left\lceil \frac{1}{2} \right\rceil = 1$. Substituting into the outer ceiling gives:

$$\left\lceil 0 + 1 + \frac{1}{2} \right\rceil = \lceil 1.5 \rceil = \mathbf{2}$$

---

**Set 2 — Find these values:**

- **a)** $\left\lceil \frac{3}{4} \right\rceil$: Since $0 < 0.75 \leq 1$, we have $\left\lceil \frac{3}{4} \right\rceil = \mathbf{1}$.
- **b)** $\left\lfloor \frac{7}{8} \right\rfloor$: Since $0 \leq 0.875 < 1$, we have $\left\lfloor \frac{7}{8} \right\rfloor = \mathbf{0}$.
- **c)** $\left\lceil -\frac{3}{4} \right\rceil$: Since $-1 < -0.75 \leq 0$, we have $\left\lceil -\frac{3}{4} \right\rceil = \mathbf{0}$.
- **d)** $\left\lfloor -\frac{7}{8} \right\rfloor$: Since $-1 \leq -0.875 < 0$, we have $\left\lfloor -\frac{7}{8} \right\rfloor = \mathbf{-1}$.
- **e)** $\lceil \mathbf{3} \rceil$: Since $3 \in \mathbb{Z}$ is already an integer, $\lceil \mathbf{3} \rceil = \mathbf{3}$.
- **f)** $\lfloor \mathbf{-1} \rfloor$: Since $-1 \in \mathbb{Z}$ is already an integer, $\lfloor \mathbf{-1} \rfloor = \mathbf{-1}$.
- **g)** $\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor$: First evaluate the inner ceiling: $\left\lceil \frac{3}{2} \right\rceil = \lceil 1.5 \rceil = 2$. Then:

$$\left\lfloor \frac{1}{2} + \left\lceil \frac{3}{2} \right\rceil \right\rfloor = \lfloor 0.5 + 2 \rfloor = \lfloor 2.5 \rfloor = \mathbf{2}$$

- **h)** $\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor$: First evaluate the inner floor: $\left\lfloor \frac{5}{2} \right\rfloor = \lfloor 2.5 \rfloor = 2$. Multiplying by $\frac{1}{2}$ inside the outer floor gives:

$$\left\lfloor \frac{1}{2} \cdot \left\lfloor \frac{5}{2} \right\rfloor \right\rfloor = \left\lfloor \frac{1}{2} \cdot 2 \right\rfloor = \lfloor 1 \rfloor = \mathbf{1}$$

## 5.2 Recursive Functions

Sometimes it is difficult or awkward to define a sequence or function directly with a closed-form formula, whereas it is natural to describe how each value builds on earlier values. This approach is called **recursion**.

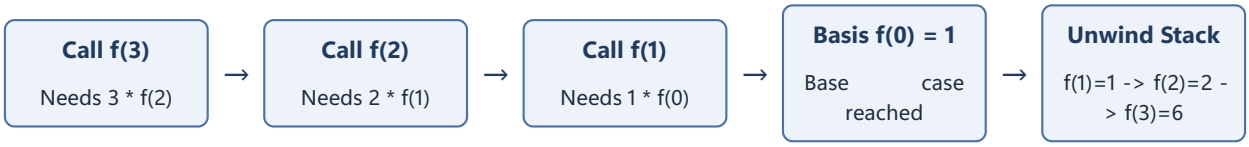
**DEFINITION** **Recursive Function**

A function $f$ (typically with domain $\mathbb{N} = \{0, 1, 2, \ldots\}$ or $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$) is called a **recursive function** if its definition refers to itself. Every well-defined recursive function consists of two essential parts:

1. **Basis Step**: Specifies the initial value(s) of the function directly, **without** referring to the function itself (for example, $f(0)$ or $f(1)$).
2. **Recursive (or Inductive) Step**: Defines the value of the function at the current input $n$ in terms of its own values at smaller inputs (such as $f(n-1)$, $f(n-2)$, ...).

**INTUITION & DEEP DIVE** **Visualizing Recursion: Self-Similarity and Call Stack Unwinding**

Recursion is like a picture-within-a-picture (the **Droste effect**), where a scene contains a smaller copy of the exact same scene, which in turn contains an even smaller copy. In programming and mathematics, however, an endless chain of self-references without a stopping condition causes infinite recursion (a stack overflow). The **Basis Step** acts as the innermost frame that holds a concrete value instead of another recursive call, allowing the evaluation to stop and unwind back to the original call.




*Evaluating f(3) for a recursive function with a base case at f(0).*

## 5.2.1 Classic Recursive Functions

### 5.2.1.1 The Factorial Function

Let $f(n) = n!$ denote the factorial function on $n \in \mathbb{N}$. Because $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1 = n \cdot (n-1)!$ for $n \geq 1$, we can write $f(n)$ recursively as:

$$f(n) = \begin{cases} 1 & \text{, for } n = 0 \quad \text{(Basis)} \\ n \cdot f(n-1) & \text{, for } n > 0 \quad \text{(Recurrence)} \end{cases}$$

**EXAMPLE & WALKTHROUGH** **Step-by-Step Evaluation of the Factorial Function**

Starting from the basis at $n = 0$ and stepping forward for $n = 0, 1, 2, 3, \ldots$:

- $n = 0 \implies f(0) = 1 = 0!$ (by the basis step)
- $n = 1 \implies f(1) = 1 \cdot f(0) = 1 \cdot 1 = 1 = 1!$
- $n = 2 \implies f(2) = 2 \cdot f(1) = 2 \cdot 1 = 2 = 2!$
- $n = 3 \implies f(3) = 3 \cdot f(2) = 3 \cdot 2 \cdot 1 = 6 = 3!$
- $n = 4 \implies f(4) = 4 \cdot f(3) = 4 \cdot 6 = 24 = 4!$, and so on.

### 5.2.1.2 Chebyshev Polynomials

A recursive definition can depend on **two** (or more) previous terms. When a recurrence depends on both $n-1$ and $n-2$ (a second-order recurrence), we must provide **two basis cases** ($n = 0$ and $n = 1$) so that the recurrence has enough starting information at $n = 2$.

The **Chebyshev polynomials of the first kind**, denoted $T(n, x)$ (used widely in numerical analysis and approximation theory because they satisfy $T(n, \cos\theta) = \cos(n\theta)$), are defined recursively by:

$$T(n, x) = \begin{cases} 1 & , \quad n = 0 \\ x & , \quad n = 1 \\ 2xT(n-1, x) - T(n-2, x) & , \quad n > 1 \end{cases}$$

**EXAMPLE & WALKTHROUGH** **Evaluating Chebyshev Polynomials**

This function has two basis cases: $n = 0$ and $n = 1$. Computing the first few polynomials:

- $n = 0 \implies T(0, x) = 1$
- $n = 1 \implies T(1, x) = x$
- $n = 2 \implies$ Substituting $n = 2$ into the recurrence:

$$\begin{aligned} T(2, x) &= 2xT(1, x) - T(0, x) \\ &= 2x(x) - 1 \\ &= 2x^2 - 1 \end{aligned}$$

- $n = 3 \implies$ Substituting $n = 3$ into the recurrence:

$$T(3, x) = 2xT(2, x) - T(1, x) = 2x(2x^2 - 1) - x = 4x^3 - 3x$$

### 5.2.1.3 The Fibonacci Sequence

The **Fibonacci function** is another classic second-order recurrence relation with two basis values ($n = 0$ and $n = 1$):

$$f(n) = \begin{cases} 0 & , \quad n = 0 \\ 1 & , \quad n = 1 \\ f(n-1) + f(n-2) & , \quad n > 1 \end{cases}$$

**EXAMPLE & WALKTHROUGH** **Evaluating Fibonacci Numbers for n = 0, 1, 2, 3, 4, 5**

**Problem:** Find the values of the Fibonacci function $f(n)$ for $n = 0, 1, 2, 3, 4, 5$.

**Solution:** Starting from the basis values $f(0) = 0$ and $f(1) = 1$, each subsequent term is the sum of the two preceding terms:

- $n = 0$: $f(0) = \mathbf{0}$ **(Basis 1)**
- $n = 1$: $f(1) = \mathbf{1}$ **(Basis 2)**
- $n = 2$: $f(2) = f(1) + f(0) = 1 + 0 = \mathbf{1}$
- $n = 3$: $f(3) = f(2) + f(1) = 1 + 1 = \mathbf{2}$
- $n = 4$: $f(4) = f(3) + f(2) = 2 + 1 = \mathbf{3}$
- $n = 5$: $f(5) = f(4) + f(3) = 3 + 2 = \mathbf{5}$

## 5.3 Expressing Closed-Form Functions Recursively

To convert an explicit formula $f(n)$ or sequence $a_n$ into a recursive definition, we follow two steps:

1. **Formulate the Recurrence**: Express the current term $f(n)$ (or $a_n$) using the previous term $f(n-1)$ (or $a_{n-1}$), either by expanding and grouping repeated operations or by algebraic substitution.
2. **Specify the Basis**: Evaluate the formula at the starting index of the domain (typically $n = 0$ or $n = 1$).

### 5.3.1 Recursive Definition of Powers $f(n) = a^n$

Suppose we want to express $f(n) = a^n$ (for $a \neq 0$ and $n \in \mathbb{N}$) recursively:

1. **Express $f(n)$ in terms of $f(n-1)$**:

$$f(n) = a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ factors}} = a \cdot \underbrace{(a \cdot a \cdots a)}_{n-1 \text{ factors}} = a \cdot a^{n-1} = a \cdot f(n-1)$$

2. **Determine the basis**: At $n = 0$, we have $f(0) = a^0 = 1$.
3. **Complete Recursive Definition**:

$$f(n) = \begin{cases} 1 & , \ n = 0 \\ a \cdot f(n-1) & , \ n > 0 \end{cases}$$

### 5.3.2 Recursive Definition of $a_n = 6n$ via Two Methods

Suppose we want to express the linear sequence $a_n = 6n$, for $n = 1, 2, 3, \ldots$, recursively. We can derive the recurrence in two ways.

**EXAMPLE & WALKTHROUGH** **Deriving the Recurrence for a_n = 6n**

**Method 1 — Expansion and Grouping:**

- **Recurrence**: Write multiplication by $n$ as repeated addition of 6 ($n$ times), then group the last $n-1$ terms together:

$$\begin{aligned} a_n = 6n &= \underbrace{6 + 6 + \ldots + 6}_{n \text{ times}} \\ &= 6 + \underbrace{(6 + \ldots + 6)}_{n-1 \text{ times}} \\ &= 6 + 6(n-1) \\ &= 6 + a_{n-1} \end{aligned}$$

- **Basis**: Because $n$ starts at $n = 1$, the basis is:

$$a_1 = 6(1) = 6$$

---

**Method 2 — Algebraic Substitution from the Closed Form:**

- Write down the closed-form formula at index $n$ and at index $n-1$:

$$a_n = 6n \quad \text{and} \quad a_{n-1} = 6(n-1) = 6n - 6$$

- Substitute $a_{n-1} = 6n - 6$ into the expression for $a_n$:

$$a_n = 6n = (6n - 6) + 6 = a_{n-1} + 6$$

- **Complete Recursive Definition**:

$$a_n = \begin{cases} 6 & , \ n = 1 \\ 6 + a_{n-1} & , \ n > 1 \end{cases}$$

**INTUITION & DEEP DIVE** **Using Finite Differences to Formulate Any Recurrence**

Method 2 generalizes to **any** sequence $a_n$:

- **Additive (Difference) Form**: Compute the difference between consecutive terms, $\Delta a_n = a_n - a_{n-1}$. Once simplified, rewrite it as:

$$a_n = a_{n-1} + (a_n - a_{n-1})$$

  For example, if $a_n = n^2$, then $a_n - a_{n-1} = n^2 - (n-1)^2 = 2n - 1$, which immediately gives $a_n = a_{n-1} + 2n - 1$.
- **Multiplicative (Ratio) Form**: For geometric sequences where $\frac{a_n}{a_{n-1}} = r$, rewrite as $a_n = r \cdot a_{n-1}$.

## 5.4 Worked Practice Exercises

Below are complete, step-by-step solutions to the practice problems on evaluating and formulating recursive functions.

### 5.4.1 First-Order Recurrence Evaluations

**EXAMPLE & WALKTHROUGH** **Evaluating First-Order Recurrences (Set 1)**

**Problem:** Find $f(1), f(2), f(3)$, and $f(4)$ if $f(n)$ is defined recursively by $\boldsymbol{f(0) = 1}$ and for $n = 0, 1, 2, \ldots$:

- **a) $\boldsymbol{f(n+1) = f(n) + 2}$ (Arithmetic progression with common difference $+2$)**:
  - $f(1) = f(0) + 2 = 1 + 2 = \mathbf{3}$
  - $f(2) = f(1) + 2 = 3 + 2 = \mathbf{5}$
  - $f(3) = f(2) + 2 = 5 + 2 = \mathbf{7}$
  - $f(4) = f(3) + 2 = 7 + 2 = \mathbf{9}$
- **b) $\boldsymbol{f(n+1) = 3f(n)}$ (Geometric progression with common ratio 3)**:
  - $f(1) = 3f(0) = 3(1) = \mathbf{3}$
  - $f(2) = 3f(1) = 3(3) = \mathbf{9}$
  - $f(3) = 3f(2) = 3(9) = \mathbf{27}$
  - $f(4) = 3f(3) = 3(27) = \mathbf{81}$
- **c) $\boldsymbol{f(n+1) = 2^{f(n)}}$ (Iterated exponentiation)**:
  - $f(1) = 2^{f(0)} = 2^1 = \mathbf{2}$
  - $f(2) = 2^{f(1)} = 2^2 = \mathbf{4}$
  - $f(3) = 2^{f(2)} = 2^4 = \mathbf{16}$
  - $f(4) = 2^{f(3)} = 2^{16} = \mathbf{65{,}536}$
- **d) $\boldsymbol{f(n+1) = f(n)^2 + f(n) + 1}$ (Quadratic recurrence)**:
  - $f(1) = f(0)^2 + f(0) + 1 = 1^2 + 1 + 1 = \mathbf{3}$
  - $f(2) = f(1)^2 + f(1) + 1 = 3^2 + 3 + 1 = 9 + 3 + 1 = \mathbf{13}$
  - $f(3) = f(2)^2 + f(2) + 1 = 13^2 + 13 + 1 = 169 + 13 + 1 = \mathbf{183}$
  - $f(4) = f(3)^2 + f(3) + 1 = 183^2 + 183 + 1 = 33,489 + 183 + 1 = \mathbf{33{,}673}$

**EXAMPLE & WALKTHROUGH** **Evaluating First-Order Recurrences (Set 2)**

**Problem:** Find $f(1), f(2), f(3), f(4)$, and $f(5)$ if $f(n)$ is defined recursively by $\boldsymbol{f(0) = 3}$ and for $n = 0, 1, 2, \ldots$:

- **a)** $\boldsymbol{f(n+1) = -2f(n)}$:
  - $f(1) = -2f(0) = -2(3) = \mathbf{-6}$
  - $f(2) = -2f(1) = -2(-6) = \mathbf{12}$
  - $f(3) = -2f(2) = -2(12) = \mathbf{-24}$
  - $f(4) = -2f(3) = -2(-24) = \mathbf{48}$
  - $f(5) = -2f(4) = -2(48) = \mathbf{-96}$
- **b)** $\boldsymbol{f(n+1) = 3f(n) + 7}$:
  - $f(1) = 3(3) + 7 = 9 + 7 = \mathbf{16}$
  - $f(2) = 3(16) + 7 = 48 + 7 = \mathbf{55}$
  - $f(3) = 3(55) + 7 = 165 + 7 = \mathbf{172}$
  - $f(4) = 3(172) + 7 = 516 + 7 = \mathbf{523}$
  - $f(5) = 3(523) + 7 = 1569 + 7 = \mathbf{1576}$
- **c)** $\boldsymbol{f(n+1) = f(n)^2 - 2f(n) - 2}$:
  - $f(1) = 3^2 - 2(3) - 2 = 9 - 6 - 2 = \mathbf{1}$
  - $f(2) = 1^2 - 2(1) - 2 = 1 - 2 - 2 = \mathbf{-3}$
  - $f(3) = (-3)^2 - 2(-3) - 2 = 9 + 6 - 2 = \mathbf{13}$
  - $f(4) = 13^2 - 2(13) - 2 = 169 - 26 - 2 = \mathbf{141}$
  - $f(5) = 141^2 - 2(141) - 2 = 19,881 - 282 - 2 = \mathbf{19{,}597}$
- **d)** $\boldsymbol{f(n+1) = 3^{\frac{f(n)}{3}}}$:
  - $f(1) = 3^{\frac{f(0)}{3}} = 3^{\frac{3}{3}} = 3^1 = \mathbf{3}$
  - $f(2) = 3^{\frac{f(1)}{3}} = 3^{\frac{3}{3}} = 3^1 = \mathbf{3}$
  - $f(3) = 3^{\frac{f(2)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$
  - $f(4) = 3^{\frac{f(3)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$
  - $f(5) = 3^{\frac{f(4)}{3}} = 3^{\frac{3}{3}} = \mathbf{3}$

**(Notice that $f(0) = 3$ is a fixed point of the map $x \mapsto 3^{\frac{x}{3}}$, so $f(n) = 3$ for all $n \geq 0$.)**

### 5.4.2 Second-Order Recurrence Evaluations

**EXAMPLE & WALKTHROUGH** **Evaluating Second-Order Recurrences (Set 3)**

**Problem:** Find $f(2), f(3), f(4)$, and $f(5)$ if $f$ is defined recursively by $\boldsymbol{f(0) = -1, f(1) = 2}$, and for $n = 1, 2, \ldots$:

- **a)** $\boldsymbol{f(n+1) = f(n) + 3f(n-1)}$:
  - $n = 1 \Longrightarrow f(2) = f(1) + 3f(0) = 2 + 3(-1) = 2 - 3 = \mathbf{-1}$
  - $n = 2 \Longrightarrow f(3) = f(2) + 3f(1) = -1 + 3(2) = -1 + 6 = \mathbf{5}$
  - $n = 3 \Longrightarrow f(4) = f(3) + 3f(2) = 5 + 3(-1) = 5 - 3 = \mathbf{2}$
  - $n = 4 \Longrightarrow f(5) = f(4) + 3f(3) = 2 + 3(5) = 2 + 15 = \mathbf{17}$
- **b)** $\boldsymbol{f(n+1) = f(n)^2 f(n-1)}$:
  - $n = 1 \Longrightarrow f(2) = f(1)^2 f(0) = (2)^2(-1) = \mathbf{-4}$
  - $n = 2 \Longrightarrow f(3) = f(2)^2 f(1) = (-4)^2(2) = 16 \cdot 2 = \mathbf{32}$
  - $n = 3 \Longrightarrow f(4) = f(3)^2 f(2) = (32)^2(-4) = 1024 \cdot (-4) = \mathbf{-4096}$
  - $n = 4 \Longrightarrow f(5) = f(4)^2 f(3) = (-4096)^2(32) = 16,777,216 \cdot 32 = \mathbf{536{,}870{,}912}$ (or $2^{29}$)
- **c)** $\boldsymbol{f(n+1) = 3f(n)^2 - 4f(n-1)^2}$:

- $n = 1 \Longrightarrow f(2) = 3f(1)^2 - 4f(0)^2 = 3(2)^2 - 4(-1)^2 = 12 - 4 = \mathbf{8}$
- $n = 2 \Longrightarrow f(3) = 3f(2)^2 - 4f(1)^2 = 3(8)^2 - 4(2)^2 = 192 - 16 = \mathbf{176}$
- $n = 3 \Longrightarrow f(4) = 3f(3)^2 - 4f(2)^2 = 3(176)^2 - 4(8)^2 = 3(30,976) - 256 = 92,928 - 256 = \mathbf{92,672}$
- $n = 4 \Longrightarrow f(5) = 3f(4)^2 - 4f(3)^2 = 3(92,672)^2 - 4(176)^2 = 25,764,298,752 - 123,904 = \mathbf{25,764,174,848}$

- **d)** $\boldsymbol{f(n+1) = \frac{f(n-1)}{f(n)}}$:
  - $n = 1 \Longrightarrow f(2) = \frac{f(0)}{f(1)} = \frac{-1}{2} = \mathbf{-1/2}$
  - $n = 2 \Longrightarrow f(3) = \frac{f(1)}{f(2)} = \frac{2}{-\frac{1}{2}} = \mathbf{-4}$
  - $n = 3 \Longrightarrow f(4) = \frac{f(2)}{f(3)} = \frac{-\frac{1}{2}}{-4} = \mathbf{1/8}$
  - $n = 4 \Longrightarrow f(5) = \frac{f(3)}{f(4)} = \frac{-4}{\frac{1}{8}} = \mathbf{-32}$

**EXAMPLE & WALKTHROUGH** **Evaluating Second-Order Recurrences (Set 4)**

**Problem:** Find $f(2), f(3), f(4)$, and $f(5)$ if $f$ is defined recursively by $\boldsymbol{f(0) = f(1) = 1}$ and for $n = 1, 2, \ldots$:

- **a)** $\boldsymbol{f(n+1) = f(n) - f(n-1)}$:
  - $n = 1 \Longrightarrow f(2) = f(1) - f(0) = 1 - 1 = \mathbf{0}$
  - $n = 2 \Longrightarrow f(3) = f(2) - f(1) = 0 - 1 = \mathbf{-1}$
  - $n = 3 \Longrightarrow f(4) = f(3) - f(2) = -1 - 0 = \mathbf{-1}$
  - $n = 4 \Longrightarrow f(5) = f(4) - f(3) = -1 - (-1) = \mathbf{0}$

  **(This sequence repeats with period 6: $\mathbf{1, 1, 0, -1, -1, 0, 1, 1, \ldots}$)**

- **b)** $\boldsymbol{f(n+1) = f(n)f(n-1)}$:
  - $n = 1 \Longrightarrow f(2) = f(1)f(0) = 1 \cdot 1 = \mathbf{1}$
  - $n = 2 \Longrightarrow f(3) = f(2)f(1) = 1 \cdot 1 = \mathbf{1}$
  - $n = 3 \Longrightarrow f(4) = f(3)f(2) = 1 \cdot 1 = \mathbf{1}$
  - $n = 4 \Longrightarrow f(5) = f(4)f(3) = 1 \cdot 1 = \mathbf{1}$

- **c)** $\boldsymbol{f(n+1) = f(n)^2 + f(n-1)^3}$:
  - $n = 1 \Longrightarrow f(2) = f(1)^2 + f(0)^3 = 1^2 + 1^3 = \mathbf{2}$
  - $n = 2 \Longrightarrow f(3) = f(2)^2 + f(1)^3 = 2^2 + 1^3 = 4 + 1 = \mathbf{5}$
  - $n = 3 \Longrightarrow f(4) = f(3)^2 + f(2)^3 = 5^2 + 2^3 = 25 + 8 = \mathbf{33}$
  - $n = 4 \Longrightarrow f(5) = f(4)^2 + f(3)^3 = 33^2 + 5^3 = 1089 + 125 = \mathbf{1214}$

- **d)** $\boldsymbol{f(n+1) = \frac{f(n)}{f(n-1)}}$:
  - $n = 1 \Longrightarrow f(2) = \frac{f(1)}{f(0)} = \frac{1}{1} = \mathbf{1}$
  - $n = 2 \Longrightarrow f(3) = \frac{f(2)}{f(1)} = \frac{1}{1} = \mathbf{1}$
  - $n = 3 \Longrightarrow f(4) = \frac{f(3)}{f(2)} = \frac{1}{1} = \mathbf{1}$
  - $n = 4 \Longrightarrow f(5) = \frac{f(4)}{f(3)} = \frac{1}{1} = \mathbf{1}$

### 5.4.3 Formulating Recursive Sequence Definitions

**EXAMPLE & WALKTHROUGH** **Writing Recursive Definitions for Sequences**

**Set 1:** Give a recursive definition of the sequence $\{a_n\}, n = 1, 2, 3, \ldots$ if:

- **a)** $\boldsymbol{a_n = 6n}$:
  - **Basis**: $a_1 = 6(1) = 6$.
  - **Recurrence**: Since $a_n - a_{n-1} = 6n - 6(n-1) = 6$, we have $a_n = a_{n-1} + 6$ for $n > 1$.

$$a_n = \begin{cases} 6 & , \ n = 1 \\ a_{n-1} + 6 & , \ n > 1 \end{cases}$$

- **b) $a_n = 2n + 1$**:
  - **Basis**: $a_1 = 2(1) + 1 = 3$.
  - **Recurrence**: Since $a_n - a_{n-1} = (2n + 1) - (2(n - 1) + 1) = 2$, we have $a_n = a_{n-1} + 2$ for $n > 1$.

$$a_n = \begin{cases} 3 & , \ n = 1 \\ a_{n-1} + 2 & , \ n > 1 \end{cases}$$

- **c) $a_n = 10^n$**:
  - **Basis**: $a_1 = 10^1 = 10$.
  - **Recurrence**: Since $a_n = 10 \cdot 10^{n-1} = 10a_{n-1}$, we have $a_n = 10a_{n-1}$ for $n > 1$.

$$a_n = \begin{cases} 10 & , \ n = 1 \\ 10a_{n-1} & , \ n > 1 \end{cases}$$

- **d) $a_n = 5$ (Constant sequence)**:
  - **Basis**: $a_1 = 5$.
  - **Recurrence**: Every term equals the previous term, so $a_n = a_{n-1}$ for $n > 1$.

$$a_n = \begin{cases} 5 & , \ n = 1 \\ a_{n-1} & , \ n > 1 \end{cases}$$

---

**Set 2:** Give a recursive definition of the sequence $\{a_n\}, n = 1, 2, 3, \ldots$ if:

- **a) $a_n = 4n - 2$**:
  - **Basis**: $a_1 = 4(1) - 2 = 2$.
  - **Recurrence**: Computing the difference gives $a_n - a_{n-1} = (4n - 2) - (4(n - 1) - 2) = 4$. Thus $a_n = a_{n-1} + 4$ for $n > 1$.

$$a_n = \begin{cases} 2 & , \ n = 1 \\ a_{n-1} + 4 & , \ n > 1 \end{cases}$$

- **b) $a_n = 1 + (-1)^n$**: Checking the first few terms: for odd $n$, $a_n = 1 - 1 = 0$; for even $n$, $a_n = 1 + 1 = 2$. Thus the sequence alternates $0, 2, 0, 2, 0, 2, \ldots$
  - **Method A (First-Order Complement)**: Notice that $a_n + a_{n-1} = (1 + (-1)^n) + (1 + (-1)^{n-1}) = 2 + 0 = 2$. Thus $a_n = 2 - a_{n-1}$ (or $a_n = a_{n-1} + 2(-1)^n$) with basis $a_1 = 0$:

$$a_n = \begin{cases} 0 & , \ n = 1 \\ 2 - a_{n-1} & , \ n > 1 \end{cases}$$

  - **Method B (Second-Order Periodic)**: Alternatively, using two basis cases $a_1 = 0, a_2 = 2$, the recurrence is $a_n = a_{n-2}$ for $n > 2$.

- **c) $a_n = n(n + 1)$**:
  - **Basis**: $a_1 = 1(1 + 1) = 2$.
  - **Recurrence**: Compute the first difference $a_n - a_{n-1}$:

$$a_n - a_{n-1} = n(n + 1) - (n - 1)n = n((n + 1) - (n - 1)) = 2n$$

    Therefore, $a_n = a_{n-1} + 2n$ for $n > 1$:

$$a_n = \begin{cases} 2 & , \ n = 1 \\ a_{n-1} + 2n & , \ n > 1 \end{cases}$$

- **d) $a_n = n^2$**:
  - **Basis**: $a_1 = 1^2 = 1$.
  - **Recurrence**: Compute the first difference $a_n - a_{n-1}$:

    $$a_n - a_{n-1} = n^2 - (n-1)^2 = n^2 - (n^2 - 2n + 1) = 2n - 1$$

    Therefore, $a_n = a_{n-1} + 2n - 1$ for $n > 1$ (showing that the $n$-th square is the sum of the first $n$ odd integers):

    $$a_n = \begin{cases} 1 & , \ n = 1 \\ a_{n-1} + 2n - 1 & , \ n > 1 \end{cases}$$

# 6. Mathematical Induction

Well-Ordering, Weak & Strong Induction, Summation Identities, Divisibility, and Combinatorial Proofs

## 6.1 The Concept of Mathematical Induction

In discrete mathematics and computer science, we often want to prove statements that claim something is true for every integer starting from some baseline—most commonly for all positive integers $\mathbb{Z}^+ = \{1, 2, 3, ...\}$ or all non-negative integers $\mathbb{N} = \{0, 1, 2, ...\}$. Testing individual values like $n = 1, 2, 3, ...$ can help us spot patterns, but it can never prove a universal statement $\forall n \geq n_0, P(n)$, because no finite number of test cases can cover an infinite domain.

**Mathematical induction** is a proof technique designed specifically for propositions that depend on integers. Instead of checking infinitely many cases one by one, mathematical induction shows that truth carries forward automatically from one integer to the next, starting from an initial base case.

**DEFINITION** **Mathematical Induction**

**Mathematical induction** is a formal method of proof used to establish that a proposition (predicate) $P(n)$ holds for all integers $n$ greater than or equal to a fixed starting integer $n_0$ (typically $n_0 = 1$ or $n_0 = 0$).

Two classic examples of integer-indexed propositions that can be proved by induction are:

1. **Divisibility Proposition**: For all integers $n \geq 1$, the expression $n^3 + 2n$ is a multiple of 3, written $3 \mid (n^3 + 2n)$.
2. **Combinatorial / Set Cardinality Proposition**: The number of subsets formed from a finite set containing $n$ elements is $2^n$; that is, if $|S| = n$, then $|\mathcal{P}(S)| = 2^n$.

### 6.1.1 Deductive vs. Inductive Reasoning in Mathematics

Despite the shared name, **mathematical induction** is fundamentally different from the **inductive reasoning** used in the empirical sciences:

- In empirical science, *inductive reasoning* generalizes a likely hypothesis from finite observations, which could later be overturned by a new counterexample.
- In mathematics, *mathematical induction* is a **deductive proof technique** whose conclusion holds with complete certainty. Once the base case and the step-by-step transition are established, there is no possibility of a counterexample at any finite integer $n$.

**INTUITION & DEEP DIVE** **Climbing an Infinite Ladder**

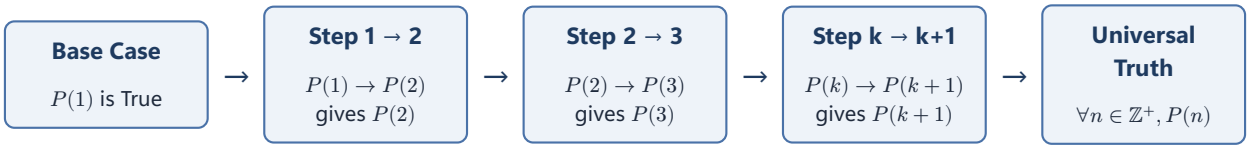
A helpful way to picture mathematical induction is the classic metaphor of **climbing an infinite ladder** whose rungs are numbered $1, 2, 3, ..., k, k+1, ...$ stretching upward without end. How can we be sure a climber can reach **every** rung on the ladder?

We only need to verify two things:

1. **First Rung Reachability (Base Step)**: We can reach rung 1 ($P(1)$ is true).
2. **Rung-to-Rung Transition (Inductive Step)**: Whenever we can reach rung $k$, we can also reach the next rung $k+1$ ($\forall k \geq 1, P(k) \implies P(k+1)$).

Once both rules hold, a chain reaction guarantees that every positive integer rung is reachable:

- Since we can reach rung 1 ($P(1)$), and reaching rung 1 lets us reach rung 2 ($P(1) \implies P(2)$), we can reach rung 2 ($P(2)$).
- Since we can reach rung 2 ($P(2)$), and reaching rung 2 lets us reach rung 3 ($P(2) \implies P(3)$), we can reach rung 3 ($P(3)$).
- Repeating this transition $n - 1$ times guarantees that we reach any rung $n \in \mathbb{Z}^+$ in a finite number of steps.




*Domino / Infinite Ladder Chain Reaction: How truth propagates from P(1) to any integer n via Modus Ponens.*

### 6.1.2 Theoretical Foundation: The Well-Ordering Principle

Why does mathematical induction work as a valid rule of inference? Its soundness comes from a foundational property of the integers called the **Well-Ordering Principle**.

**CORE CONCEPT** **Equivalence to the Well-Ordering Principle**

The **Well-Ordering Principle** states that every non-empty subset of non-negative integers (or positive integers) has a **least element**.

We can see why mathematical induction follows from the Well-Ordering Principle by arguing by contradiction:

1. Suppose that $P(1)$ is true and $\forall k \geq 1, (P(k) \implies P(k+1))$ is true, yet $P(n)$ fails for at least one positive integer.
2. Let $S = \{n \in \mathbb{Z}^+ \mid P(n) \text{ is false}\}$ be the set of all counterexamples. By our assumption, $S \neq \varnothing$.
3. By the Well-Ordering Principle, $S$ must have a smallest element $m \in S$.
4. Since $P(1)$ is true, $1 \notin S$, which means $m > 1$. Thus, $m - 1$ is a positive integer strictly smaller than $m$.
5. Because $m$ is the **smallest** element in $S$, $m - 1$ cannot be in $S$, meaning $P(m-1)$ must be **true**.
6. Applying the inductive implication $P(k) \implies P(k+1)$ at $k = m - 1$ forces $P(m)$ to be **true**, which contradicts $m \in S$. Therefore, $S$ must be empty, and $P(n)$ holds for all $n \in \mathbb{Z}^+$.

## 6.2 Principle of Simple Mathematical Induction

The standard form of mathematical induction is often called **simple induction** (or **weak induction**).

**DEFINITION** **Principle of Simple Induction**

Let $P(n)$ be a proposition about positive integers $n \in \mathbb{Z}^+$. To prove that $P(n)$ is true for all positive integers $n \geq 1$, we complete three steps:

1. **Basis Step**: Show that $P(1)$ is true.
2. **Inductive Hypothesis**: Assume that $P(n)$ is true for an arbitrary positive integer $n \geq 1$. *(Some authors write $P(k)$ for the hypothesis and $P(k+1)$ for the next step to distinguish the specific integer $k$ from the general variable $n$; both conventions mean the exact same thing.)*
3. **Inductive Step**: Using the assumption that $P(n)$ holds, show that $P(n+1)$ must also be true.

Expressed in predicate logic, the principle of simple induction states:

$$(P(1) \land \forall n \in \mathbb{Z}^+ (P(n) \implies P(n+1))) \implies \forall n \in \mathbb{Z}^+ P(n)$$

**Algorithm: Blueprint for a Proof by Simple Induction** `3 Logical Steps`

**Input:** A universal claim $\forall n \geq n_0, P(n)$ over integers $n$
**Output:** A complete proof of $\forall n \geq n_0, P(n)$

1. **Define the Proposition $P(n)$**: State clearly what $P(n)$ claims for an integer $n$, along with its starting integer $n \geq n_0$.
2. **Verify the Base Case $P(n_0)$**:

- Plug $n = n_0$ into the Left-Hand Side (LHS) and Right-Hand Side (RHS) separately (or check the property directly).
- Confirm that LHS = RHS (or that the condition holds), establishing $P(n_0)$.

3. **State the Inductive Hypothesis (IH)**:
   - Write: *"Assume that $P(n)$ holds for some arbitrary integer $n \geq n_0$."*
   - Write out the explicit equation or statement assumed for $n$.
4. **Carry Out the Inductive Step ($P(n) \implies P(n+1)$)**:
   - State the target: *"We want to show that $P(n+1)$ holds, namely [write target formula for $n+1$]."*
   - Start with one side of $P(n+1)$ (usually the LHS), group the part that matches $P(n)$, **substitute the Inductive Hypothesis**, and simplify algebraically until you reach the RHS of $P(n+1)$.
5. **Conclude**:
   - State that by mathematical induction, $P(n)$ holds for all integers $n \geq n_0$.

**PITFALL / WARNING** **Why Assuming P(n) Is Not Circular Reasoning**

At first glance, Step 2 ("Assume $P(n)$ is true") can feel like circular reasoning, since $P(n)$ is the very statement we want to prove!

- **Why it is valid**: In the inductive step, we are **not** assuming that $P(n)$ is already true for every integer. Instead, we are proving the **conditional statement** $P(n) \implies P(n+1)$ for a fixed, arbitrary integer $n$.
- To prove any conditional statement $A \implies B$ directly, we assume $A$ ($P(n)$) and deduce $B$ ($P(n+1)$). Combined with the base case $P(1)$, this implication powers the chain of steps without any circularity.

## 6.3 Fundamental Worked Examples

Let us work through several core proofs—covering arithmetic sums, odd-integer sums, divisibility, and power-set size—while highlighting algebraic habits that keep inductive proofs clean.

### 6.3.1 Sum of the First $n$ Positive Integers

**EXAMPLE & WALKTHROUGH** **Gauss's Arithmetic Sum Formula**

**Proposition**: Show that for all integers $n \geq 1$,

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$.

1. **Basis Step ($P(1)$)**: For $n = 1$, the left-hand side is simply 1, and the right-hand side evaluates to:

$$\frac{1(1+1)}{2} = \frac{1 \cdot 2}{2} = 1$$

   Since LHS = 1 = RHS, the base case $P(1)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for some arbitrary positive integer $n \geq 1$:

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We need to show that $P(n+1)$ holds, namely:

$$1 + 2 + 3 + \ldots + n + (n+1) = \frac{(n+1)((n+1)+1)}{2} = \frac{(n+1)(n+2)}{2}$$

Starting from the left-hand side of $P(n+1)$ and grouping the first $n$ terms:

$$\begin{aligned}1+2+\ldots+n+(n+1) &= \underbrace{(1+2+\ldots+n)}_{\text{apply IH}}+(n+1)\\ &= \frac{n(n+1)}{2}+(n+1)\\ &= \frac{n(n+1)+2(n+1)}{2}\\ &= \frac{n^2+n+2n+2}{2}\\ &= \frac{n^2+3n+2}{2}\\ &= \frac{(n+1)(n+2)}{2}\\ &= \frac{(n+1)((n+1)+1)}{2}\end{aligned}$$

This matches the right-hand side of $P(n+1)$. Thus, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, $1+2+3+\ldots+n=\frac{n(n+1)}{2}$ for all integers $n \geq 1$. □

**INTUITION & DEEP DIVE** **Algebraic Tip: Factor Early Instead of Expanding**

In the calculation above, we expanded $n(n+1)+2(n+1)$ into $n^2+3n+2$ and then factored it back into $(n+1)(n+2)$. While expanding works fine for a quadratic, doing that on degree-3 or degree-4 polynomials makes the algebra much harder than it needs to be.

A much cleaner habit is to **factor out common terms immediately**:

$$\frac{n(n+1)+2(n+1)}{2}=\frac{(n+1)(n+2)}{2}$$

By factoring out $(n+1)$ right away, we reach the target expression in a single step without expanding and re-factoring.

### 6.3.2 Sum of the First $n$ Positive Odd Integers

**EXAMPLE & WALKTHROUGH** **Sum of the First n Positive Odd Integers**

**Proposition**: Show that the sum of the first $n$ positive odd integers is $n^2$.

**Proof by Mathematical Induction**: Notice that the $k$-th positive odd integer is $2k-1$ (giving 1 for $k=1$, 3 for $k=2$, 5 for $k=3$, and $2n-1$ for $k=n$). Let $P(n)$ be the proposition that for $n \geq 1$:

$$1+3+5+\ldots+(2n-1)=n^2$$

1. **Basis Step ($P(1)$)**: For $n=1$, the left-hand side is $2(1)-1=1$, and the right-hand side is $1^2=1$. Since $1=1$, $P(1)$ is true.
2. **Inductive Hypothesis**: Assume that $P(n)$ is true for an arbitrary positive integer $n \geq 1$:

$$1+3+5+\ldots+(2n-1)=n^2 \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \Longrightarrow P(n+1)$)**: We want to show that $P(n+1)$ is true. The $(n+1)$-st odd positive integer is $2(n+1)-1=2n+1$, so the target equation for $P(n+1)$ is:

$$1+3+5+\ldots+(2n-1)+(2n+1)=(n+1)^2$$

Grouping the first $n$ odd integers on the left-hand side and applying the Inductive Hypothesis:

$$\begin{aligned} 1+3+5+\ldots+(2n-1)+(2n+1) &= \underbrace{(1+3+5+\ldots+(2n-1))}_{\text{apply IH}}+(2n+1) \\ &= n^2+2n+1 \\ &= (n+1)^2 \end{aligned}$$

Thus, $P(n+1)$ is true whenever $P(n)$ is true.

**Conclusion**: By mathematical induction, the sum of the first $n$ positive odd integers is $n^2$ for all $n \geq 1$. □

### 6.3.3 Divisibility and Set Cardinality Proofs

Mathematical induction is equally effective outside summation formulas. Let us prove the two motivating propositions introduced in Section 1.

**EXAMPLE & WALKTHROUGH** **Divisibility by 3: $n^3 + 2n$ is a Multiple of 3**

**Proposition**: For all integers $n \geq 1$, $n^3 + 2n$ is a multiple of 3.

**Proof by Mathematical Induction**: Let $P(n)$ be the statement "$3 \mid (n^3 + 2n)$", meaning there exists an integer $k \in \mathbb{Z}$ such that $n^3 + 2n = 3k$.

1. **Basis Step ($P(1)$)**: For $n = 1$, we have:

   $$1^3 + 2(1) = 1 + 2 = 3 = 3 \cdot 1$$

   which is a multiple of 3. Thus, $P(1)$ holds.

2. **Inductive Hypothesis**: Assume $P(n)$ holds for some arbitrary integer $n \geq 1$. That is, assume there exists an integer $k \in \mathbb{Z}$ such that:

   $$n^3 + 2n = 3k \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We must show that $(n+1)^3 + 2(n+1)$ is also a multiple of 3. Expanding the expression and regrouping terms so that $(n^3 + 2n)$ appears explicitly:

   $$\begin{aligned} (n+1)^3 + 2(n+1) &= (n^3 + 3n^2 + 3n + 1) + (2n + 2) \\ &= \underbrace{(n^3 + 2n)}_{\text{apply IH}} + 3n^2 + 3n + 3 \\ &= 3k + 3(n^2 + n + 1) \\ &= 3(k + n^2 + n + 1) \end{aligned}$$

   Because $k$ and $n$ are integers, $m = k + n^2 + n + 1$ is also an integer, so $(n+1)^3 + 2(n+1) = 3m$ is a multiple of 3. Therefore, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, $n^3 + 2n$ is a multiple of 3 for all integers $n \geq 1$. □

**EXAMPLE & WALKTHROUGH** **Power Set Cardinality: A Set of n Elements Has $2^n$ Subsets**

**Proposition**: For every integer $n \geq 0$, a finite set containing $n$ elements has exactly $2^n$ subsets.

**Proof by Mathematical Induction**: Let $P(n)$ be the statement: "Every finite set $S$ with $|S| = n$ elements has $2^n$ subsets."

1. **Basis Step ($P(0)$)**: For $n = 0$, the only set with 0 elements is the empty set $S = \varnothing$. Its only subset is $\varnothing$ itself, giving $1 = 2^0$ subset. (If starting at $n = 1$, a singleton set $S = \{x_1\}$ has two subsets, $\varnothing$ and $\{x_1\}$, which is $2 = 2^1$.) Thus, the base case holds.
2. **Inductive Hypothesis**: Assume $P(n)$ holds for some arbitrary non-negative integer $n \geq 0$: any set of $n$ elements has $2^n$ subsets.
3. **Inductive Step ($P(n) \implies P(n+1)$)**: Let $T$ be an arbitrary set with $n+1$ elements, say $T = \{a_1, a_2, ..., a_n, a_{n+1}\}$. Remove the last element to form $S = \{a_1, a_2, ..., a_n\} = T \setminus \{a_{n+1}\}$, which has $|S| = n$ elements. Every subset $A \subseteq T$ belongs to one of two disjoint categories:
   - **Category 1 ($a_{n+1} \notin A$)**: $A$ is a subset of $S$ alone. By the Inductive Hypothesis applied to $S$, there are $2^n$ such subsets.
   - **Category 2 ($a_{n+1} \in A$)**: $A$ is formed by taking a subset $B \subseteq S$ and including $a_{n+1}$, so $A = B \cup \{a_{n+1}\}$. Because each of the $2^n$ subsets $B \subseteq S$ produces a distinct subset $B \cup \{a_{n+1}\}$ of $T$, there are $2^n$ subsets in Category 2 as well.

   Adding the two disjoint categories together gives the total number of subsets of $T$:

$$|\text{Subsets of } T| = 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}$$

   Thus, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, any set with $n$ elements has $2^n$ subsets for all $n \geq 0$. □

## 6.4 Worked Practice Problems

To build fluency with different series patterns—including sums of squares, factorials, geometric progressions, alternating series, and products of consecutive integers—we work through seven classic practice problems below.

### 6.4.1 Sums of Squares, Factorials, and Geometric Series

**EXAMPLE & WALKTHROUGH** **Problem 1: Sum of Squares of Odd Integers**

**Problem Statement**: Prove that whenever $n$ is a non-negative integer ($n \geq 0$),

$$1^2 + 3^2 + 5^2 + ... + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=0}^{n} (2k+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3}$ for $n \geq 0$.

1. **Basis Step ($P(0)$)**: Since the statement applies to all non-negative integers ($n \geq 0$), our base case is $n = 0$:
   - LHS $= (2(0)+1)^2 = 1^2 = 1$.
   - RHS $= \frac{(0+1)(2(0)+1)(2(0)+3)}{3} = \frac{1 \cdot 1 \cdot 3}{3} = 1$.

   Since LHS = RHS = 1, $P(0)$ holds.
2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary integer $n \geq 0$:

$$1^2 + 3^2 + 5^2 + ... + (2n+1)^2 = \frac{(n+1)(2n+1)(2n+3)}{3} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We must show that $P(n+1)$ holds. Replacing $n$ with $n+1$, the next term on the LHS is $(2(n+1)+1)^2 = (2n+3)^2$, and the target RHS is:

$$\frac{((n+1)+1)(2(n+1)+1)(2(n+1)+3)}{3} = \frac{(n+2)(2n+3)(2n+5)}{3}$$

   Starting from the LHS of $P(n+1)$ and applying the Inductive Hypothesis:

$$1^2 + 3^2 + \ldots + (2n+1)^2 + (2n+3)^2 = \frac{(n+1)(2n+1)(2n+3)}{3} + (2n+3)^2$$
$$= \frac{(n+1)(2n+1)(2n+3) + 3(2n+3)^2}{3}$$

Factoring out the common term $(2n+3)$ in the numerator:

$$= \frac{(2n+3)[(n+1)(2n+1) + 3(2n+3)]}{3}$$
$$= \frac{(2n+3)\big[(2n^2+3n+1) + (6n+9)\big]}{3}$$
$$= \frac{(2n+3)(2n^2+9n+10)}{3}$$

Factoring the quadratic $2n^2 + 9n + 10 = (n+2)(2n+5)$ yields:

$$= \frac{(n+2)(2n+3)(2n+5)}{3}$$

which equals the target RHS of $P(n+1)$.

**Conclusion**: By mathematical induction, the identity holds for all non-negative integers $n \geq 0$. □

EXAMPLE & WALKTHROUGH **Problem 2: Factorial Series Identity**

**Problem Statement**: Prove that whenever $n$ is a positive integer ($n \geq 1$),

$$1 \cdot 1! + 2 \cdot 2! + \ldots + n \cdot n! = (n+1)! - 1$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=1}^{n} k \cdot k! = (n+1)! - 1$ for $n \geq 1$.

1. **Basis Step ($P(1)$)**: For $n = 1$:
   - LHS $= 1 \cdot 1! = 1 \cdot 1 = 1$.
   - RHS $= (1+1)! - 1 = 2! - 1 = 2 - 1 = 1$.

   Since LHS = RHS = 1, $P(1)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary integer $n \geq 1$:

   $$1 \cdot 1! + 2 \cdot 2! + \ldots + n \cdot n! = (n+1)! - 1 \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $P(n+1)$ holds:

   $$1 \cdot 1! + 2 \cdot 2! + \ldots + n \cdot n! + (n+1)(n+1)! = (n+2)! - 1$$

   Starting from the LHS of $P(n+1)$ and substituting the Inductive Hypothesis:

   $$\underbrace{(1 \cdot 1! + 2 \cdot 2! + \ldots + n \cdot n!)}_{\text{apply IH}} + (n+1)(n+1)! = ((n+1)! - 1) + (n+1)(n+1)!$$
   $$= (n+1)! \cdot [1 + (n+1)] - 1$$
   $$= (n+1)! \cdot (n+2) - 1$$
   $$= (n+2)! - 1$$

   where the last step uses the factorial recurrence $(n+2)! = (n+2) \cdot (n+1)!$. This matches the RHS of $P(n+1)$.

**Conclusion**: By mathematical induction, $\sum_{k=1}^{n} k \cdot k! = (n+1)! - 1$ for all positive integers $n \geq 1$. □

**EXAMPLE & WALKTHROUGH** **Problem 3: Geometric Series with Ratio 5**

**Problem Statement**: Prove that whenever $n$ is a non-negative integer ($n \geq 0$),

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \ldots + 3 \cdot 5^n = \frac{3(5^{n+1} - 1)}{4}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=0}^{n} 3 \cdot 5^k = \frac{3(5^{n+1}-1)}{4}$ for $n \geq 0$.

1. **Basis Step ($P(0)$)**: For $n = 0$:
   - LHS $= 3 \cdot 5^0 = 3 \cdot 1 = 3$.
   - RHS $= \frac{3(5^{0+1}-1)}{4} = \frac{3(5-1)}{4} = \frac{3 \cdot 4}{4} = 3$.

   Since LHS $=$ RHS $= 3$, $P(0)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary integer $n \geq 0$:

$$3 + 3 \cdot 5 + 3 \cdot 5^2 + \ldots + 3 \cdot 5^n = \frac{3(5^{n+1} - 1)}{4} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $P(n+1)$ holds:

$$3 + 3 \cdot 5 + \ldots + 3 \cdot 5^n + 3 \cdot 5^{n+1} = \frac{3(5^{n+2} - 1)}{4}$$

   Starting from the LHS of $P(n+1)$ and applying the Inductive Hypothesis:

$$\begin{aligned}
(3 + 3 \cdot 5 + \ldots + 3 \cdot 5^n) + 3 \cdot 5^{n+1} &= \frac{3(5^{n+1} - 1)}{4} + 3 \cdot 5^{n+1} \\
&= \frac{3 \cdot 5^{n+1} - 3 + 4 \cdot 3 \cdot 5^{n+1}}{4} \\
&= \frac{3 \cdot 5^{n+1}(1 + 4) - 3}{4} \\
&= \frac{3 \cdot 5^{n+1} \cdot 5 - 3}{4} \\
&= \frac{3 \cdot 5^{n+2} - 3}{4} \\
&= \frac{3(5^{n+2} - 1)}{4}
\end{aligned}$$

   Thus, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, the formula holds for all non-negative integers $n \geq 0$. □

**EXAMPLE & WALKTHROUGH** **Problem 4: Alternating Geometric Series with Ratio -7**

**Problem Statement**: Prove that whenever $n$ is a non-negative integer ($n \geq 0$),

$$2 - 2 \cdot 7 + 2 \cdot 7^2 - \ldots + 2(-7)^n = \frac{1 - (-7)^{n+1}}{4}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=0}^{n} 2(-7)^k = \frac{1-(-7)^{n+1}}{4}$ for $n \geq 0$.

1. **Basis Step ($P(0)$)**: For $n = 0$:
   - LHS $= 2(-7)^0 = 2(1) = 2$.
   - RHS $= \frac{1-(-7)^{0+1}}{4} = \frac{1-(-7)}{4} = \frac{8}{4} = 2$.

Since LHS = RHS = 2, $P(0)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary integer $n \geq 0$:

$$2 - 2 \cdot 7 + 2 \cdot 7^2 - \ldots + 2(-7)^n = \frac{1 - (-7)^{n+1}}{4} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $P(n+1)$ holds:

$$2 - 2 \cdot 7 + \ldots + 2(-7)^n + 2(-7)^{n+1} = \frac{1 - (-7)^{n+2}}{4}$$

Starting from the LHS of $P(n+1)$ and substituting the Inductive Hypothesis:

$$\begin{aligned}(2 - 2 \cdot 7 + \ldots + 2(-7)^n) + 2(-7)^{n+1} &= \frac{1 - (-7)^{n+1}}{4} + 2(-7)^{n+1} \\ &= \frac{1 - (-7)^{n+1} + 8(-7)^{n+1}}{4} \\ &= \frac{1 + 7(-7)^{n+1}}{4} \\ &= \frac{1 - (-7)(-7)^{n+1}}{4} \\ &= \frac{1 - (-7)^{n+2}}{4}\end{aligned}$$

Notice how $+7(-7)^{n+1}$ rewrites cleanly as $-(-7)(-7)^{n+1} = -(-7)^{n+2}$, matching the RHS of $P(n+1)$.

**Conclusion**: By mathematical induction, the identity holds for all non-negative integers $n \geq 0$. □

### 6.4.2 Alternating Squares and Consecutive Factor Products

**EXAMPLE & WALKTHROUGH** **Problem 5: Alternating Sum of Squares**

**Problem Statement**: Prove that whenever $n$ is a positive integer ($n \geq 1$),

$$1^2 - 2^2 + 3^2 - \ldots + (-1)^{n-1} n^2 = (-1)^{n-1} \frac{n(n+1)}{2}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=1}^{n} (-1)^{k-1} k^2 = (-1)^{n-1} \frac{n(n+1)}{2}$ for $n \geq 1$.

1. **Basis Step ($P(1)$)**: For $n = 1$:
   - LHS $= (-1)^{1-1} 1^2 = (-1)^0 \cdot 1 = 1$.
   - RHS $= (-1)^{1-1} \frac{1(1+1)}{2} = 1 \cdot \frac{2}{2} = 1$.

   Since LHS = RHS = 1, $P(1)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary positive integer $n \geq 1$:

$$1^2 - 2^2 + 3^2 - \ldots + (-1)^{n-1} n^2 = (-1)^{n-1} \frac{n(n+1)}{2} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $P(n+1)$ holds:

$$1^2 - 2^2 + \ldots + (-1)^{n-1} n^2 + (-1)^n (n+1)^2 = (-1)^n \frac{(n+1)(n+2)}{2}$$

Starting from the LHS of $P(n+1)$ and substituting the Inductive Hypothesis:

$$\underbrace{(1^2 - 2^2 + \ldots + (-1)^{n-1}n^2)}_{\text{apply IH}} + (-1)^n(n+1)^2 = (-1)^{n-1}\frac{n(n+1)}{2} + (-1)^n(n+1)^2$$

Using $(-1)^{n-1} = -(-1)^n$ and factoring out $(-1)^n(n+1)$ from both terms:

$$\begin{aligned} &= -(-1)^n\frac{n(n+1)}{2} + (-1)^n\frac{2(n+1)^2}{2} \\ &= (-1)^n\frac{(n+1)[-n+2(n+1)]}{2} \\ &= (-1)^n\frac{(n+1)(n+2)}{2} \end{aligned}$$

which matches the target RHS of $P(n+1)$.

**Conclusion**: By mathematical induction, the identity holds for all positive integers $n \geq 1$. □

**EXAMPLE & WALKTHROUGH** **Problem 6: Sum of Consecutive Pair Products**

**Problem Statement**: Prove that for every positive integer $n \geq 1$,

$$1 \cdot 2 + 2 \cdot 3 + \ldots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$ for $n \geq 1$.

1. **Basis Step ($P(1)$)**: For $n = 1$:
   - LHS $= 1(1+1) = 1 \cdot 2 = 2$.
   - RHS $= \frac{1(1+1)(1+2)}{3} = \frac{1 \cdot 2 \cdot 3}{3} = 2$.

   Since LHS $=$ RHS $= 2$, $P(1)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary positive integer $n \geq 1$:

$$1 \cdot 2 + 2 \cdot 3 + \ldots + n(n+1) = \frac{n(n+1)(n+2)}{3} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \Longrightarrow P(n+1)$)**: We want to show that $P(n+1)$ holds:

$$1 \cdot 2 + 2 \cdot 3 + \ldots + n(n+1) + (n+1)(n+2) = \frac{(n+1)(n+2)(n+3)}{3}$$

   Starting from the LHS of $P(n+1)$ and applying the Inductive Hypothesis:

$$\begin{aligned} \underbrace{(1 \cdot 2 + 2 \cdot 3 + \ldots + n(n+1))}_{\text{apply IH}} + (n+1)(n+2) &= \frac{n(n+1)(n+2)}{3} + (n+1)(n+2) \\ &= \frac{n(n+1)(n+2) + 3(n+1)(n+2)}{3} \\ &= \frac{(n+1)(n+2)(n+3)}{3} \end{aligned}$$

   by factoring out $(n+1)(n+2)$. Thus, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, the identity holds for all positive integers $n \geq 1$. □

**EXAMPLE & WALKTHROUGH** **Problem 7: Sum of Consecutive Triple Products**

**Problem Statement**: Prove that for every positive integer $n \geq 1$,

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \ldots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

**Proof by Mathematical Induction**: Let $P(n)$ be the proposition that $\sum_{k=1}^{n} k(k+1)(k+2) = \frac{n(n+1)(n+2)(n+3)}{4}$ for $n \geq 1$.

1. **Basis Step ($P(1)$)**: For $n = 1$:
   - LHS $= 1 \cdot 2 \cdot 3 = 6$.
   - RHS $= \frac{1(1+1)(1+2)(1+3)}{4} = \frac{1 \cdot 2 \cdot 3 \cdot 4}{4} = 6$.

   Since LHS $=$ RHS $= 6$, $P(1)$ holds.

2. **Inductive Hypothesis**: Assume that $P(n)$ holds for an arbitrary positive integer $n \geq 1$:

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \ldots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4} \quad \text{(IH)}$$

3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $P(n+1)$ holds:

$$1 \cdot 2 \cdot 3 + \ldots + n(n+1)(n+2) + (n+1)(n+2)(n+3) = \frac{(n+1)(n+2)(n+3)(n+4)}{4}$$

   Starting from the LHS of $P(n+1)$ and substituting the Inductive Hypothesis:

$$\begin{aligned} &\underbrace{(1 \cdot 2 \cdot 3 + \ldots + n(n+1)(n+2))}_{\text{apply IH}} + (n+1)(n+2)(n+3) \\ &= \frac{n(n+1)(n+2)(n+3)}{4} + (n+1)(n+2)(n+3) \\ &= \frac{n(n+1)(n+2)(n+3) + 4(n+1)(n+2)(n+3)}{4} \\ &= \frac{(n+1)(n+2)(n+3)(n+4)}{4} \end{aligned}$$

   by factoring out the common product $(n+1)(n+2)(n+3)$. Thus, $P(n+1)$ holds.

**Conclusion**: By mathematical induction, the identity holds for all positive integers $n \geq 1$. □

## 6.5 Inequalities, Strong Induction, and Common Pitfalls

Beyond summation identities, induction is a workhorse for proving **inequalities**, analyzing **recurrence relations**, and establishing **number-theoretic properties**.

### 6.5.1 Induction with Shifted Base Cases ($n \geq n_0$) and Inequalities

Not every true integer proposition starts at $n = 1$ or $n = 0$. Some properties fail for the first few integers and only become true starting from a threshold $n_0 > 1$. When that happens:

- The **Basis Step** verifies $P(n_0)$.
- The **Inductive Step** proves $\forall n \geq n_0, (P(n) \implies P(n+1))$.

**EXAMPLE & WALKTHROUGH** **Induction on Inequalities with Shifted Base Case: $2^n < n!$ for $n \geq 4$**

**Proposition**: Prove that $2^n < n!$ for every integer $n \geq 4$.

**(For $n = 1, 2, 3$, the inequality does not hold: $2^1 = 2 > 1!$, $2^2 = 4 > 2!$, and $2^3 = 8 > 3! = 6$. It first becomes true at $n_0 = 4$.)**

**Proof by Mathematical Induction**: Let $P(n)$ be the inequality $2^n < n!$.

1. **Basis Step ($P(4)$)**: For $n = 4$, LHS $= 2^4 = 16$ and RHS $= 4! = 24$. Since $16 < 24$, $P(4)$ is true.
2. **Inductive Hypothesis**: Assume $P(n)$ holds for an arbitrary integer $n \geq 4$, so $2^n < n!$.
3. **Inductive Step ($P(n) \implies P(n+1)$)**: We want to show that $2^{n+1} < (n+1)!$. Starting from the left-hand side:

$$\begin{aligned} 2^{n+1} &= 2 \cdot 2^n \\ &< 2 \cdot n! && \text{by the Inductive Hypothesis } (2^n < n!) \\ &< (n+1) \cdot n! && \text{since } n \geq 4 \implies 2 < n+1 \\ &= (n+1)! \end{aligned}$$

Thus, $2^{n+1} < (n+1)!$, so $P(n+1)$ holds.

**Conclusion**: By mathematical induction, $2^n < n!$ for all integers $n \geq 4$. □

### 6.5.2 Strong Mathematical Induction

In simple induction, proving $P(n+1)$ relies **only** on the single immediately preceding case $P(n)$. However, in many recurrence relations (like the Fibonacci sequence $F_{n+1} = F_n + F_{n-1}$), number-theoretic factorizations, and divide-and-conquer algorithms (like Binary Search or Merge Sort where problem size drops from $n$ to $\lfloor \frac{n}{2} \rfloor$), proving $P(n+1)$ requires knowing that the property holds for **smaller integers earlier than $n$**.

**DEFINITION** **Principle of Strong Induction**

Let $P(n)$ be a predicate defined for integers $n \geq n_0$. To prove $\forall n \geq n_0, P(n)$ by **strong induction**:

1. **Basis Step**: Prove that $P(n_0)$ (and any additional initial cases $P(n_0+1), ..., P(n_0+r)$ required by the recurrence) is true.
2. **Strong Inductive Hypothesis**: Assume that $P(j)$ is true for **all** integers $j$ in the range $n_0 \leq j \leq n$:

$$P(n_0) \land P(n_0+1) \land ... \land P(n)$$

3. **Inductive Step**: Prove that $P(n+1)$ is true using any combination of the assumptions $P(n_0), ..., P(n)$.

| Feature | Simple (Weak) Induction | Strong Induction |
|---|---|---|
| **Base Step** | Verify $P(n_0)$ (single starting point) | Verify $P(n_0), ..., P(n_0+r)$ (one or more initial cases) |
| **Inductive Hypothesis** | Assume $P(n)$ holds for a **single** integer $n \geq n_0$ | Assume $P(j)$ holds for **all** integers $n_0 \leq j \leq n$ |
| **Typical Use Cases** | Series sums ($\sum a_k$), single-step recurrences ($T(n+1) = f(T(n))$), divisibility | Prime factorization, multi-term recurrences (Fibonacci), divide-and-conquer ($T(n) = 2T(\frac{n}{2}) + n$) |
| **Logical Power** | Equivalent to Strong Induction and Well-Ordering | Equivalent to Simple Induction and Well-Ordering |

**EXAMPLE & WALKTHROUGH** **Strong Induction Classic: Existence of Prime Factorizations**

**Proposition**: Every integer $n \geq 2$ can be written as a prime or a product of primes.

**Proof by Strong Induction**: Let $P(n)$ be the statement "$n$ can be written as a product of one or more prime numbers."

1. **Basis Step ($P(2)$)**: For $n = 2$, 2 is itself prime, so $P(2)$ is true.

2. **Strong Inductive Hypothesis**: Assume that for some integer $n \geq 2$, $P(j)$ holds for all integers $2 \leq j \leq n$ (every integer from 2 up to $n$ is prime or a product of primes).
3. **Inductive Step**: Consider the integer $n+1$. There are two cases:
   - **Case 1 ($n+1$ is prime)**: Then $n+1$ is already a prime, so $P(n+1)$ holds.
   - **Case 2 ($n+1$ is composite)**: By definition of a composite number, $n+1 = a \cdot b$ for two integers $a, b$ satisfying $2 \leq a \leq n$ and $2 \leq b \leq n$. Simple induction ($P(n)$) cannot help us here because neither $a$ nor $b$ is equal to $n$. However, by our **Strong Inductive Hypothesis**, since $2 \leq a, b \leq n$, both $P(a)$ and $P(b)$ hold: $a$ is a product of primes and $b$ is a product of primes. Multiplying them together shows that $n+1 = a \cdot b$ is also a product of primes.

**Conclusion**: By strong induction, every integer $n \geq 2$ is a prime or a product of primes. □

### 6.5.3 Common Pitfalls & Flawed Inductive Proofs

#### Four Common Pitfalls in Mathematical Induction Proofs

**1. Omitting or Mismatching the Base Case**

The inductive step $P(n) \implies P(n+1)$ can hold even for a **false** formula! For instance, if $P(n)$ claims "$1 + 2 + \ldots + n = \frac{n(n+1)}{2} + 99$", the inductive step still works algebraically because adding $n+1$ to both sides preserves $+99$. Only the base case $P(1)$ catches the error ($1 \neq 100$). Also pay close attention to whether the domain starts at $n = 0$ or $n = 1$.

**2. The 'All Horses Are the Same Color' Fallacy**

George Pólya's famous paradoxical proof claims that in any set of $n$ horses, all horses have the same color. The base case $n = 1$ is trivially true. In the step $n \to n+1$, removing the first horse leaves $n$ horses of the same color, and removing the last horse leaves $n$ horses of the same color; the proof assumes the two groups overlap. **Flaw**: When going from $n = 1$ to $n + 1 = 2$, the two subsets of size 1 have **no overlap at all**, so $P(1) \not\Rightarrow P(2)$.

**3. Working Backwards From P(n+1) to True**

Writing down the target equation $\text{LHS}_{n+1} = \text{RHS}_{n+1}$ and manipulating both sides until reaching $0 = 0$ is a **logical fallacy** ($Q \implies$ True does not prove $Q$ unless every step is reversible $\iff$). Always start on one side (such as $\text{LHS}_{n+1}$) and derive the other side ($\text{RHS}_{n+1}$) in a single chain of equalities.

**4. Confusing the Predicate P(n) with a Number**

$P(n)$ is a **proposition** (a true/false statement), not a number! Writing "$P(n+1) = P(n) + (n+1)$" is a type error because you cannot add an integer $n+1$ to a truth value $P(n)$. Instead, write out the left-hand side of the sum and substitute the inductive hypothesis into the numerical expression.

**PITFALL / WARNING** **Practical Checklist for Writing Induction Proofs**

When writing a proof by mathematical induction, double-check these five details:

1. **Check the Starting Integer**: Does the statement begin at $n = 1$ (positive integers) or $n = 0$ (non-negative integers), or some larger threshold $n_0$? Make sure your base case matches the stated domain.
2. **Highlight Where the Inductive Hypothesis Is Used**: Clearly mark the step where you substitute the Inductive Hypothesis (IH), since that is the logical heart of the proof.
3. **Find the Correct $(n+1)$-st Term**: When moving from $n$ to $n+1$ in a sum like $\sum_{k=1}^{n}(2k-1)$, replace $n$ with $(n+1)$ inside parentheses to get $2(n+1) - 1 = 2n + 1$ (do not simply add 1 to $2n - 1$).

4. **Factor Before Expanding**: Factor out common terms like $(n+1)$ or $(2n+3)$ early when combining fractions so you do not have to expand and re-factor cubic polynomials.
5. **Write a Clear Conclusion**: Finish with a brief closing sentence stating that the proposition holds for all $n \geq n_0$ by mathematical induction.